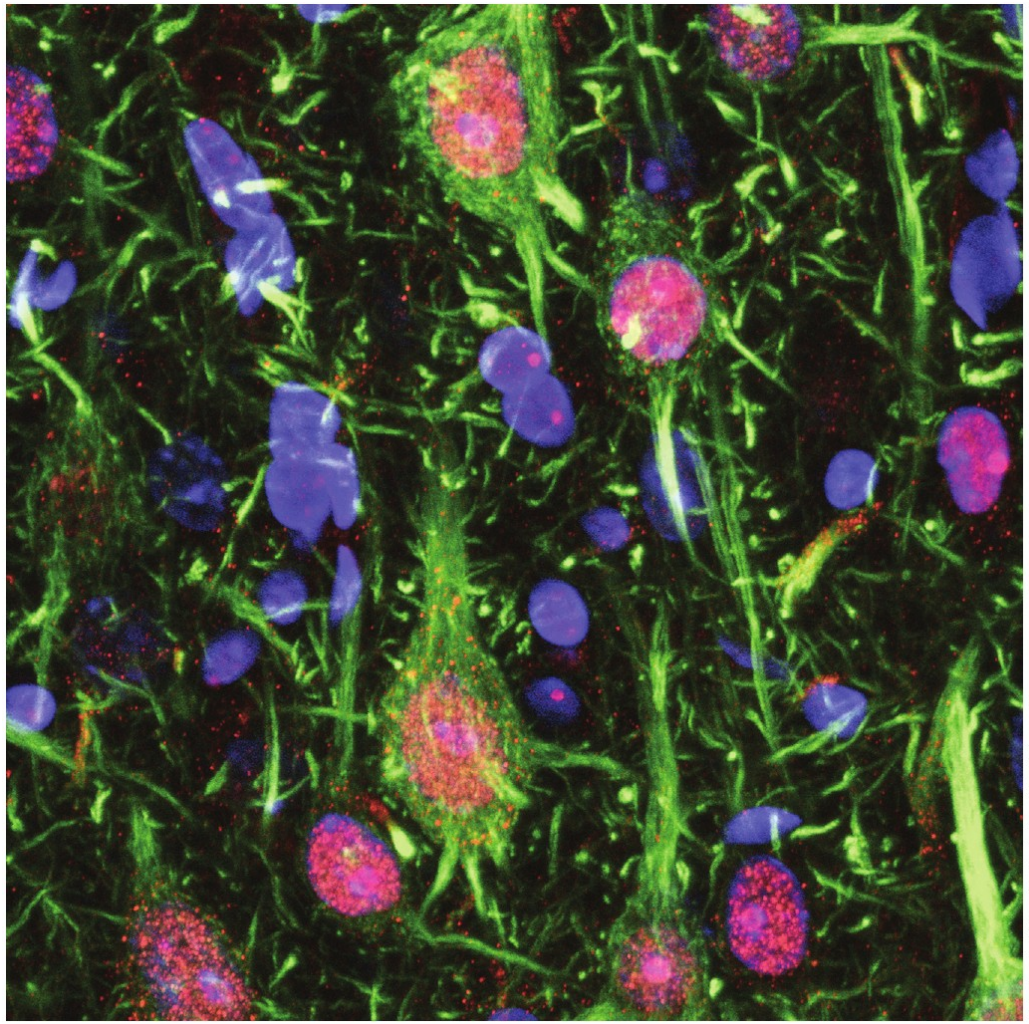


Carlson and Birkett (2022), chapter opener, p. 91. Photo: C. J. Guerin / Science Source.

BEHAVIORAL NEUROSCIENCE 392

Drugs, Receptors, and Neural Signaling

Chemical Control of Behavior: Drugs and Neural Signaling

Fall 2026

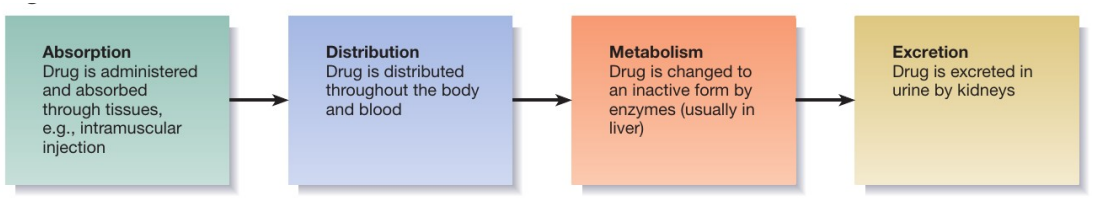
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Chemicals change behavior by acting on living cells

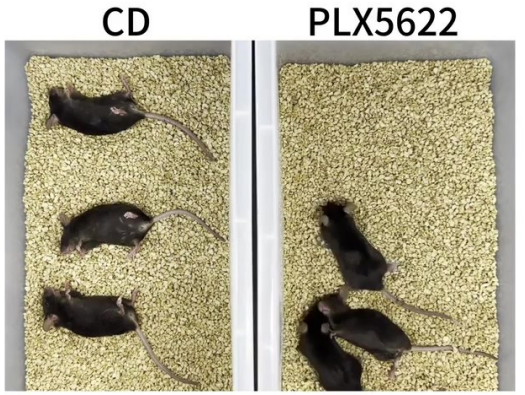
Psychopharmacology examines how drugs change nervous-system function and behavior. A **drug** is a chemical introduced from outside the body that alters cellular activity at relatively low amounts. Its behavioral effect depends on the cells it reaches, the molecules it acts on, and the signals those cells send to other targets.

A drug's **site of action** is the molecular or cellular location where it produces a change. Its **effect** is the resulting alteration in physiology or behavior. A sleeping animal, a slower heart, and a reduced pain response are observable outcomes; the proteins that produced those outcomes occupy particular places inside the body.

The same chemical can produce different responses in different preparations. General anesthetics provide a concrete example: their effects depend on neural circuits and supporting cells as well as the administered compound. In mice, changing microglial populations alters the transition into and out of anesthesia, placing a drug response within its living tissue environment.



Carlson and Birkett (2022), Fig. 4.2, p. 94.



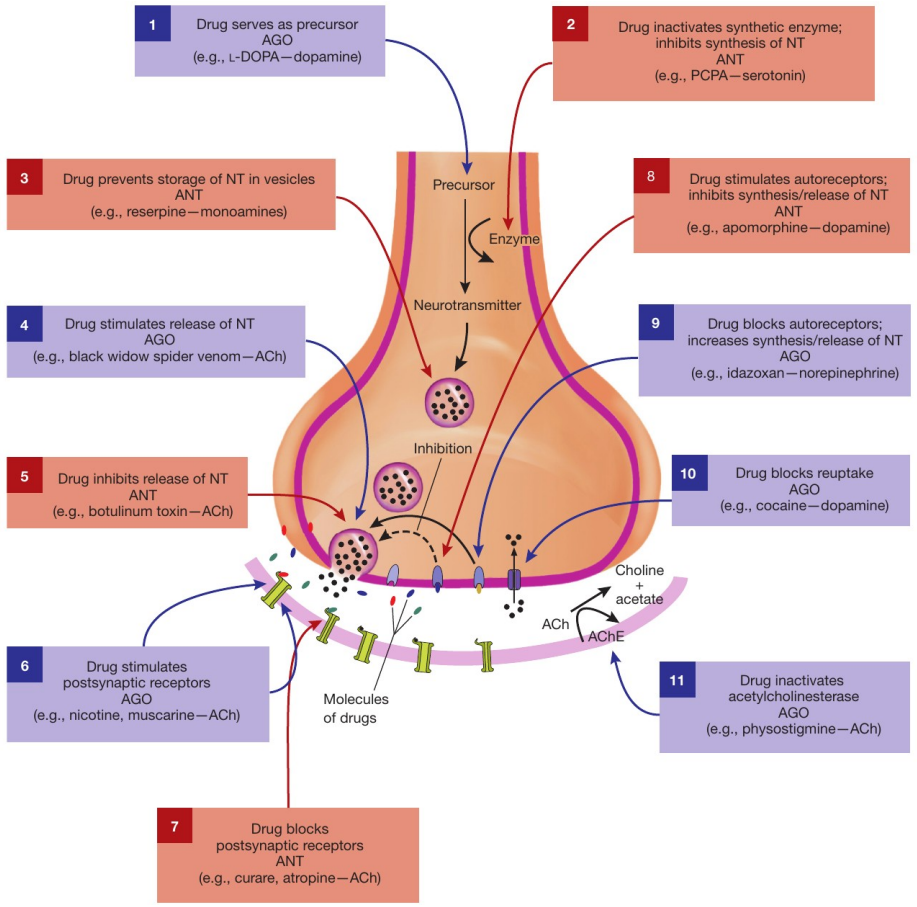
He et al. (2023), video1. Loss of the righting reflex after anesthesia.

A drug and a neurotransmitter can act on the same receptor

A **neurotransmitter** is a chemical released by a neuron that changes the activity of a receiving cell. A **receptor** is a protein that recognizes a chemical signal and initiates a response. A drug can activate or block a receptor normally used by a messenger produced within the nervous system.

Endogenous substances originate within the body, whereas **exogenous** substances enter from outside it. Dopamine released from a nerve terminal is endogenous. A compound administered to activate dopamine receptors is exogenous. Their origin differs even when both affect the same receptor population and influence some of the same cellular processes.

A receptor does not recognize a chemical according to its purpose as medicine or recreational drug. Binding depends on molecular properties, and the response depends on the receptor and receiving cell. Explaining a drug effect therefore requires identifying the target protein and the cellular changes that follow its activation or inhibition.



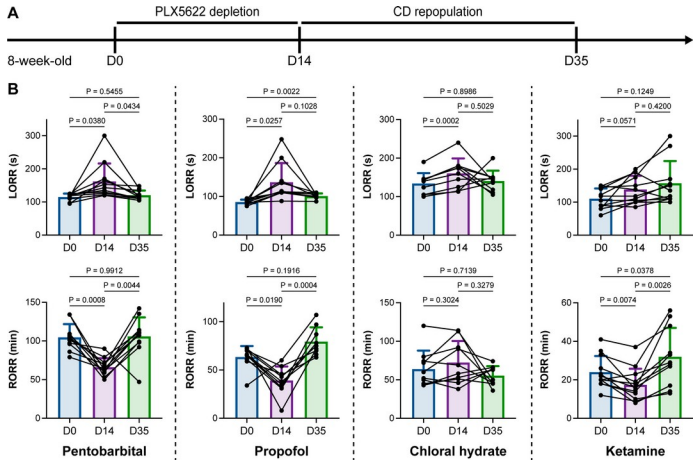
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Microglial depletion changes the response to anesthesia

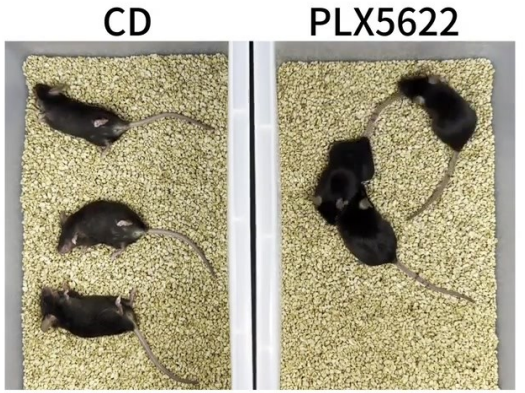
In a mouse study, the investigators reduced microglial populations with **PLX5622**, a compound that inhibits a receptor needed for microglial survival. They then measured responses to general anesthesia. The intervention changed a supporting-cell population before the anesthetic was given, allowing the contribution of that tissue environment to be examined directly.

Microglia-depleted mice took longer to lose the **righting reflex**, the movement that normally restores an animal from its back to an upright position. They also recovered the reflex sooner during emergence. These measurements provided defined behavioral endpoints for comparing the response to anesthesia across the two experimental conditions.

The result separates the administered anesthetic from the state of the nervous system receiving it. Changing microglia altered induction and recovery even though the experiment still used general anesthetic treatment. The investigators then examined neuronal activity and microglial signaling to explain how nonneuronal cells participate in the response of central networks.



He et al. (2023), Fig. 1 (A–B). Anesthesia induction and emergence.



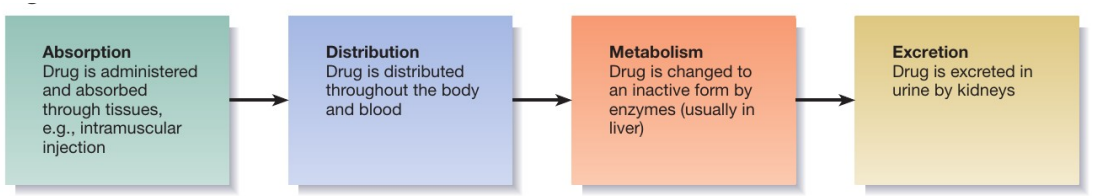
He et al. (2023), video2. Recovery of the righting reflex.

Absorption determines how a drug enters circulation

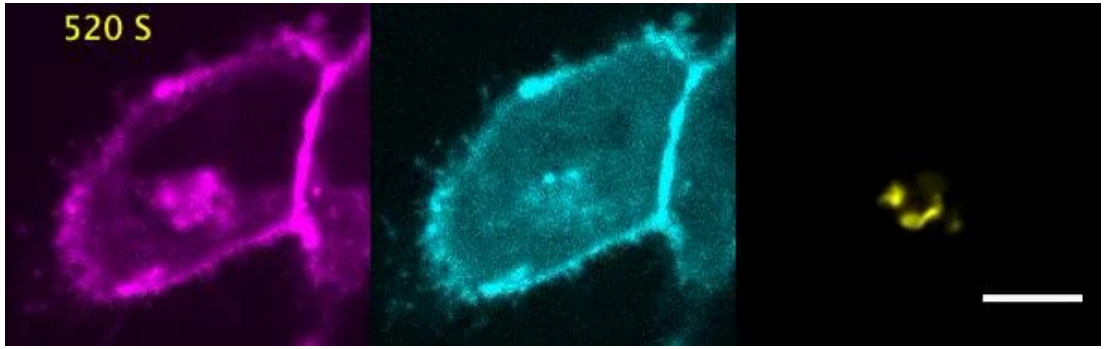
Pharmacokinetics describes how a drug moves through the body and is removed from it. Its major processes are **absorption**, **distribution**, **metabolism**, and **excretion**. Absorption carries a substance from its administration site into the bloodstream, beginning the route by which it can reach tissues distant from where it entered.

The **route of administration** changes the tissues a drug encounters before reaching the blood. An oral drug must be released and absorbed from the digestive tract, while an intravenous injection enters circulation directly. Inhaled substances cross lung surfaces, and some compounds pass gradually through skin from a controlled delivery system.

After oral absorption, blood from the digestive tract passes through the liver before reaching much of the body. **Metabolism** chemically modifies a drug, sometimes producing an active metabolite. This **first-pass metabolism** changes **bioavailability**, the fraction of the administered substance reaching systemic circulation, so swallowed and injected amounts need not produce the same exposure.



Carlson and Birkett (2022), Fig. 4.2, p. 94.



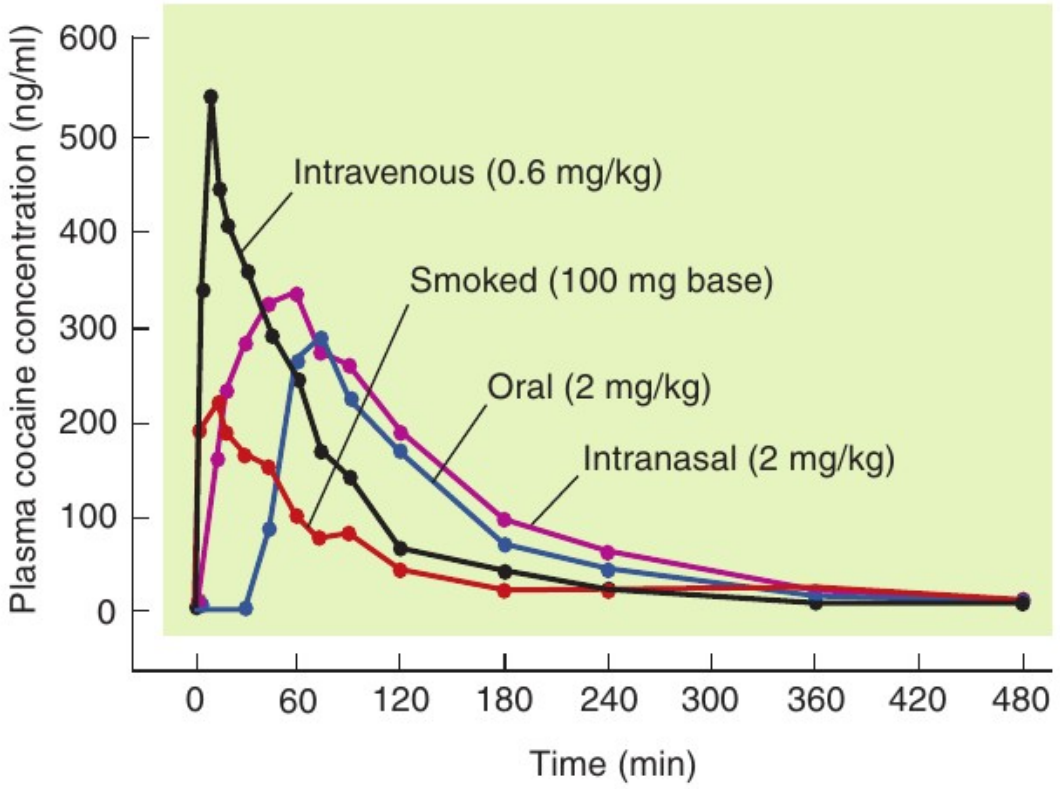
Puri et al. (2022), fig1-video1. Dopamine applied to D1 receptor-expressing HeLa cells.

Distribution determines which tissues a drug reaches

Distribution is the movement of a drug from circulation into tissues. Blood delivers a substance around the body, but entry into a tissue depends on molecular properties, transport proteins, and the barriers at that location. A high concentration in plasma does not guarantee an equally high concentration beside a receptor in the brain.

The **blood–brain barrier** restricts passage from brain capillaries into neural tissue. Many lipid-soluble molecules enter more readily than strongly charged molecules, while selected transporters carry particular compounds across membranes. Binding to blood proteins also changes the fraction available to leave circulation and reach other tissues during a given interval.

Excretion removes a drug or its metabolites from the body, often through the kidneys. Metabolism and excretion together influence **clearance**, the removal of active substance from circulation. A **half-life** is the time required for concentration to fall by half during a specified elimination phase, helping describe how exposure changes after administration stops.



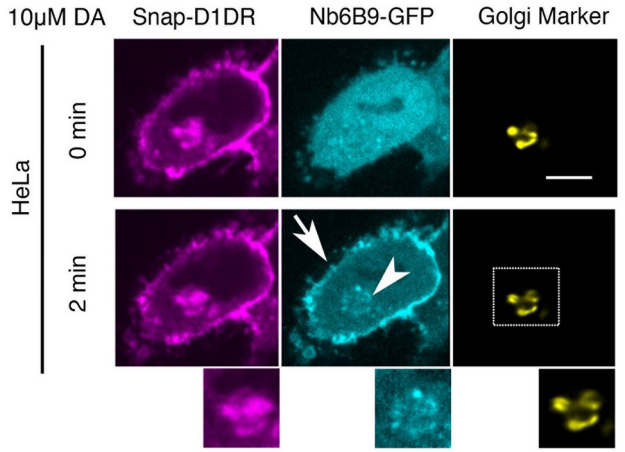
Carlson and Birkett (2022), Fig. 4.3, p. 95.

Drug access differs between surface and internal receptors

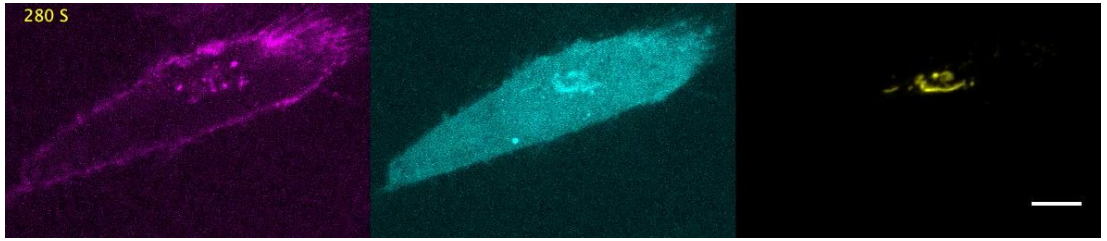
A dopamine-receptor study examined **D1 receptors**, proteins that activate intracellular signaling after binding an appropriate chemical. The investigators expressed labeled receptors and fluorescent activity detectors in cultured cells. They distinguished receptors at the **plasma membrane**, the cell's outer boundary, from receptors at the **Golgi apparatus**, an internal membrane compartment.

An **agonist** activates a receptor. The researchers compared dopamine with a synthetic D1 agonist that could reach internal compartments more readily. The two compounds acted on the same receptor class, but access across cellular membranes affected where receptor activation occurred within the cell during the exposure.

This experiment applies distribution at a smaller spatial scale. Reaching a tissue and reaching a protein inside one of its cells are separate steps. A receptor can be present and capable of responding while remaining inaccessible to a particular extracellular chemical unless that chemical can cross the intervening membrane or use a transporter.



Puri et al. (2022), Fig. 1 (b, HeLa). D1 receptor activation at two cellular locations.



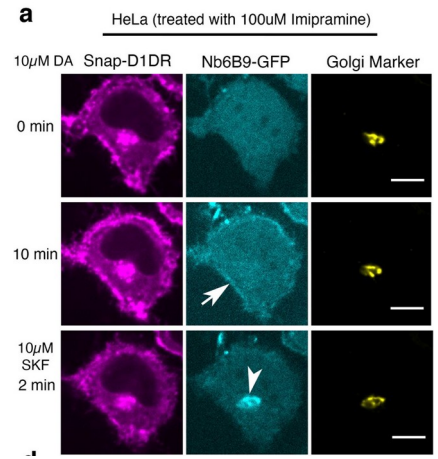
Puri et al. (2022), fig1-video3. A membrane-permeant D1 agonist in HeLa cells.

A transporter gives dopamine access to an internal target

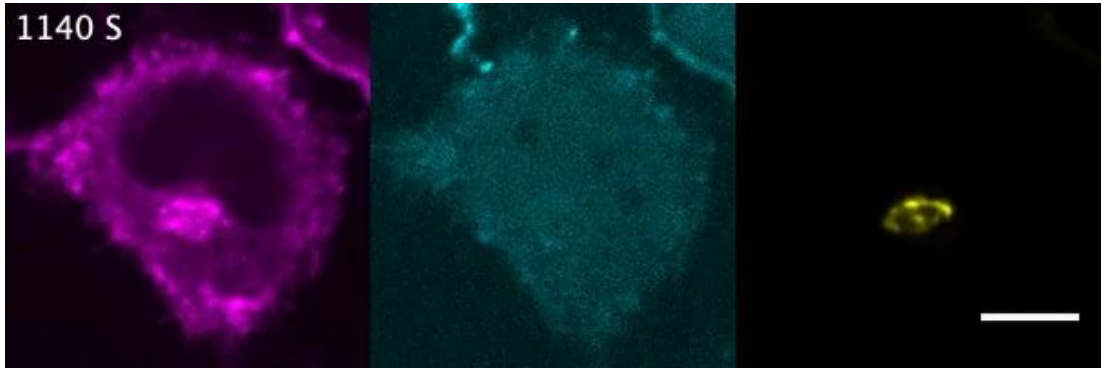
The D1-receptor investigators examined **organic cation transporter 2**, or **OCT2**, a membrane protein that carries certain positively charged compounds. Dopamine crosses lipid membranes poorly on its own. The experiment tested whether transport through OCT2 allowed dopamine applied outside a cell to activate a receptor population associated with the internal Golgi compartment.

In cells that used this transport route, dopamine activated Golgi-associated D1 receptors. Pretreatment with a transporter-inhibiting compound reduced that response, whereas the membrane-permeant synthetic agonist could still activate the internal population. This comparison separated impaired chemical access from a complete loss of the receptor's ability to respond.

The inhibitor was used as an experimental tool to change one step between application and signaling. Its presence altered the route available to the ligand, the chemical that binds the receptor. The findings explain why membrane permeability and transport must be considered alongside receptor binding when comparing the cellular actions of different agonists.



Puri et al. (2022), Fig. 2 (a). Dopamine response during transporter inhibition.



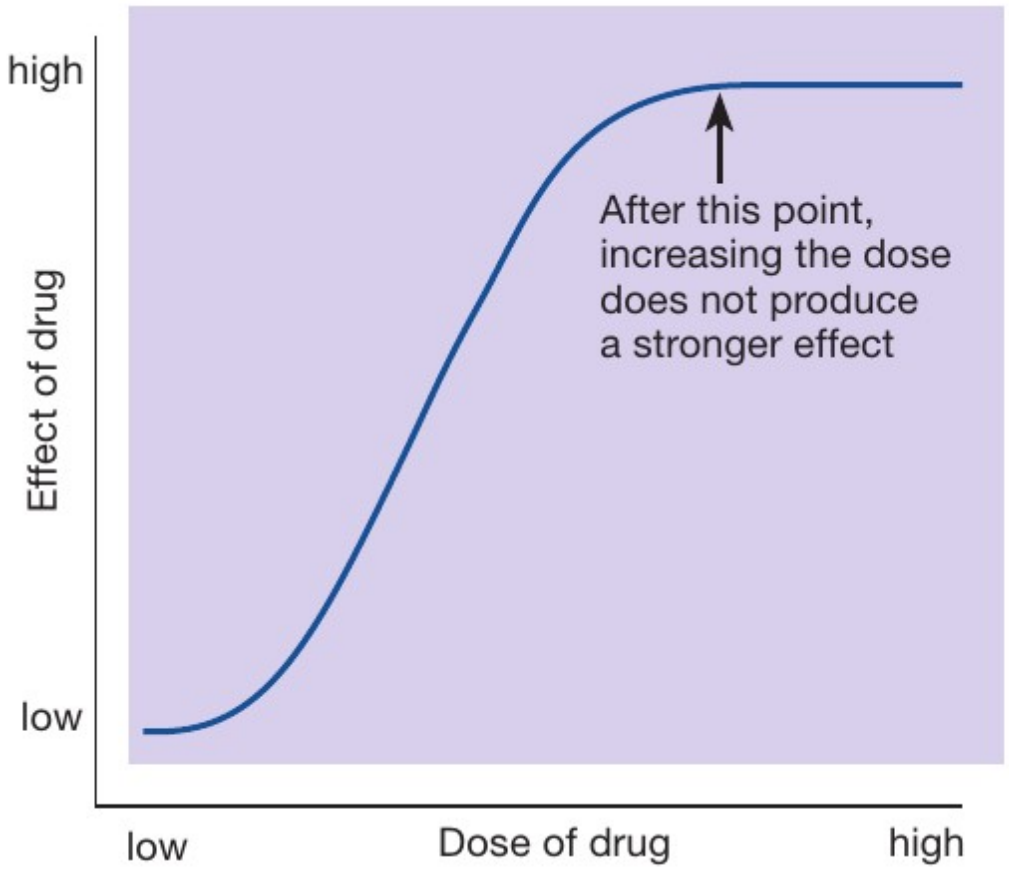
Puri et al. (2022), fig2-video1. D1 receptor response during OCT2 inhibition.

Potency and efficacy describe different properties of a drug

A **dose–response curve** relates an administered amount or concentration to a measured effect. The rising portion indicates that increasing exposure produces a larger response within that range. The curve can level off when the measured system approaches its maximum response, even though adding more drug may still affect other targets.

Potency describes how much drug is required to produce a specified effect. A more potent compound reaches that effect at a lower dose under the tested conditions. **Efficacy** describes the maximum effect a compound can produce in that system, so greater potency does not necessarily mean a larger maximum response.

Affinity describes a ligand's tendency to bind a receptor, while potency also depends on exposure and how receptor activation produces the measured effect. Two compounds can differ in potency for movement, pain, or cellular signaling. Comparing dose–response curves therefore requires keeping the preparation, measurement, and exposure conditions explicit rather than calling one compound stronger in every respect.



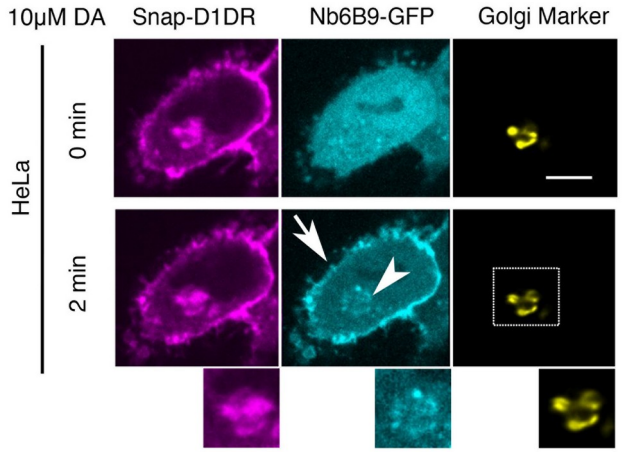
Carlson and Birkett (2022), Fig. 4.4, p. 96.

The same receptor class operates in different compartments

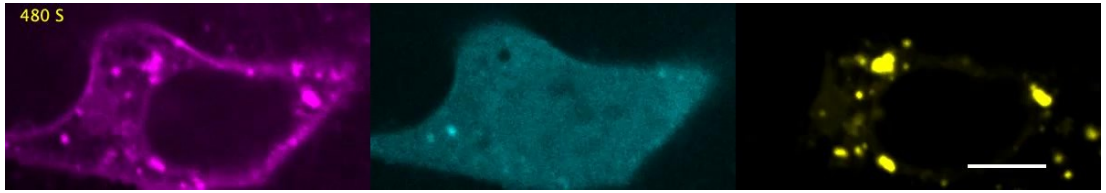
The D1 study used fluorescent detectors that recognize an activated receptor state. When dopamine or a synthetic agonist reached responsive D1 receptors, the detectors moved toward the activated population. Their changing position made it possible to examine signaling at the surface and at the Golgi within the same cellular preparation.

Experiments in two cell lines provided different cellular settings for the receptor. The proteins were expressed experimentally, so the study could compare access and activation under controlled conditions. The resulting images identified where the detector accumulated, separating the location of receptor protein from the location of detectable agonist-induced activity.

A named receptor class therefore does not imply one universal signaling location. Membrane compartments contain different proteins and transport routes, and their local environments influence what an applied compound can activate. The experiment defines the receiving compartment before asking how activity there contributes to the cell's broader chemical response.



Puri et al. (2022), Fig. 1 (b, HeLa). D1 receptor activation at two cellular locations.



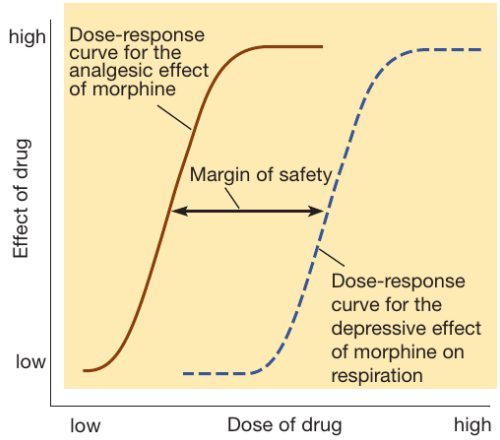
Puri et al. (2022), fig1-video2. Dopamine applied to D1 receptor-expressing HEK293 cells.

Desired and adverse effects can rise at different doses

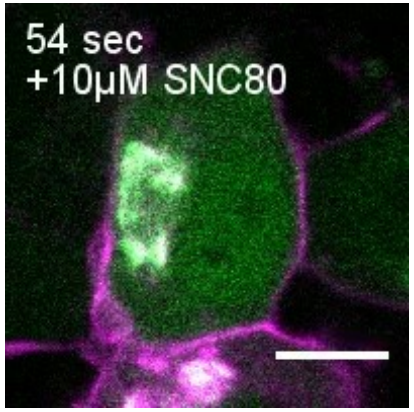
A **therapeutic effect** is a desired change produced by treatment, while an **adverse effect** is an unwanted change. The same drug can produce both because its targets occur in several tissues or because one target regulates several functions. Separate dose–response curves describe how each measured outcome changes with exposure.

A **therapeutic window** is the exposure range in which useful effects can occur without unacceptable adverse effects. The separation depends on the drug, the person or preparation, and the outcomes being compared. A compound can strongly activate its target while offering little separation between its desired and unwanted physiological consequences.

Receptor location helps explain this problem. Opioid-sensitive pathways influence pain processing, breathing, and other functions, so activating the broader system produces more than one outcome. Comparing responses across doses and tissues is necessary to understand the available separation between effects, rather than assuming that selective receptor binding guarantees selective behavior.



Carlson and Birkett (2022), Fig. 4.5, p. 97.



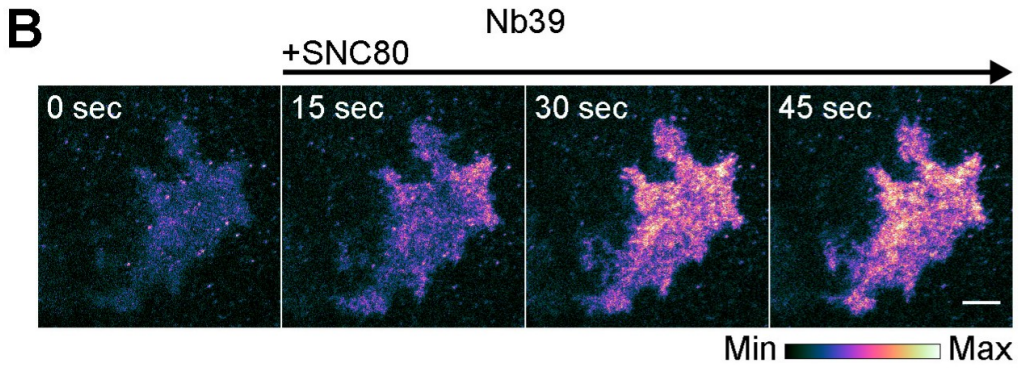
Crilly et al. (2021), video1. Nb39 recruitment to intracellular opioid receptors.

An opioid receptor's location changes its active state

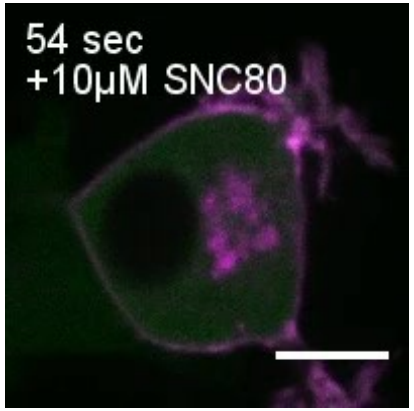
An opioid-receptor study examined the **delta-opioid receptor**, or **DOR**, in cultured PC12 cells. DOR is a **G protein-coupled receptor**, meaning it communicates through intracellular G proteins after activation. The investigators compared receptors at the cell surface with receptors in the Golgi while applying the same receptor-activating drug.

Two fluorescent detectors recognized different features of receptor activation. One accumulated at both surface and internal receptors after the agonist was added, while the other preferentially reported the surface population. The difference indicated that receptors exposed to the same ligand did not present identical active conformations in the two compartments.

A **conformation** is a protein's three-dimensional arrangement. Local membrane composition and nearby proteins can influence which arrangements and interactions occur after binding. Drug action therefore depends on the environment around the receptor, providing a cellular explanation for why one ligand can recruit different downstream processes at different locations.



Crilly et al. (2021), Fig. 1 (B). Opioid receptor biosensor recruitment.



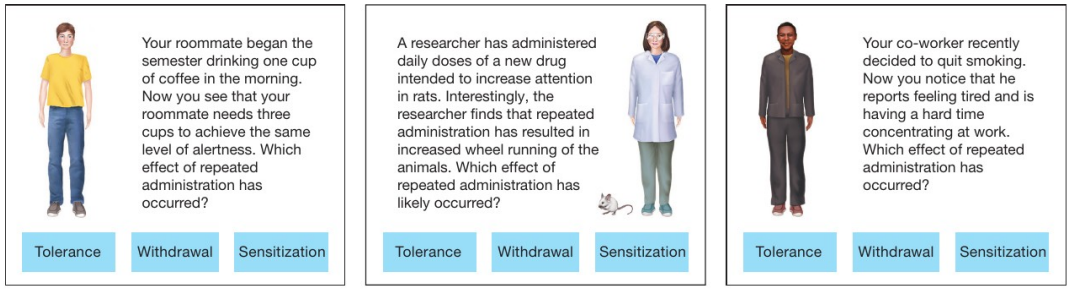
Crilly et al. (2021), video2. A second biosensor tests intracellular opioid receptors.

Repeated exposure can decrease or increase a response

Tolerance develops when repeated exposure produces a smaller response, so a previously effective dose has less effect. Cells can change receptor availability, signaling proteins, or other processes during continued treatment. Metabolism can also change, altering the exposure produced by the same administered amount before it reaches its targets.

Sensitization is an increase in a response after repeated exposure. A drug can produce tolerance in one measured effect and sensitization in another because the underlying circuits and adaptations differ. Describing repeated administration therefore requires naming the particular outcome rather than assigning one universal direction of change to the whole drug.

Withdrawal occurs when stopping or reducing a repeatedly used drug exposes adaptations that developed during its presence. Compensatory processes can temporarily push function away from its usual range when the compound is removed. The resulting symptoms depend on the affected systems and the history of exposure, rather than being identical for every psychoactive substance.



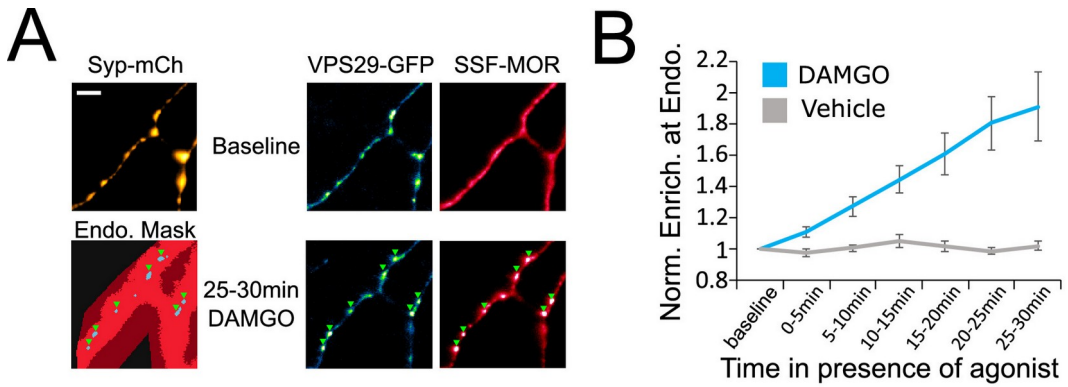
Carlson and Birkett (2022), Fig. 4.6, p. 98.

Opioid exposure removes receptors from an axon's surface

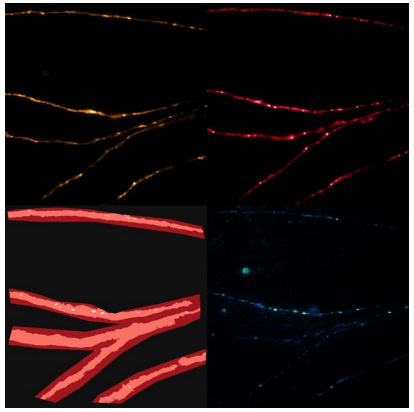
An opioid-tolerance study used cultured rat **medium spiny neurons**, a major striatal neuron type, to examine receptors in axons. The investigators labeled surface **mu-opioid receptors**, or **MORs**, and followed their movement after adding a receptor agonist. This preparation made the membrane location of receptors measurable during sustained chemical exposure.

Activated receptors entered the cell through **endocytosis**, the uptake of surface membrane and its proteins into internal compartments. Repeated cycles of uptake and return gradually depleted receptors available on the axonal surface. The cellular response to prolonged agonist exposure therefore changed as the receptors moved between membrane locations.

The study measured declining opioid inhibition of transmitter release alongside these trafficking events. Its account places tolerance within a defined presynaptic preparation: fewer available surface receptors reduced the effect of continued activation. This cellular mechanism explains a measured adaptation in the cultured neurons rather than equating that preparation with every component of whole-animal opioid tolerance.



Jullié et al. (2022), Fig. 2 (A–B). Mu opioid receptor endosomal recruitment.



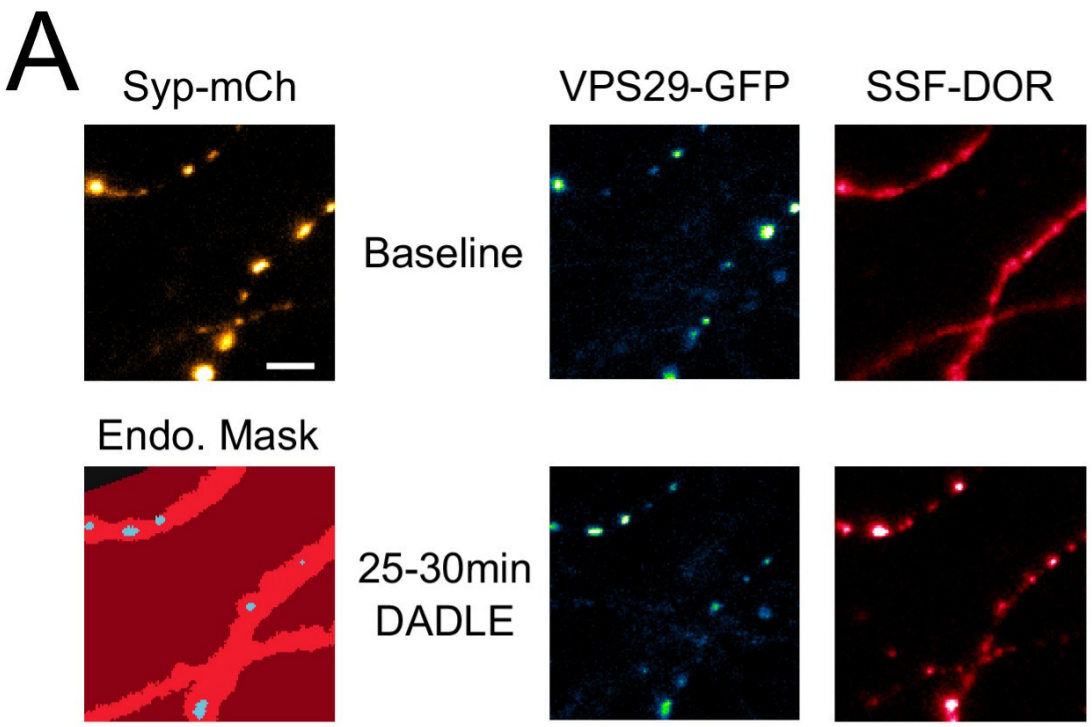
Jullié et al. (2022), fig2-video1. Mu opioid receptor internalization in axons.

Opioid receptor subtypes differ in their recycling

The tolerance investigators compared mu- and delta-opioid receptors in axons. Both underwent agonist-induced internalization, but their return to the surface differed. **Recycling** sends internalized membrane proteins back to the plasma membrane, allowing receptors to become available again rather than remaining within an intracellular compartment after the initial activation.

Delta receptors recycled less efficiently than mu receptors in the tested neurons and developed stronger presynaptic tolerance. The comparison placed receptor movement beside the changing physiological response. Two receptors can respond to related drug classes yet show different adaptation because the cell handles their activated populations differently over time.

The mechanisms initiating uptake also differed between subtypes. Mu-receptor internalization required selected phosphorylation events, the addition of phosphate groups to the receptor, whereas delta-receptor uptake could occur without the same requirement. The effect of prolonged drug exposure therefore depended on subtype-specific cellular machinery as well as agonist binding.



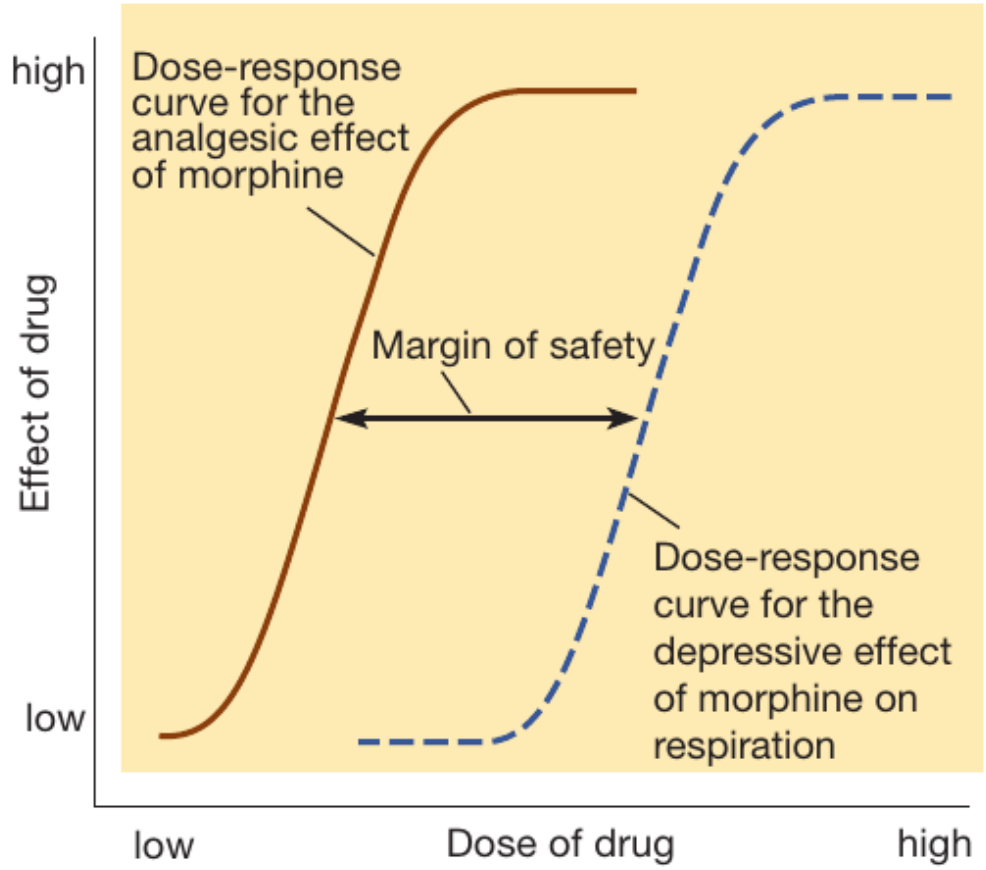
Jullié et al. (2022), Fig. 4 (A). Delta opioid receptor endosomal recruitment.

Expectations and conditioning change measured responses

A **placebo** resembles a treatment but lacks the active component whose effect is being tested. A **placebo response** is a change occurring in that treatment setting, which may include expectation, learned associations, and other processes. Such changes can involve measurable physiology rather than requiring a person to invent an experience.

An expected reduction in pain can recruit endogenous systems that regulate pain processing. Prior experience with treatment can also make a contextual cue influence later responses. The setting therefore contributes to the measured outcome alongside any direct molecular action of the administered compound, especially when the outcome includes perception or reported symptoms.

A treatment comparison must account for these influences to identify what the active compound contributes. A blinded procedure limits differences in expectations between conditions, while a matched placebo preserves much of the treatment context. The resulting contrast estimates the drug's added effect within that setting rather than treating every observed improvement as its direct chemical action.



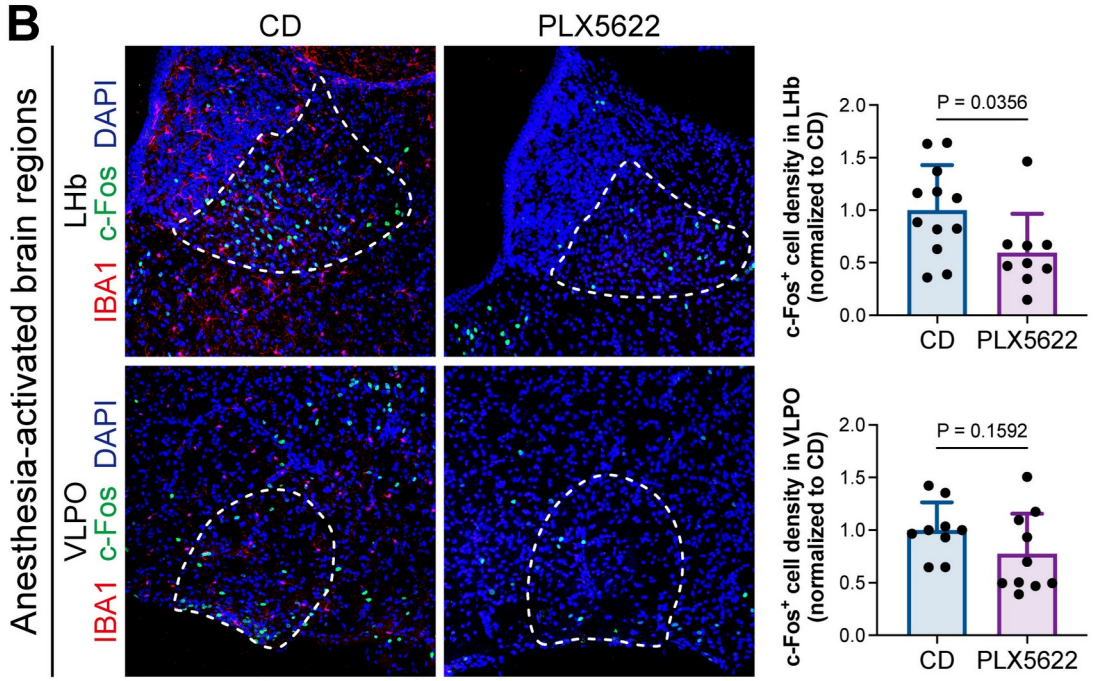
Carlson and Birkett (2022), Fig. 4.5, p. 97.

A drug response depends on the state of neural networks

The anesthesia investigators recorded neuronal activity after microglial depletion. Different brain regions changed in different directions, including populations recruited during anesthesia and populations associated with emergence. The manipulation therefore altered the network receiving the anesthetic, providing a route from a change in supporting cells to a change in the behavioral response.

The researchers also examined **P2Y12**, a receptor used by microglia to detect extracellular nucleotide signals. Its activation and subsequent calcium signaling contributed to the anesthetic response in the study. These observations assigned a signaling mechanism to the supporting cells rather than describing microglia only as a passive marker of treatment.

The findings place pharmacology across interacting cell types. An anesthetic can alter neuronal operation while microglial signaling changes how the network enters or maintains the affected state. The physiological response emerges from those interacting components, explaining why cellular composition and ongoing activity can change the outcome of a chemical exposure.



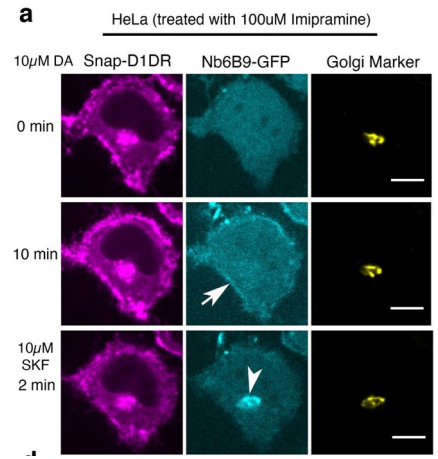
He et al. (2023), Fig. 7 (B, LHb and SON). Regional neuronal activation during anesthesia.

Transporters control access to an internal receptor

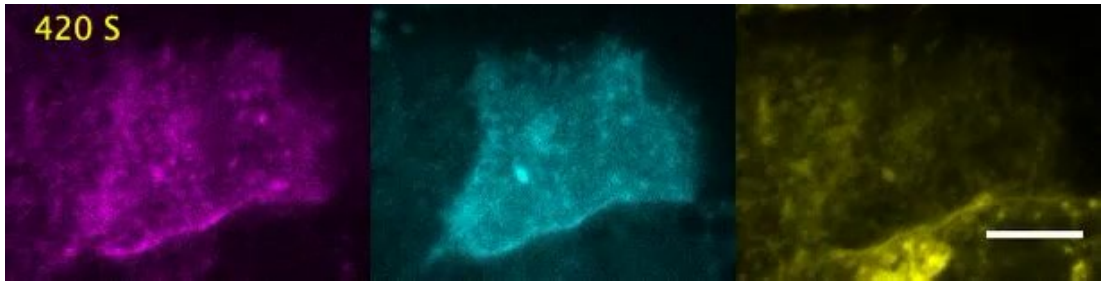
The D1-receptor investigators increased OCT2 expression in a cell preparation and examined dopamine-dependent activation. Cells that acquired the transport route showed stronger access to the internal receptor population. Adding a transporter therefore changed what an externally applied chemical could reach without requiring a different chemical identity for the receptor-binding ligand.

The comparison used labeled transporter, receptor, and activity-detector proteins to identify their locations. A receptor image established the protein's presence, while movement of the activity detector identified activation after exposure. These measurements separated having an internal receptor from being able to deliver an extracellular ligand to that receptor during the experiment.

The same principle helps explain differences among tissues. Cells vary in the transport proteins they express, so a charged messenger may enter one cell population more readily than another. A drug that changes transport can therefore alter local signaling by changing chemical access, in addition to any direct action it has at a receptor.



Puri et al. (2022), Fig. 2 (a). Dopamine response during transporter inhibition.



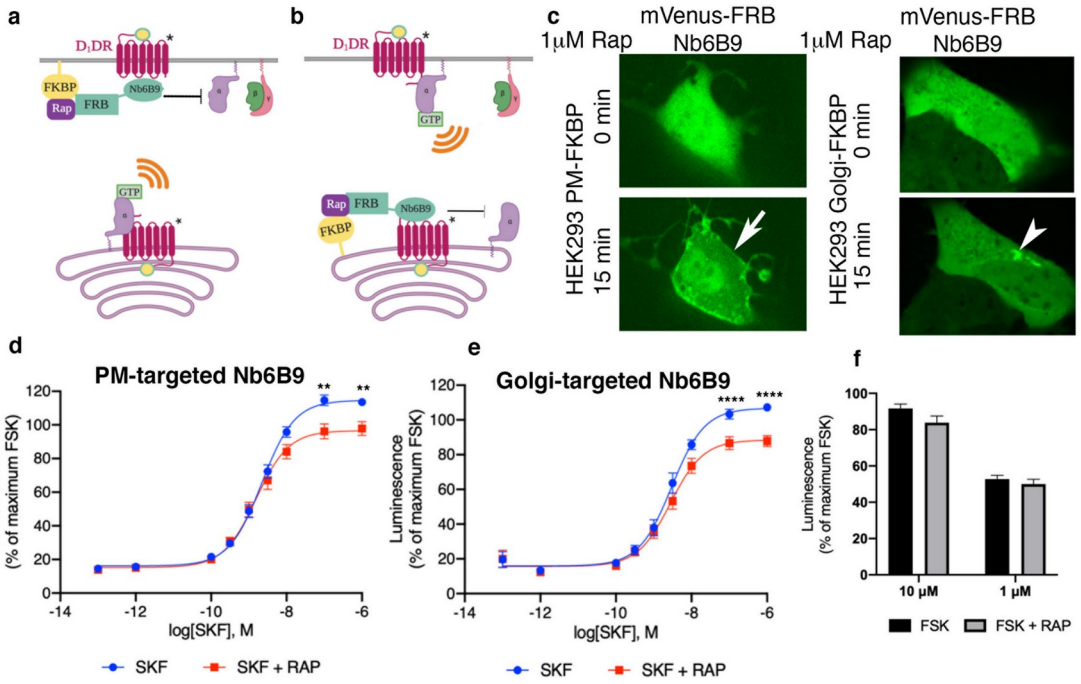
Puri et al. (2022), fig2-video2. Dopamine response in cells expressing OCT2.

Internal D1 receptors contribute to a second messenger

The D1-receptor study measured **cyclic AMP**, or **cAMP**, a small intracellular messenger made after selected G protein signals activate an enzyme. A **second messenger** carries a signal inside the cell after the extracellular ligand has activated its receptor. The receptor and the messenger therefore occupy different steps in the response.

The investigators selectively interfered with G protein coupling at the Golgi-associated receptor population. This changed the contribution of that compartment to cAMP signaling while allowing other receptor locations to be considered separately. The experiment tested whether internal receptor activation produced a functional chemical output rather than only recruiting a fluorescent detector.

Local cAMP signaling also influenced **protein kinase A**, an enzyme that modifies other proteins by phosphorylation. A ligand-receptor interaction could therefore initiate a sequence extending from a membrane compartment to downstream cellular machinery. Identifying the location of the first step clarified how the same receptor class contributed to spatially organized intracellular responses.



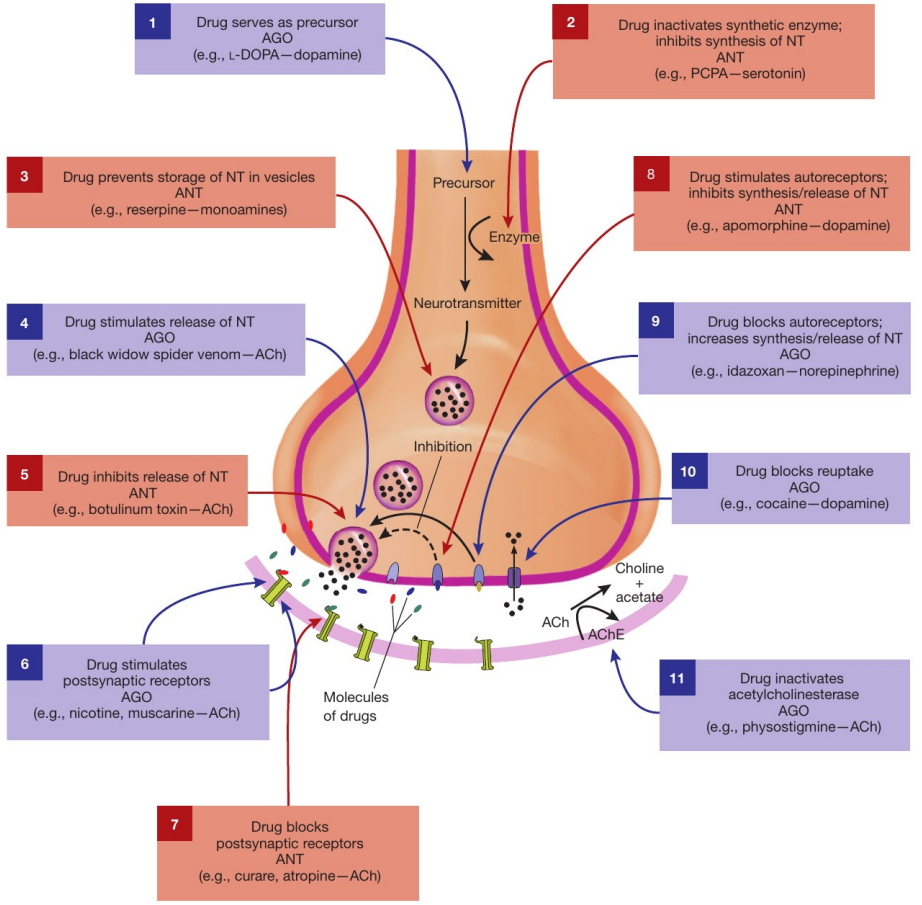
Puri et al. (2022), Fig. 4 (a–f). Compartment-specific D1 receptor inhibition.

Drugs can alter several stages of synaptic communication

At a chemical synapse, a neuron synthesizes a transmitter, stores it, and releases it toward a receiving cell. Receptors detect the released chemical, and transporters or enzymes help terminate its action. Each stage contains proteins that an exogenous compound can alter, creating several routes to changing the strength or duration of communication.

A drug can increase signaling without directly activating the receiving receptor. It might supply a synthetic precursor, increase release, block a presynaptic inhibitory receptor, or slow removal of the transmitter. Conversely, reducing synthesis or release can weaken transmission before the chemical ever reaches its usual postsynaptic binding site.

The resulting effect depends on the direction of the altered step. Blocking a transporter that clears a transmitter differs from blocking a receptor that detects it. Naming only the transmitter system therefore leaves the mechanism incomplete; the specific target and its position in the communication sequence determine how the compound changes signaling.



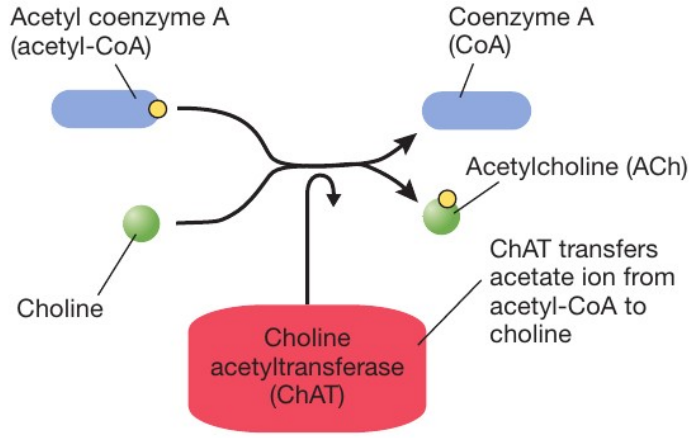
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Proteins and precursors supply chemical signaling

Cells manufacture the proteins needed for chemical signaling, including receptors and enzymes that synthesize transmitters. An **enzyme** accelerates a chemical reaction, while a **precursor** supplies material for that reaction. Changing enzyme activity or precursor availability can alter the transmitter available for later storage and release without directly activating the postsynaptic receptor.

Acetylcholine is synthesized from choline and acetyl coenzyme A by **choline acetyltransferase**. Other systems use different precursors and enzymes, so one synthesis inhibitor does not automatically reduce every transmitter. For example, catecholamine production begins with tyrosine, while **gamma-aminobutyric acid**, or **GABA**, is produced from glutamate by a different enzyme.

Receptor proteins also require synthesis and delivery to useful cellular locations. Newly made AMPA-receptor components travel within dendrites before insertion into the surface. The unstimulated transport preparation makes this ongoing supply visible, placing receptor availability beside transmitter production as another cellular process that determines how strongly a neuron can communicate through a chemical signal.



Carlson and Birkett (2022), Fig. 4.12, p. 107.



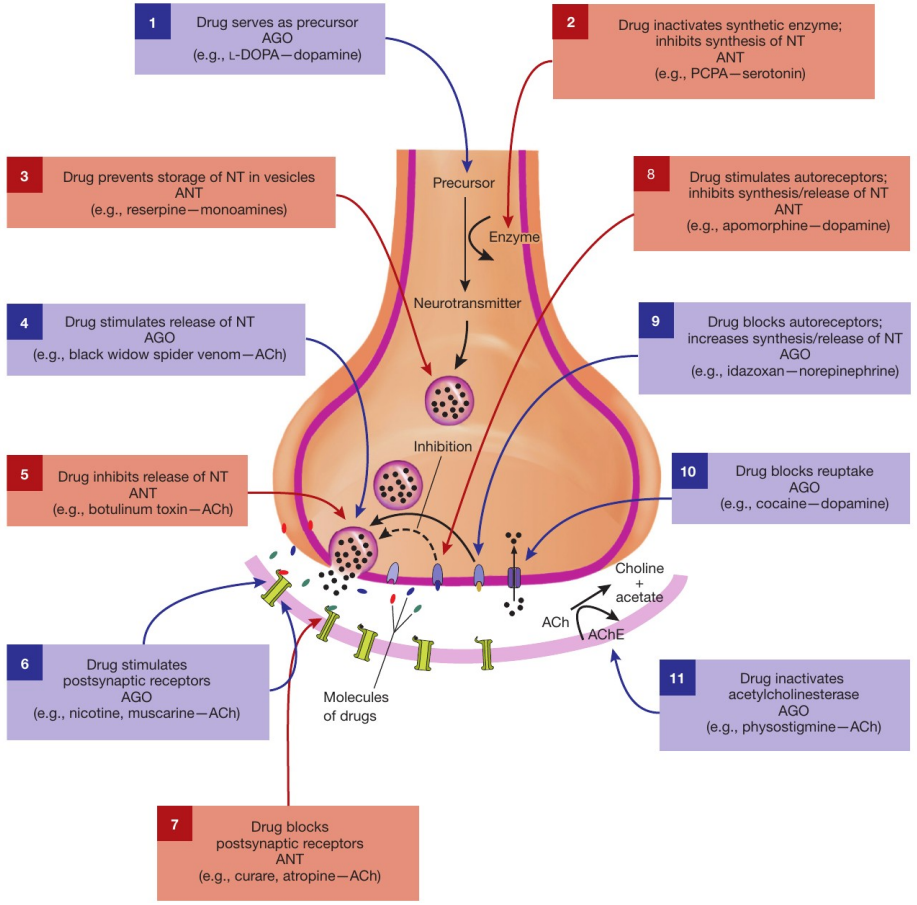
Wong et al. (2024), fig1-video5. Receptor cargo in an unstimulated dendrite.

Vesicle storage and release offer presynaptic targets

Synaptic vesicles store many classical neurotransmitters near release sites in axon terminals. Transport proteins load the vesicles using energy-dependent gradients. A drug that prevents loading can reduce the amount available for release, even when the terminal still synthesizes the transmitter and can generate its usual electrical activity.

An arriving action potential opens voltage-dependent calcium channels. Calcium then activates proteins that make selected vesicles fuse with the terminal membrane and release their contents. Blocking calcium entry or disrupting the fusion machinery can reduce transmitter output without requiring the receiving cell's receptors to be blocked directly by the drug.

Botulinum toxin interferes with proteins needed for acetylcholine release at affected terminals. Its mechanism differs from a receptor antagonist acting on the receiving muscle, although both can weaken muscle activation. Identifying whether a compound acts before or after release explains how superficially similar outcomes can arise through different cellular targets.



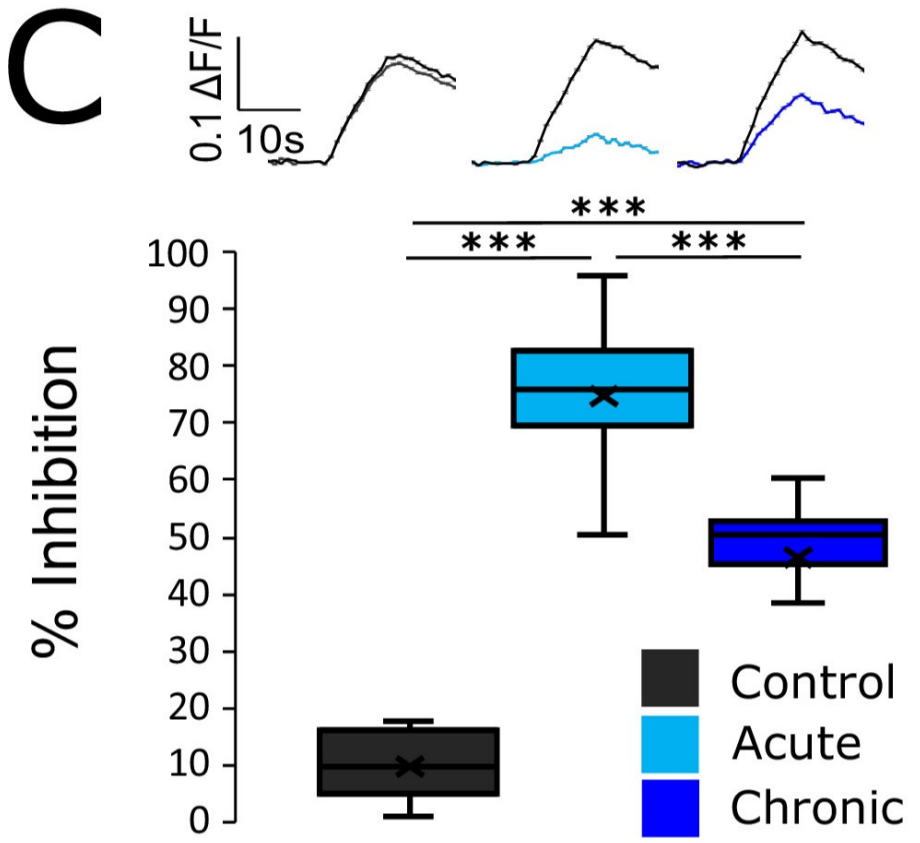
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Opioids reduce release through a presynaptic receptor

The opioid-tolerance investigators activated receptors on axons of cultured medium spiny neurons and measured synaptic communication. Opioid activation reduced transmitter release from the presynaptic population. The experiment therefore examined a receptor that changes a neuron's output rather than assuming that every drug-sensitive receptor acts only on the receiving side of a synapse.

During prolonged activation, the ability of opioid signaling to inhibit release weakened. The researchers compared this physiological change with receptor surface availability. Measuring both variables placed the changing release response beside a specific cellular process, the movement of receptors between the axonal membrane and internal compartments during continued chemical exposure.

A presynaptic receptor regulates how much messenger reaches other cells. Its adaptation can therefore alter the communication of an entire downstream pathway, even when postsynaptic receptors remain capable of responding. The cultured-neuron preparation identifies a mechanism by which sustained receptor activation changes later signaling at the output stage of a neuron.



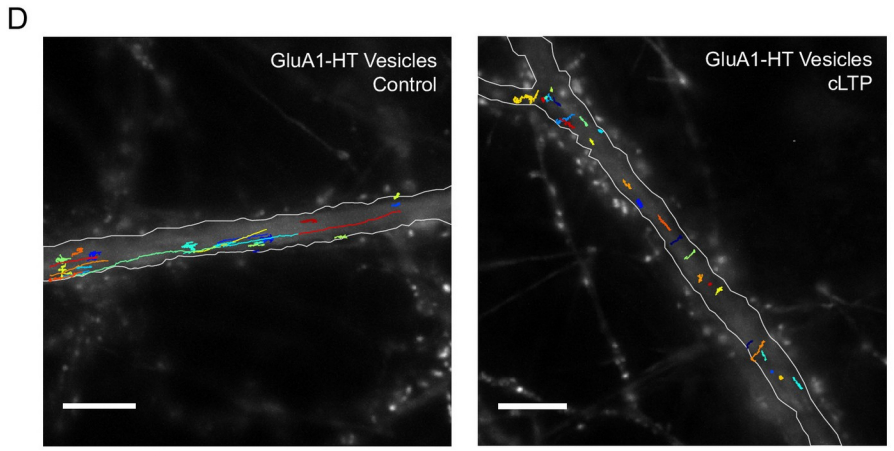
Jullié et al. (2022), Fig. 1 (C). Presynaptic opioid responses after chronic treatment.

Receptor-containing cargo moves inside dendrites

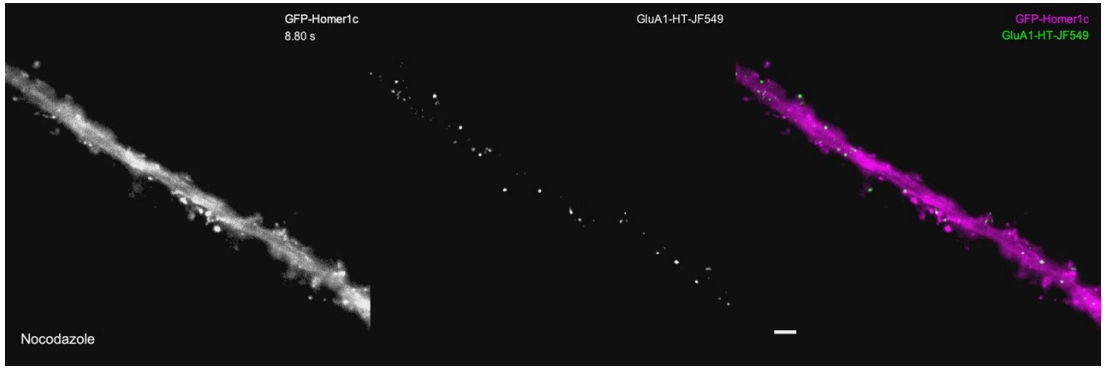
An AMPA-receptor study labeled **GluA1**, a subunit of receptors activated by the amino acid transmitter **glutamate**. The investigators followed newly synthesized receptor-containing vesicles in cultured rat hippocampal neurons. **Dendrites** are neuronal branches receiving many synaptic inputs, so transport through them helps position receptors near sites where glutamate will act.

The researchers treated neurons with **nocodazole**, a compound that disrupts microtubules, the internal tracks used by many transport processes. Receptor-containing cargo then moved differently within the dendrite. The intervention tested a cellular transport mechanism rather than changing the transmitter's binding site on the receptor directly during the recording.

A neuron's sensitivity depends partly on where functional receptors are delivered. Synthesis in the cell body does not place every new receptor immediately into every synapse. Transport and local insertion supply an additional regulatory stage, allowing cellular machinery to change the receptor population available to respond to a later chemical signal.



Wong et al. (2024), Fig. 1 (D). AMPA receptor cargo in neuronal dendrites.



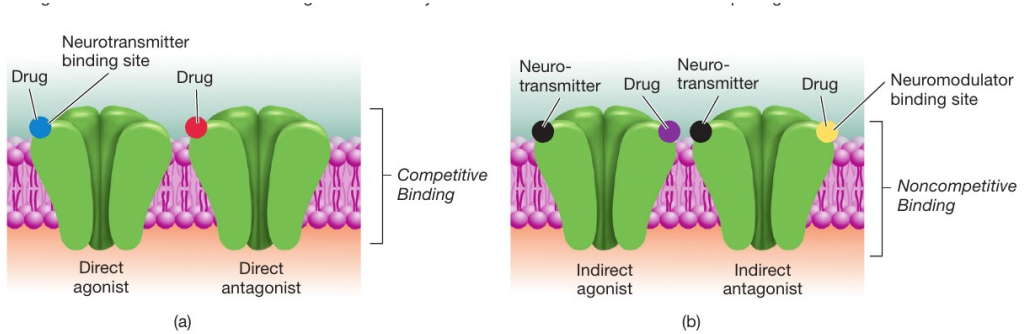
Wong et al. (2024), fig1-video2. Receptor cargo after microtubule disruption.

Agonists and antagonists alter receptor activation

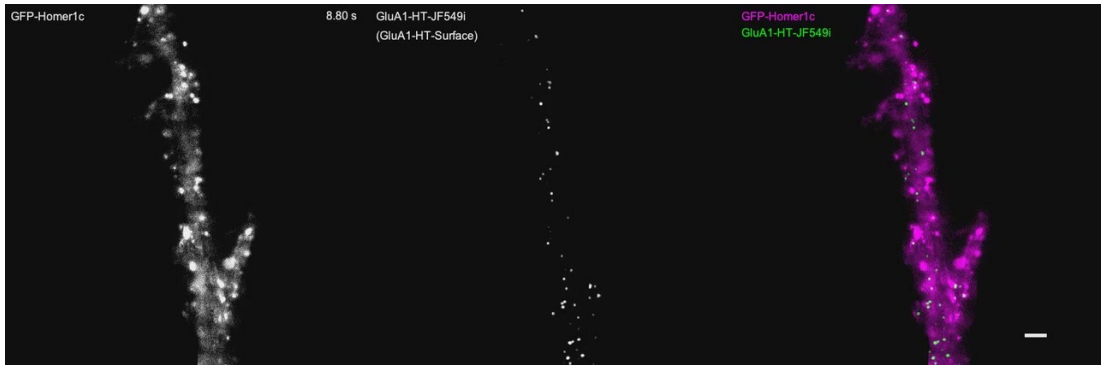
An **agonist** binds to a receptor and activates it, while an **antagonist** binds without producing the same activation and reduces the effect of an agonist. A **competitive antagonist** occupies a binding site that the agonist also needs, so the two compounds compete for access to the receptor population.

A **noncompetitive antagonist** reduces receptor function through another interaction, such as occupying a different site or interfering with channel operation. Increasing agonist concentration cannot always restore the same maximum response under those conditions. The shape of the dose–response change therefore helps identify how the interacting compounds affect the measured receptor system.

The receptor's presence and its activation are separate properties. Imaging a protein at the surface does not establish whether a ligand has activated it, and activating one receptor population does not establish that every downstream cell responds identically. The agonist, antagonist, receptor state, and receiving cell together determine the physiological result.



Carlson and Birkett (2022), Fig. 4.8, p. 102.



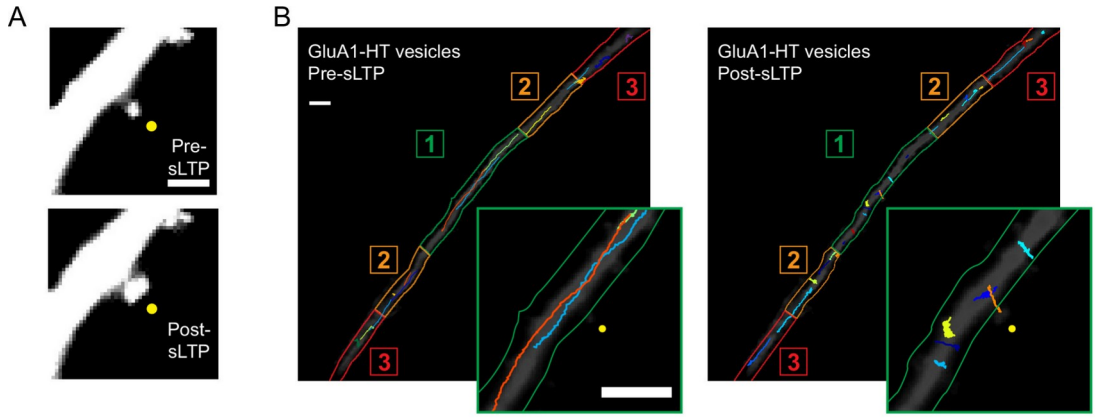
Wong et al. (2024), fig1-video3. Newly labeled receptors at the dendritic surface.

Synaptic stimulation changes local receptor transport

The AMPA-receptor investigators stimulated a selected dendritic region and followed receptor-containing vesicles before and after the event. They used **glutamate uncaging**, in which light releases an active chemical from a caged precursor, to deliver a spatially restricted stimulus. This approach placed the stimulation site beside the receptor cargo being tracked.

After the stimulation, GluA1-containing vesicles became more confined near the affected region. The change increased the local reservoir of receptor-containing cargo available around a site undergoing **plasticity**, a lasting change in cellular or synaptic properties. Transport therefore became spatially organized instead of distributing receptor material uniformly throughout the dendrite.

The experiment separates local receptor availability from receptor synthesis across the whole neuron. A cell can alter delivery and retention near an active input using material already present in its dendrite. Such redistribution changes which synapses can acquire receptors, providing a cellular mechanism through which previous activity influences later responsiveness to glutamate.



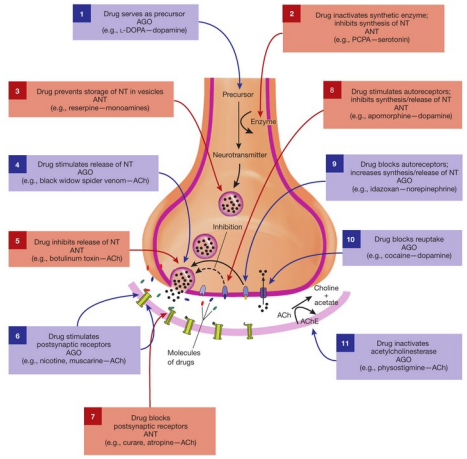
Wong et al. (2024), Fig. 2 (A–B). Receptor cargo near a stimulated dendritic spine.

Receptor delivery changes a cell's responsiveness

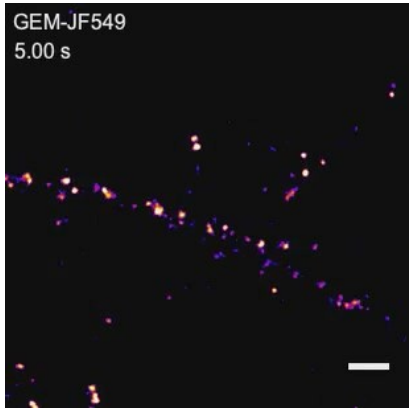
A receptor must occupy a useful membrane location before it can detect an extracellular transmitter. Cells deliver receptors from internal stores, retain them at selected sites, and remove them through endocytosis. These processes change the available population even when the amino-acid sequence of the receptor protein and the identity of its ligand remain unchanged.

The **cytoskeleton** is a network of intracellular proteins that helps organize cell structure and transport. Motor proteins move some cargo along tracks, while other particles spread by diffusion through the cytoplasm. The density and arrangement of cellular material can therefore influence how quickly receptor-containing vesicles reach different regions of a neuron.

Repeated activity can change this environment near a synapse, altering local retention and delivery. A stronger response need not require a drug to bind more tightly to each receptor; it can arise because more functional receptors become available at the relevant membrane. Cellular trafficking therefore contributes to the response produced by later transmitter release.



Carlson and Birkett (2022), Fig. 4.7, p. 100.



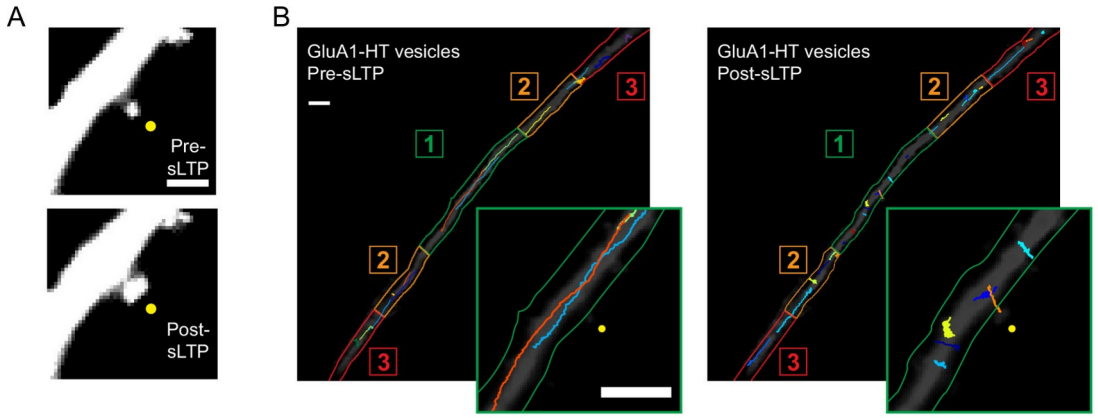
Wong et al. (2024), fig4-video1. Inert particles moving inside a dendrite.

Local stimulation concentrates receptor cargo nearby

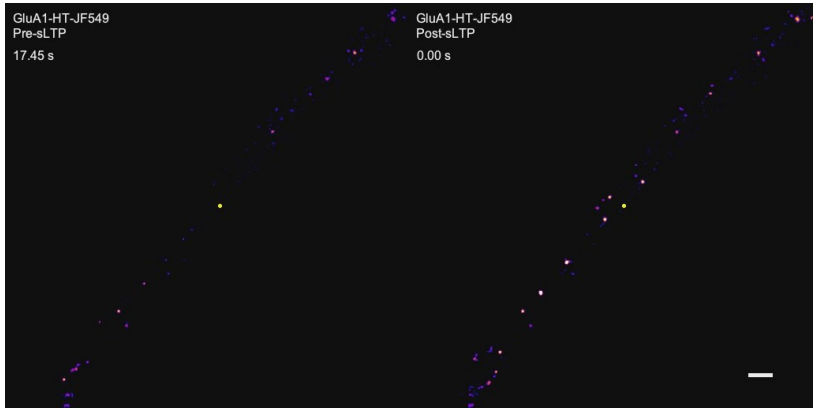
The AMPA-receptor investigators followed GluA1-containing vesicles around a region receiving localized glutamate stimulation. Before stimulation, cargo moved through the dendritic shaft by active transport and diffusion. Afterward, vesicles spent more time near the affected region, creating a spatially restricted reservoir rather than increasing receptor material uniformly throughout the branch.

The researchers measured changes in motion at defined distances from the stimulation site. This spatial comparison distinguished a local effect from a general slowing of every vesicle in the neuron. The response therefore involved the position of the input as well as the identity of the receptor protein moving through the dendrite.

An active synapse needs a supply of membrane proteins in the appropriate place. Retaining receptor-containing vesicles nearby increases the material available for insertion when local machinery triggers membrane fusion. The experiment identifies a transport step through which previous glutamate stimulation can influence how a receiving neuron responds to subsequent chemical communication.



Wong et al. (2024), Fig. 2 (A–B). Receptor cargo near a stimulated dendritic spine.



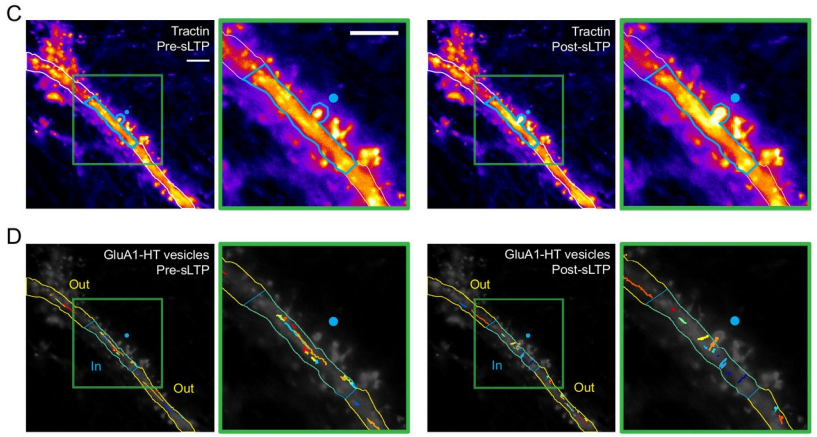
Wong et al. (2024), fig2-video1. Receptor cargo before and after local stimulation.

Actin and myosin help retain and deliver receptor cargo

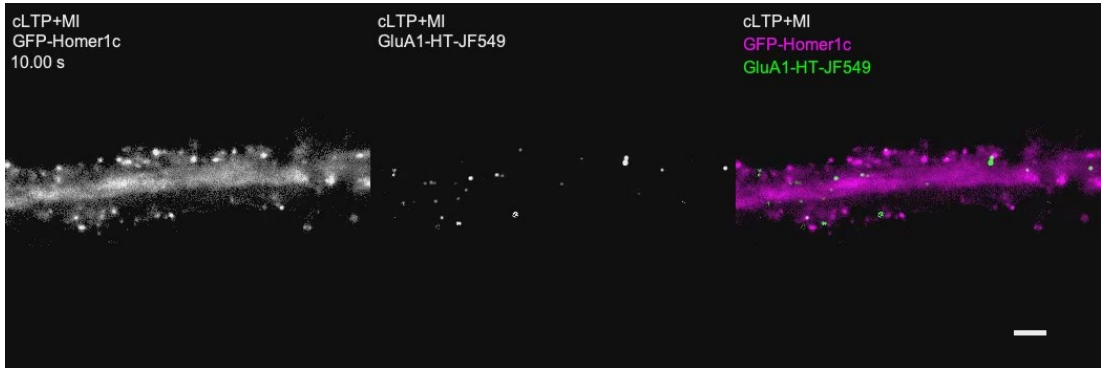
The AMPA-receptor study examined **actin**, a cytoskeletal protein that forms filaments, and **myosin**, a motor protein that interacts with those filaments. Following synaptic stimulation, actin accumulated near the affected region. The investigators tested how changing this machinery altered the movement and local delivery of receptor-containing vesicles in the dendrite.

A cocktail of myosin inhibitors changed receptor-vesicle behavior during chemically induced plasticity. Comparing inhibited and control preparations separated the contribution of motor activity from the receptor's ability to bind glutamate. The intervention acted on intracellular transport machinery, providing a way to test a downstream cellular step in the response to stimulation.

The study concluded that actin both restricts long-range motion and facilitates myosin-dependent delivery toward release sites for receptor cargo. These functions can cooperate: keeping vesicles nearby supplies a reservoir, and short-range transport positions them for fusion. The resulting membrane insertion changes the receptors available to participate in later synaptic responses.



Wong et al. (2024), Fig. 3 (C–D). Actin and receptor cargo near stimulated spines.



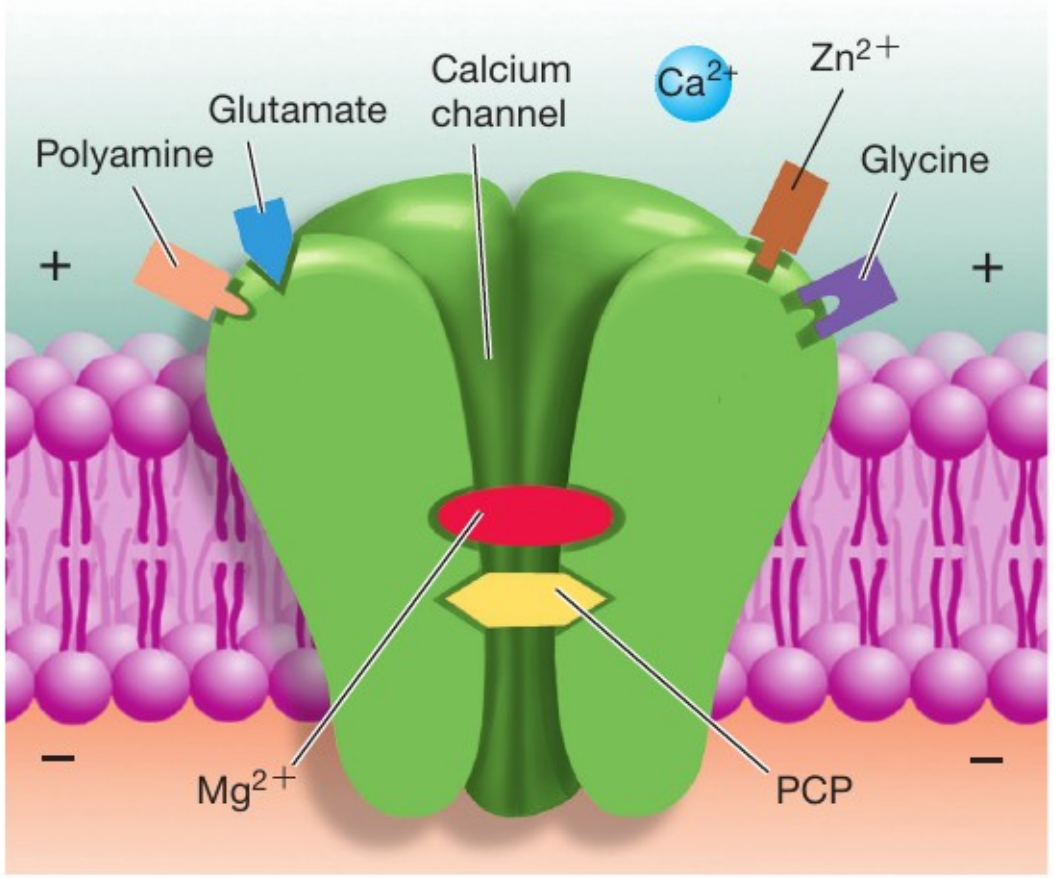
Wong et al. (2024), fig3-video3. Receptor cargo during myosin inhibition.

Glutamate activates several receptor classes

Glutamate is the major excitatory neurotransmitter in the mammalian central nervous system. **AMPA receptors** commonly produce a rapid postsynaptic response by opening a cation channel. **NMDA receptors** are another glutamate-sensitive channel class, while **metabotropic glutamate receptors** act through G proteins and intracellular signaling rather than forming the same type of ion-conducting pore.

An NMDA receptor's operation depends on chemical binding and membrane voltage. Glutamate and a co-agonist must bind, and depolarization helps remove a magnesium block from the channel. Calcium can then enter through the activated receptor, providing an intracellular signal that contributes to activity-dependent changes in synaptic function and other cellular processes.

Drugs can affect these receptor classes differently. An NMDA-channel blocker does not act in the same way as an AMPA-binding-site antagonist, even though both influence glutamatergic communication. The receptor subtype, its location, and the current activity of the receiving cell determine which component of the response the compound changes.



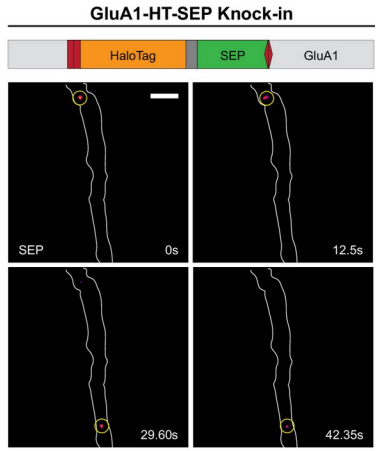
Carlson and Birkett (2022), Fig. 4.9, p. 104.

Receptor vesicles fuse near a stimulated synapse

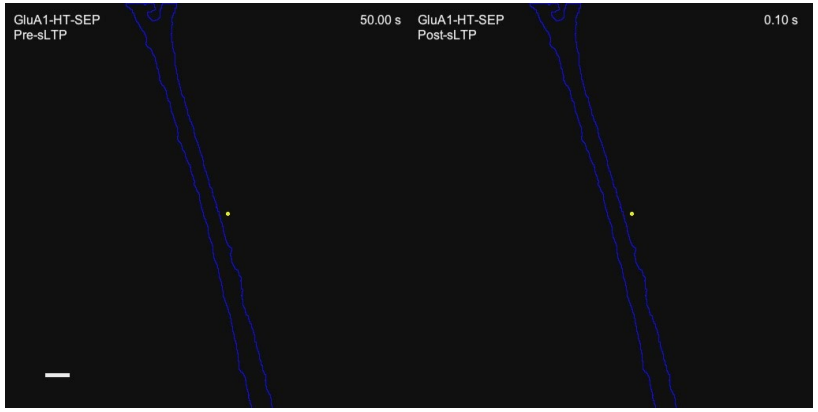
The AMPA-receptor investigators used a fluorescent tag that becomes more visible when a receptor-containing vesicle fuses with the cell membrane. This **exocytosis** moves receptor protein from an internal compartment into the surface. Individual fusion events therefore supplied a measure of delivery to the membrane, complementing measurements of cargo moving within the dendrite.

Localized stimulation increased receptor delivery near the affected region. The experiment compared positions close to and farther from the stimulation site, placing membrane insertion within the same spatial framework used for vesicle retention. A local supply of cargo was therefore accompanied by changes in where newly delivered receptor material appeared.

The sequence extends beyond a receptor simply receiving a transmitter. Glutamate-driven activity changes intracellular organization, that organization alters vesicle movement, and vesicle fusion changes surface receptor availability. These events provide a cellular route through which an earlier chemical signal can modify the strength of a later response in the same neuronal region.



Wong et al. (2024), Fig. 5 (A, image sequence). AMPA receptor insertion at the cell surface.



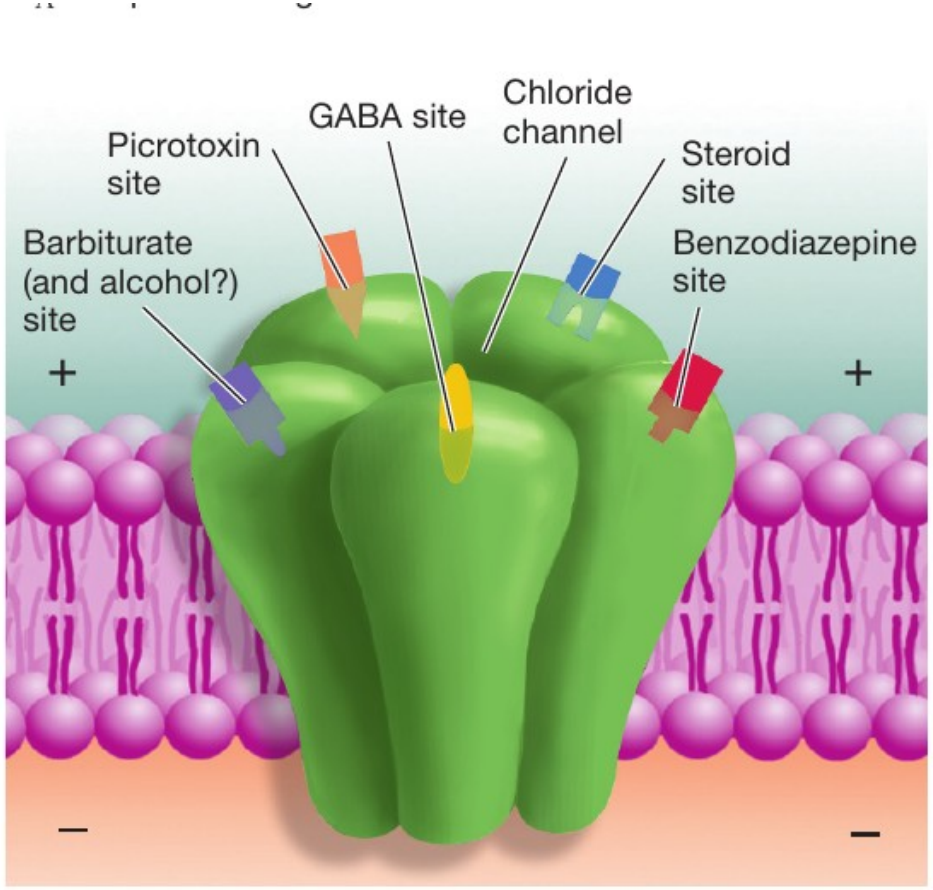
Wong et al. (2024), fig5-video3. Receptor insertion before and after local stimulation.

GABA controls excitability through channels and G proteins

GABA is a major inhibitory neurotransmitter in the mature mammalian brain. **GABA-A receptors** are ligand-gated channels that increase membrane conductance to chloride and related anions. Their effect depends on the ions' electrochemical gradients, so the channel's influence is determined by the receiving cell's conditions as well as by transmitter binding.

Increasing inhibitory conductance can pull voltage toward the anion equilibrium potential or reduce the effect of concurrent excitatory current. This reduction of voltage change is often called **shunting inhibition**. A neuron can therefore become less likely to fire without every inhibitory event producing a large visible hyperpolarization of its membrane.

GABA-B receptors use G proteins and act through slower signaling processes, including changes in potassium and calcium channels. Presynaptic GABA-B activity can reduce transmitter release, while postsynaptic activity can reduce excitability. Receptor subtype identifies the mechanism, preventing the mistaken assumption that every GABA effect is produced by one identical chloride channel.



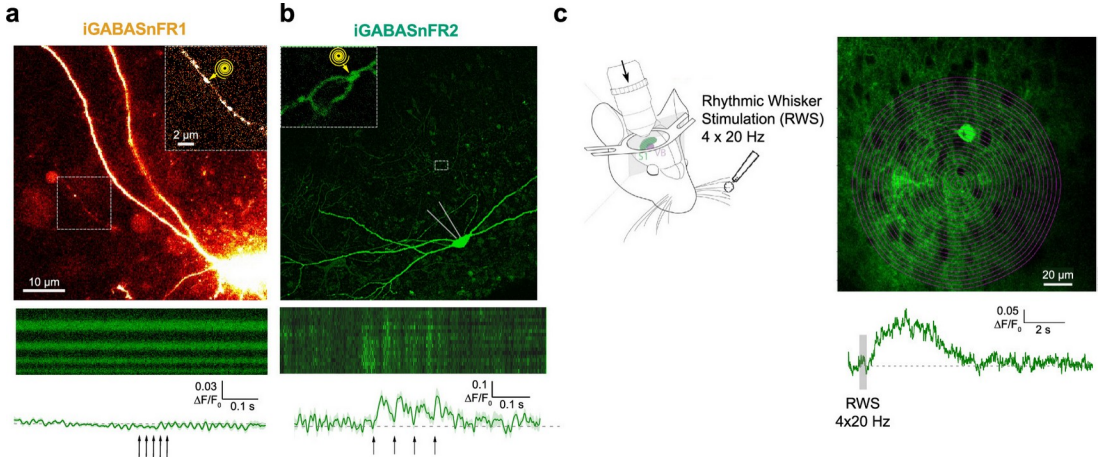
Carlson and Birkett (2022), Fig. 4.10, p. 105.

A fluorescent sensor detects extracellular GABA release

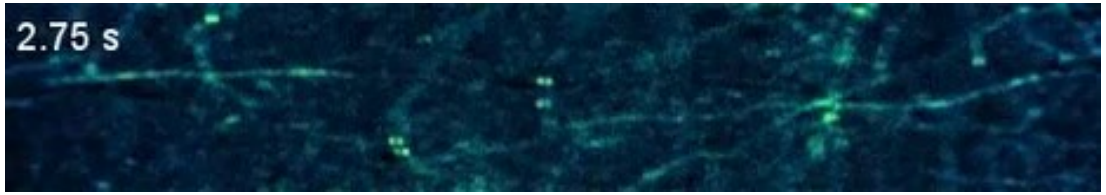
Kolb and colleagues developed **iGABASnFR2**, a fluorescent protein sensor whose brightness changes when it binds GABA. Unlike a calcium indicator, this sensor reports the availability of an extracellular transmitter. The investigators optimized its properties and tested whether it could detect GABA signals in cultured neurons, brain tissue, and living neural circuits.

In the mouse somatosensory cortex, brief whisker stimulation evoked a measurable extracellular GABA signal. The imaging experiment followed transmitter-related fluorescence over time in the responding cortical region. This observation placed a chemical signal beside a defined sensory event, making inhibitory transmission measurable without inferring it solely from the firing of presumed GABAergic neurons.

The study also used variants with different fluorescence responses to binding, allowing observations to be examined through complementary detector properties. Sensor behavior must be understood before converting brightness into a biological interpretation. The chemical-binding mechanism establishes what the measured signal represents, separating transmitter detection from calcium entry or a direct measurement of membrane voltage.



Kolb et al. (2026), Fig. 6 (a–c). Fluorescent detection of extracellular GABA.



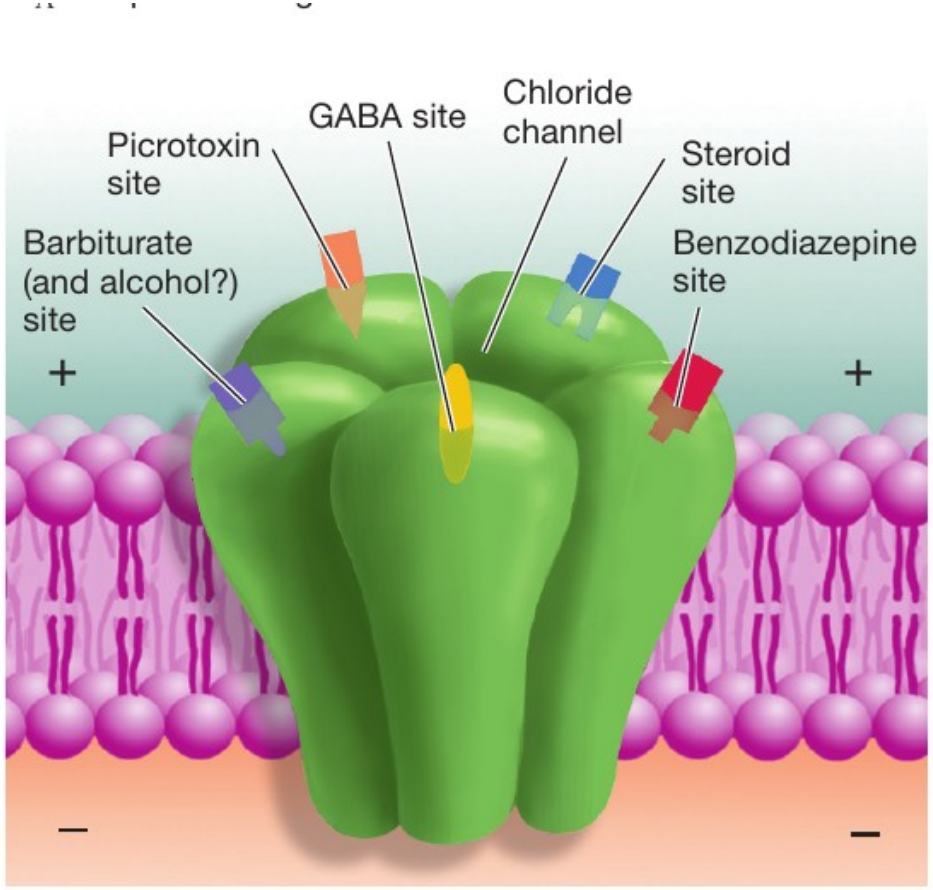
Kolb et al. (2026), fig6-video1. Whisker-evoked extracellular GABA fluorescence.

Allosteric drugs modify how a receptor responds

An **allosteric site** is a receptor location distinct from the principal binding site for its transmitter. A **positive allosteric modulator** increases receptor responses under the relevant conditions, while a **negative allosteric modulator** reduces them. Binding at a different site allows a drug to alter signaling without directly replacing the normal messenger.

Benzodiazepines bind to selected GABA-A receptor assemblies and enhance their response to GABA. The drug and transmitter occupy different sites, so the mechanism differs from a compound that directly opens the channel through the transmitter's binding site. Receptor subunit composition influences which assemblies respond to a given modulator and how they operate.

A cellular effect can produce several behavioral consequences because the receptor occurs across different circuits. Enhancing inhibitory signaling can alter arousal, anxiety-related behavior, or motor coordination depending on the affected populations and exposure. Naming the drug class identifies part of the mechanism; locating the responsive receptors explains which physiological processes can change.



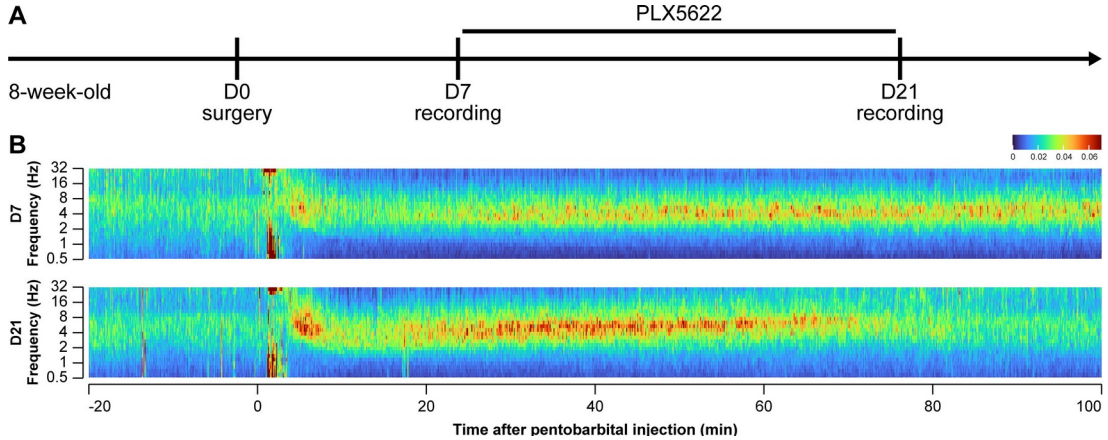
Carlson and Birkett (2022), Fig. 4.10, p. 105.

Anesthesia alters brain activity as well as movement

The microglia-anesthesia investigators measured **electroencephalography**, or **EEG**, which records electrical activity from the brain, together with **electromyography**, or **EMG**, which measures muscle-related electrical activity. These signals provided physiological information alongside the righting-reflex tests, allowing drug-induced changes to be examined beyond the visible presence or absence of a movement.

Microglial depletion altered measures associated with entry into and recovery from anesthesia. The study compared both behavioral endpoints and neural-state signals under defined treatments. The investigators therefore tested whether an altered motor response occurred in the setting of broader changes in the nervous system's response to the administered anesthetic.

General anesthesia recruits several molecular targets and interacting neural populations. The EEG and EMG measurements do not identify one receptor by themselves, but they reveal the state in which the molecular interventions operate. Combining these signals with targeted cellular manipulation places supporting-cell pharmacology within the response of an intact brain and body.



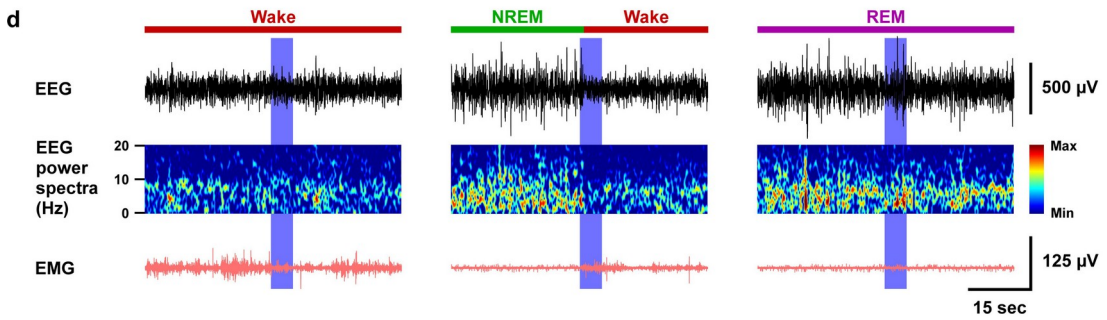
He et al. (2023), Fig. 4 (A–B). Brain electrical activity during anesthesia.

GABAergic neurons contribute to the control of wakefulness

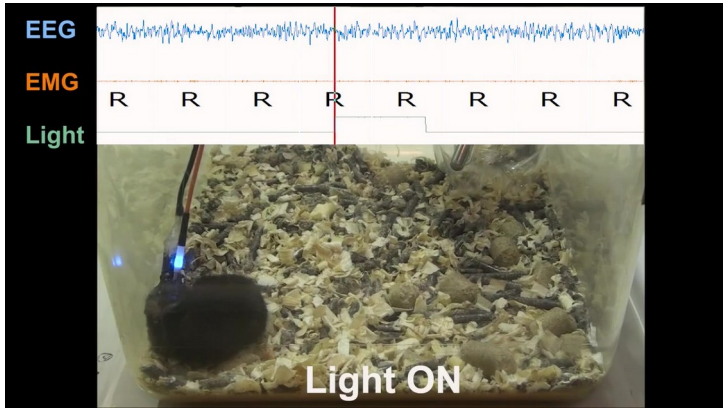
A sleep study examined GABA-producing neurons in the **ventral tegmental area**, or **VTA**, a midbrain region containing several cell populations. The investigators selected cells expressing an enzyme used to synthesize GABA. This identified a transmitter-defined population without assuming that every neuron in the anatomical region released the same chemical messenger.

Optogenetic inhibition of the selected neurons rapidly promoted wakefulness from non-REM sleep under the tested conditions. The effect differed during REM sleep, indicating that the current brain state changed the consequence of the manipulation. The original behavior and physiological recordings placed the transition beside activity in a particular GABAergic population.

The same study examined projections to wake-promoting hypothalamic orexin neurons and found inhibitory GABAergic communication along that route. This circuit explains how one transmitter-defined population can influence arousal through its targets. GABA's effect on a receiving cell and the resulting effect on the animal's state are different levels of the mechanism.



Chowdhury et al. (2019), Fig. 3 (d). State-dependent EEG and muscle responses.



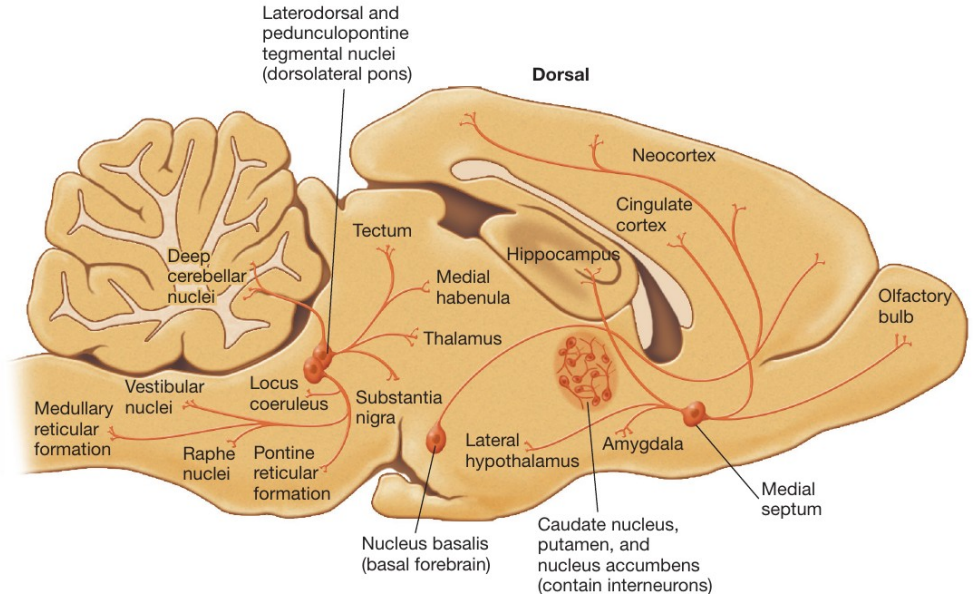
Chowdhury et al. (2019), video1. Inhibiting GABA neurons across sleep and wake states.

Acetylcholine acts through two major receptor families

Acetylcholine, or **ACh**, is released by motor neurons at skeletal muscle and by several autonomic and central pathways. In the brain, major cholinergic populations arise in the basal forebrain and brainstem. Their axons influence activity in widespread targets, including circuits involved in arousal, attention, memory, and behavioral engagement.

Nicotinic receptors are ligand-gated cation channels, while **muscarinic receptors** are G protein-coupled receptors. The same transmitter can therefore produce responses with different cellular mechanisms. Nicotine activates selected nicotinic receptors, whereas a muscarinic antagonist such as atropine blocks a different receptor family even though both act within acetylcholine-sensitive systems.

Acetylcholinesterase breaks down released acetylcholine, helping terminate its action. Drugs that inhibit this enzyme increase the duration or availability of the transmitter rather than directly mimicking it at every receptor. Synthesis, vesicle loading, release, receptor activation, and enzymatic breakdown consequently provide distinct targets for changing cholinergic communication in different tissues.



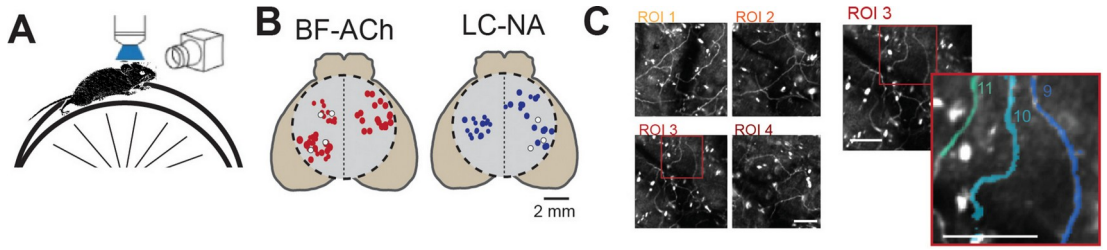
Carlson and Birkett (2022), Fig. 4.11, p. 107.

Cholinergic axons respond during behavioral engagement

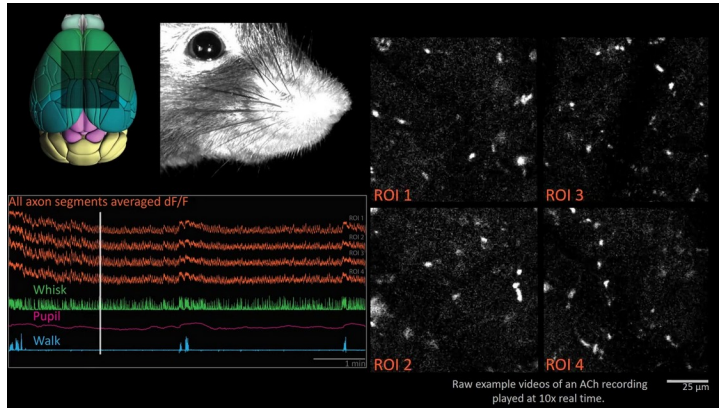
A cortical imaging study recorded labeled acetylcholine-releasing axons in awake mice. The investigators collected fluorescence signals while also measuring whisker movement, pupil diameter, and locomotion. This arrangement placed transmitter-defined projections beside observable behavior, allowing their activity to be examined during spontaneous changes in engagement rather than during only one instructed task.

Activity in many axonal segments increased around arousal and movement events across separated cortical locations. The broad coordination indicated that a cholinergic system can influence several regions during the same change in state. The signals therefore carried information beyond the narrow identity of a single sensory stimulus in one local cortical area.

The recording measured calcium within labeled axons, which reflects activity associated with the projection rather than directly counting released acetylcholine molecules. Identifying that measurement keeps the circuit account precise. The anatomical label established the transmitter-defined source, while simultaneous behavior placed the recorded activity within the animal's changing state of engagement.



Collins et al. (2023), Fig. 1 (A–C). Recording labeled cholinergic and noradrenergic axons.



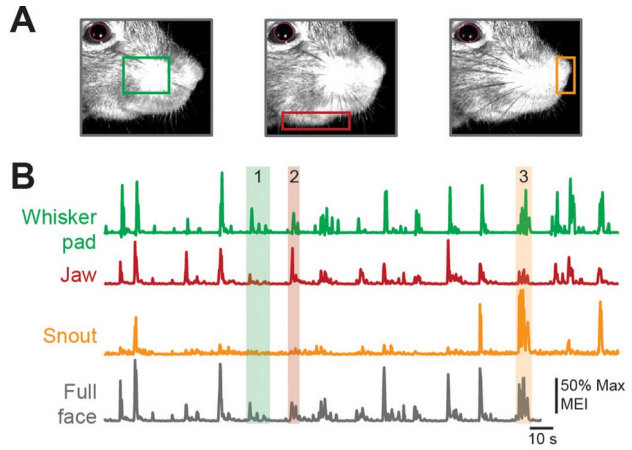
Collins et al. (2023), video1. Axon imaging alongside behavior measurements.

Two transmitter systems track facial movement

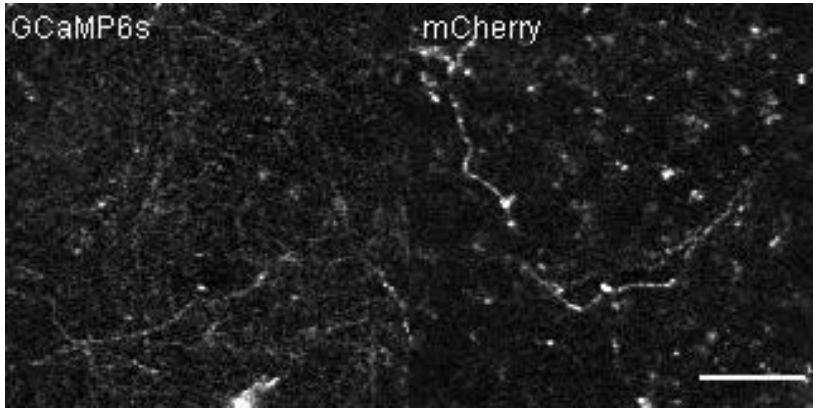
The cortical imaging investigators compared acetylcholine-releasing axons with axons from a **noradrenergic** population, which releases norepinephrine. Both projection systems showed activity associated with changes in behavioral state. Measuring the two populations addressed how broadly projecting chemical systems participate in common events while retaining different transmitter identities and cellular origins.

Whisker movements were especially informative about changes in the recorded axonal signals. The study also compared activity-dependent fluorescent indicators with a marker whose brightness did not report calcium changes. This comparison helped distinguish signals associated with cellular activity from fluorescence changes caused solely by the imaging environment or movement of the preparation.

The two systems did not operate as one undifferentiated chemical signal. Some axonal populations contained activity not fully described by the measured movements and arousal variables. Their shared timing and additional variation provide a framework for examining how different neuromodulators coordinate cortical state while still contributing distinct information through their particular projections.



Collins et al. (2023), Fig. 3 (A–B). Face tracking and behavioral measurements.



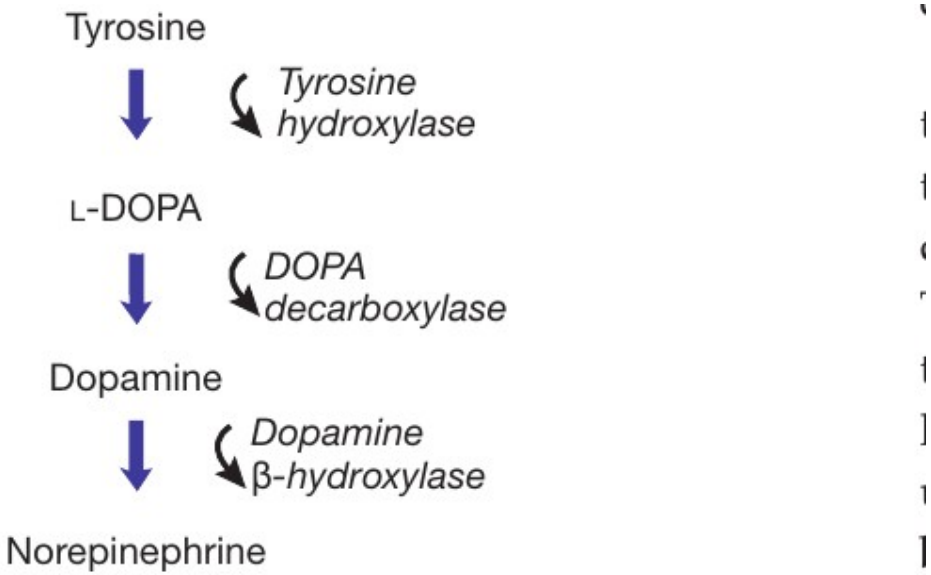
Collins et al. (2023), video2. Activity-sensitive and reference fluorophore recordings.

Catecholamine synthesis creates a sequence of drug targets

The **catecholamines** include dopamine, norepinephrine, and epinephrine. Their synthesis begins with **tyrosine**, an amino acid. **Tyrosine hydroxylase** converts tyrosine to **L-DOPA**, and another enzyme converts L-DOPA to dopamine. The identity of the enzymes expressed in a cell determines which later products that cell can manufacture from this shared pathway.

In norepinephrine-producing neurons, **dopamine beta-hydroxylase** converts dopamine to norepinephrine. Epinephrine synthesis requires an additional enzymatic step in cells equipped for it. The chemical sequence explains why blocking an early enzyme can affect several downstream products, while changing a later step can alter the balance between particular messengers.

L-DOPA is used therapeutically because it can cross the blood–brain barrier through amino-acid transport mechanisms and provide a precursor for dopamine synthesis. Dopamine itself enters the brain poorly from circulation. The treatment therefore changes chemical supply upstream of release and receptor activation, applying pharmacokinetics and enzymatic synthesis within the same intervention.



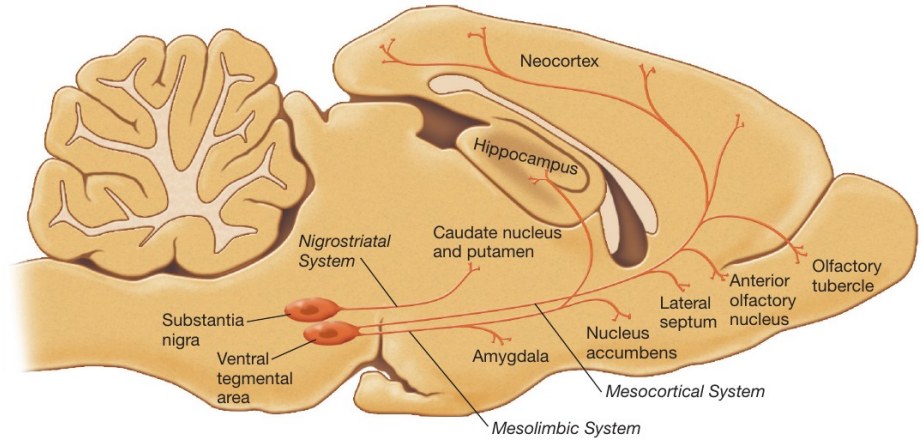
Carlson and Birkett (2022), Fig. 4.15, p. 110.

Dopamine pathways influence different functions

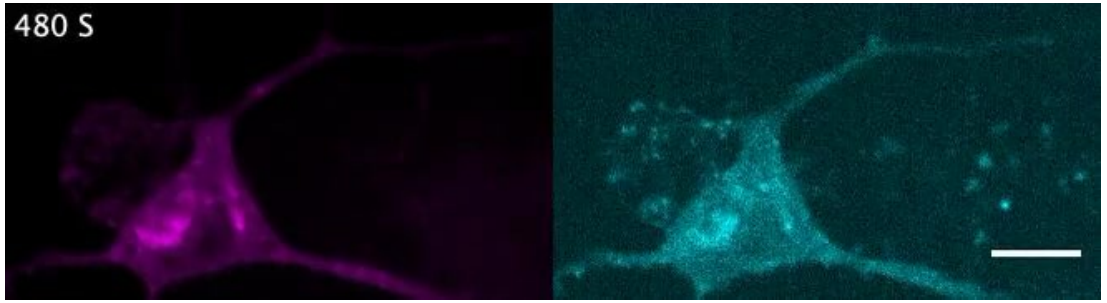
Dopamine-producing neurons arise in several central populations and send axons to distinct targets. The **nigrostriatal pathway** carries signals from substantia nigra toward striatal circuits involved in movement. **Mesolimbic** and **mesocortical pathways** arise largely from the ventral tegmental area and influence motivation, learning, cognition, and other aspects of behavioral regulation.

Dopamine receptors belong to G protein-coupled families with different signaling properties. **D1-like** receptors commonly increase cAMP signaling, while **D2-like** receptors commonly reduce it through different G proteins. These cellular tendencies must be interpreted in the receiving circuit, where activating or inhibiting one population can change the output of other neurons.

A dopamine response therefore cannot be described simply as pleasure or movement. The source population, axonal target, receptor subtype, and current activity all matter. In striatal neurons, dopamine-sensitive receptors also occupy cellular compartments with different access routes, adding another level of organization within a pathway already specialized by its anatomical position.



Carlson and Birkett (2022), Fig. 4.14, p. 109.



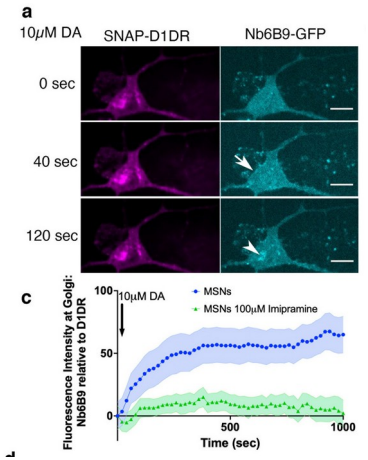
Puri et al. (2022), fig3-video1. Dopamine response in striatal neurons.

Dopamine activates Golgi receptors in striatal neurons

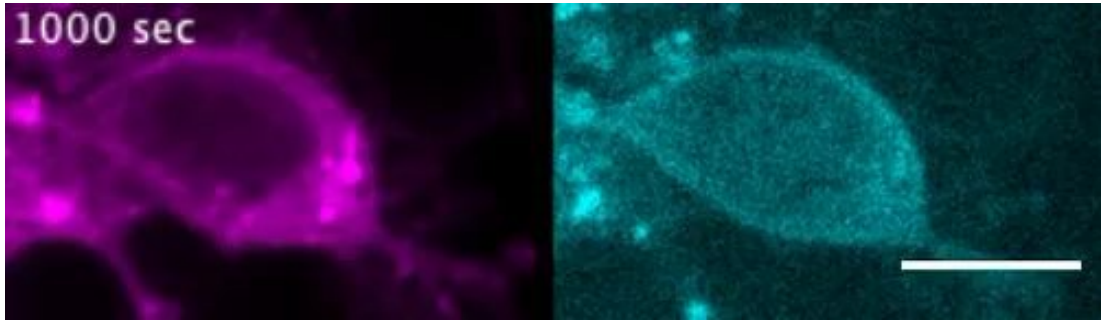
The D1-receptor investigators extended their experiments from cell lines to cultured striatal medium spiny neurons. They examined whether dopamine could activate a Golgi-associated receptor population in a neural preparation. This comparison tested the cellular mechanism in a neuron type that normally participates in dopamine-sensitive circuits, rather than relying on nonneuronal cells alone.

Dopamine-dependent activity was detected at the Golgi in these neurons. Inhibiting the OCT2 transport route reduced internal receptor activation, while an accessible synthetic agonist could still recruit the response. The comparison again separated delivery of the ligand across membranes from the internal receptor's capacity to enter an activated state.

The findings locate pharmacological access within a defined neural cell type. A striatal neuron's response can include signaling from internal membranes as well as from its outer surface. The consequence of an applied compound therefore depends on the receptor population it reaches, the transporters present, and the intracellular machinery available in the receiving compartment.



Puri et al. (2022), Fig. 3 (a, c). D1 receptor activity in striatal neurons.



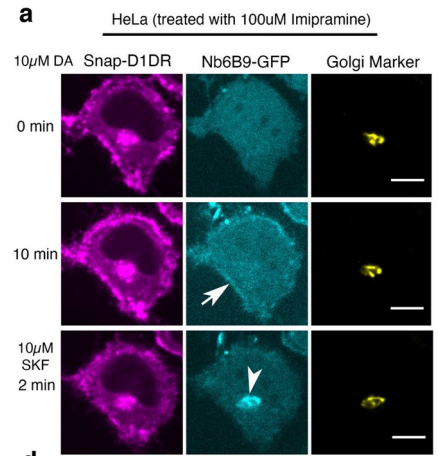
Puri et al. (2022), fig3-video2. Striatal D1 response during transporter inhibition.

Transporter reduction alters dopamine signaling

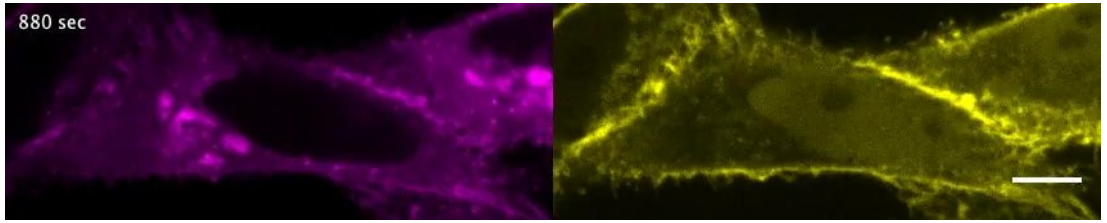
The D1-receptor study reduced OCT2 expression using a targeted genetic method and examined activation of the internal receptor population. Decreasing the transporter reduced the route available to extracellular dopamine. The intervention changed a delivery component of the signaling pathway, complementing the experiments that inhibited transport pharmacologically or increased transporter expression.

A membrane-permeant synthetic agonist could bypass part of the access problem because it entered compartments differently from dopamine. Comparing these ligands helped identify what the transporter manipulation changed. Loss of a dopamine-dependent internal response did not require the entire cell to lose its receptors or every means of activating them.

Combining genetic and chemical interventions can separate several steps in a pharmacological mechanism. A transporter carries the ligand, the receptor detects it, and downstream proteins generate the cellular response. Changing one step and testing a route that bypasses it establishes where that component contributes within the sequence from exposure to signaling.



Puri et al. (2022), Fig. 2 (a). Dopamine response during transporter inhibition.



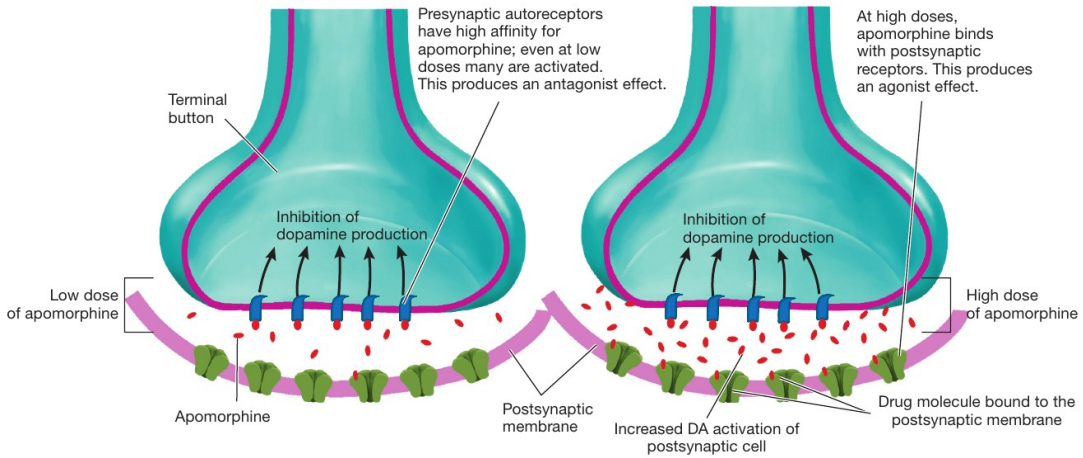
Puri et al. (2022), fig2-video3. D1 signaling after OCT2 reduction.

Autoreceptors let a neuron regulate its own output

An **autoreceptor** responds to a transmitter released by the same type of neuron on which it is located. Presynaptic dopamine autoreceptors can reduce further synthesis or release when dopamine becomes available nearby. This feedback changes the amount sent to other cells, even though the receptor belongs to the same broad chemical system as its transmitter.

A receptor agonist can therefore have different effects at presynaptic and postsynaptic sites. Activating an inhibitory autoreceptor can reduce transmitter output, while activating a postsynaptic receptor changes the receiving cell's signaling. The net behavioral response reflects the combination of sites reached and their sensitivity at the exposure being tested.

Dose can change that combination. At lower exposures, a sensitive autoreceptor population may contribute strongly; at higher exposures, additional postsynaptic effects become prominent. The apomorphine example illustrates why a dopamine agonist need not increase every dopamine-mediated outcome at every dose, because the compound can recruit feedback and target-cell mechanisms in different proportions.



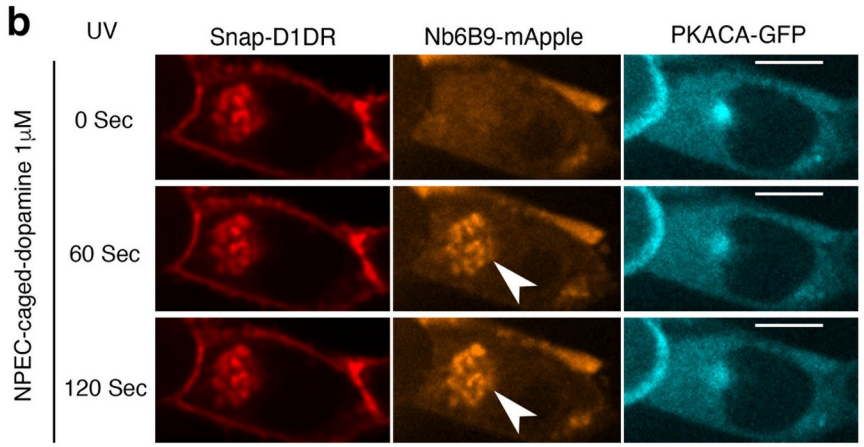
Carlson and Birkett (2022), Fig. 4.16, p. 111.

Local dopamine release activates a compartmental signal

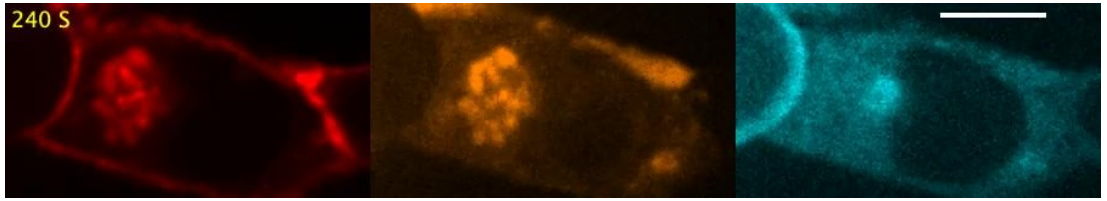
The D1-receptor investigators used **caged dopamine**, a chemically masked precursor that releases active dopamine after a light pulse. This approach changed the timing and location of ligand availability. The experiment examined how a controlled chemical event recruited receptor-associated signaling within a cell containing identifiable internal membrane compartments and fluorescent activity detectors.

The researchers followed both a receptor-activation detector and a reporter for protein kinase A signaling. Their measurements placed the initial ligand event beside a downstream response. A chemical pulse can therefore be traced through several stages, from receptor activation to second-messenger-dependent enzyme activity, without treating fluorescence at every location as the same biological variable.

Selective interference with compartmental signaling helped assign a contribution to internal D1 receptors. The resulting account combined ligand access, receptor location, and downstream output. It demonstrates how a neurotransmitter can recruit a spatially organized intracellular response, rather than producing one simultaneous chemical change throughout every region of the receiving cell.



Puri et al. (2022), Fig. 6 (b). Intracellular dopamine release and receptor activity.



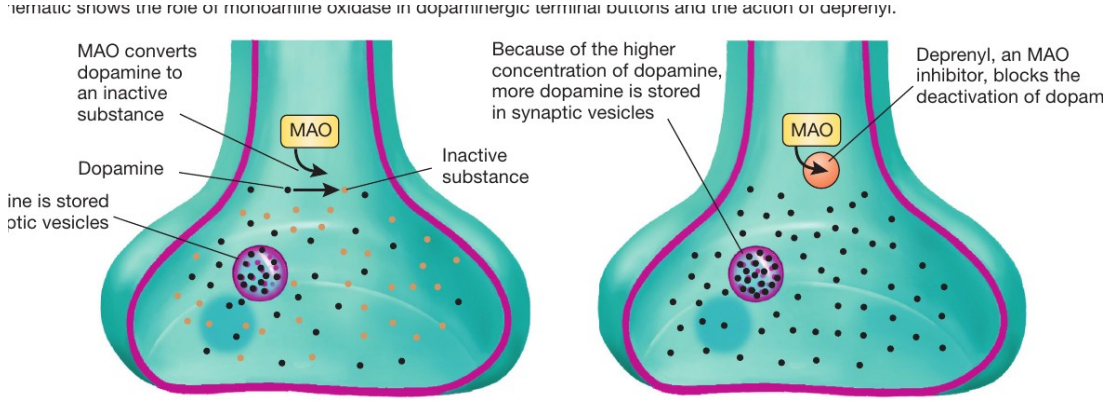
Puri et al. (2022), fig6-video1. D1 receptor and PKA responses to uncaged dopamine.

Reuptake and enzymes terminate monoamine signals

Reuptake carries released transmitter back across a membrane through a transport protein. Dopamine and other monoamines use specific transport systems that help limit extracellular signaling. Blocking a clearance transporter can prolong transmitter availability, changing the exposure of receptors to the endogenous messenger without requiring the drug to activate those receptors directly.

Cocaine blocks monoamine transporters, including the dopamine transporter, while selective serotonin reuptake inhibitors primarily inhibit the serotonin transporter. These compounds affect a removal step, but the behavioral result still depends on where transmitter is released and which receptors receive it. Transporter inhibition does not make every neuron release the same amount of chemical.

Monoamine oxidase, or **MAO**, is an enzyme that participates in monoamine breakdown within cells. MAO inhibitors act through this metabolic route rather than the same mechanism as a receptor antagonist. Reuptake, intracellular metabolism, and renewed vesicle loading interact to determine how much transmitter remains available for later release and receptor activation.



Carlson and Birkett (2022), Fig. 4.17, p. 112.

Dopamine helps coordinate several motor programs

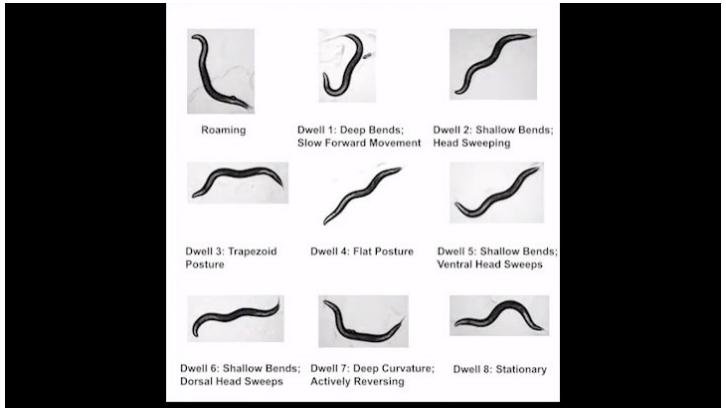
A study in **Caenorhabditis elegans**, a small nematode worm, measured locomotion, feeding-related movements, and egg laying over extended periods. The investigators classified recurring patterns of behavior rather than recording only how far an animal moved. This approach examined how a neuromodulator helps coordinate distinct motor programs within a whole behaving organism.

The worms switched between states in which the measured actions occurred in different combinations. The source footage contains examples of these actual behavioral patterns. Their coordination provides a concrete problem for chemical modulation: one messenger system can influence several circuits, changing which actions tend to occur together during a particular state.

The investigators then used dopamine-related mutations, activity measurements, and neural manipulations to examine the coupling between locomotion and egg laying. These interventions assigned a role to dopamine in coordinating different motor programs, rather than reducing its function to a universal command that increases or decreases every movement simultaneously.



Cermak et al. (2020), Fig. 1 (A–B). Worm behavior recording and animal.



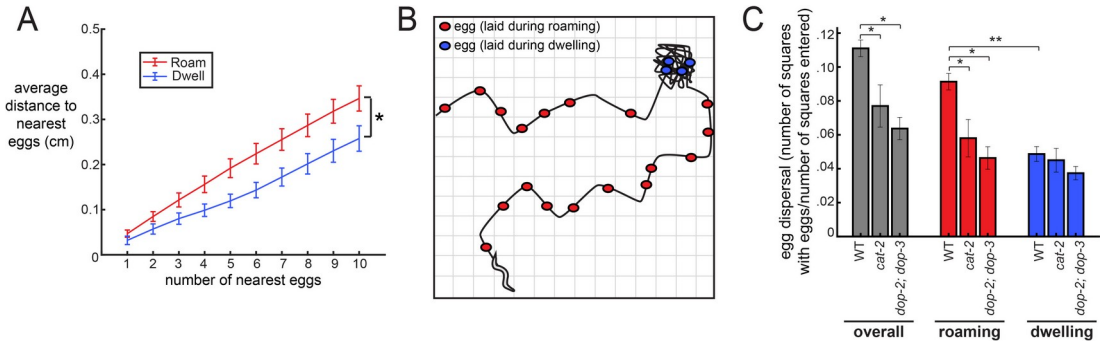
Cermak et al. (2020), video1. Worm locomotor and egg-laying states.

Dopamine couples egg laying to a locomotor state

The worm study compared how egg-laying events occurred during different locomotor states. Dopamine-system manipulations changed the coupling between those events and roaming behavior. The outcome concerned coordination across programs: when and where eggs were deposited depended on how movement and the reproductive motor circuit were organized together within the animal's behavioral state.

The investigators also manipulated dopamine neurons and examined activity in an identified neuronal population during behavior. These methods placed a transmitter-defined system within the timing of the measured actions. The circuit's contribution could therefore be tested through changed output and recorded activity, rather than inferred solely from the presence of dopamine in the organism.

A neuromodulator can alter the probability that two motor programs occur together without supplying every patterned muscle command itself. Local circuits generate the actions, while chemical modulation changes their coordination across time. The study demonstrates this principle through whole-animal measurements tied to specific dopamine-related genetic and neuronal interventions in the worm.



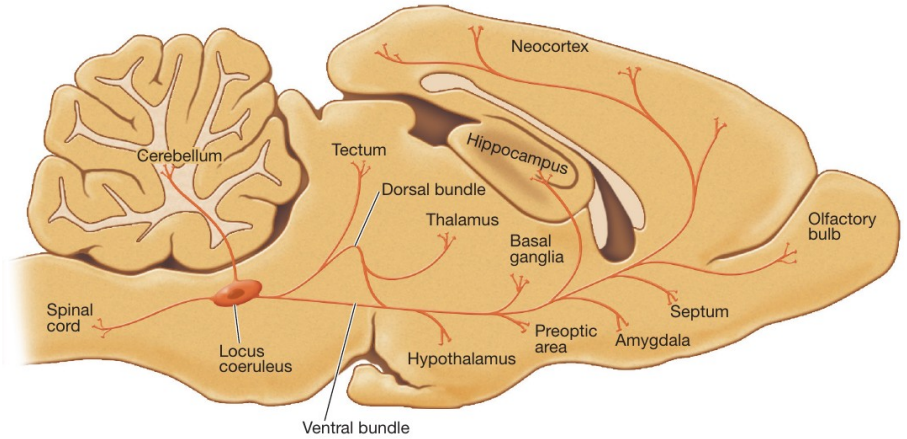
Cermak et al. (2020), Fig. 4 (A–C). Egg-laying and locomotor-state measurements.

Norepinephrine acts through several receptor types

Norepinephrine, also called **noradrenaline**, is produced by central neurons including those of the **locus coeruleus**, a brainstem nucleus. Its axons reach broad regions of the brain and spinal cord. These projections allow a transmitter-defined system to influence arousal, attention, sensory processing, and other functions through anatomically distinct output routes.

Alpha and **beta adrenergic receptors** are G protein-coupled receptor families activated by norepinephrine. Their subtypes engage different cellular signaling mechanisms. Alpha-2 receptors often reduce release when positioned presynaptically, while receptor actions in other locations can change excitability or intracellular signaling in the cells receiving the norepinephrine input.

The projection target determines which operation can be modified. Norepinephrine delivered to spinal pain-processing circuits can influence a different behavior from norepinephrine delivered to prefrontal circuits. A shared transmitter identity therefore does not require one shared psychological effect, especially when the source populations reach different cells with different receptors and functions.



Carlson and Birkett (2022), Fig. 4.18, p. 112.



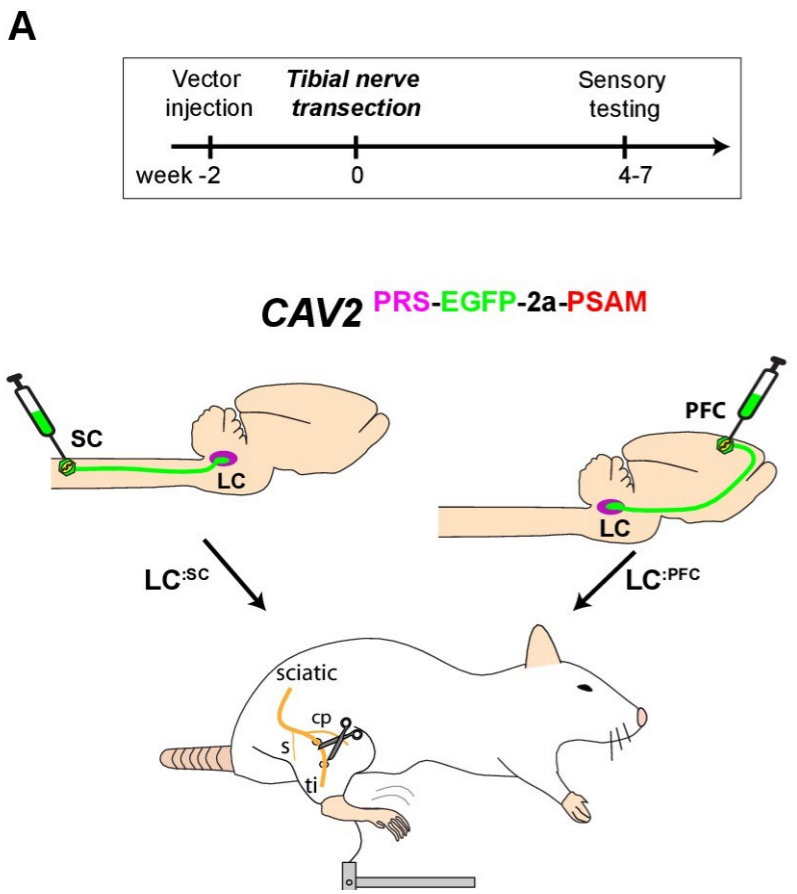
Hirschberg et al. (2017), video1. Pain behavior during projection-targeted activation.

A noradrenergic projection suppresses pain behavior

A locus-coeruleus study selected neurons according to whether their axons reached the spinal cord or prefrontal cortex. The investigators introduced an engineered ligand-sensitive channel into the chosen population. An administered compound could then activate those cells, allowing the contribution of a particular projection to be examined through a chemically controlled intervention.

Activating the spinal-projecting population reduced pain-related responses in rats, including a neuropathic-pain preparation. The study identified a contribution from spinal alpha-2 adrenergic signaling. The intervention therefore provided a route from a selected noradrenergic source, through its anatomical projection, to a receptor mechanism in the receiving region and a measurable behavioral outcome.

Activating prefrontal-projecting neurons produced a different pattern involving aversion and anxiety-related behavior. The contrast shows why a drug's neural target cannot be specified by transmitter name alone. Selecting the source population and its destination changed which circuits received the chemical influence, allowing different functions of the broader noradrenergic system to be examined separately.



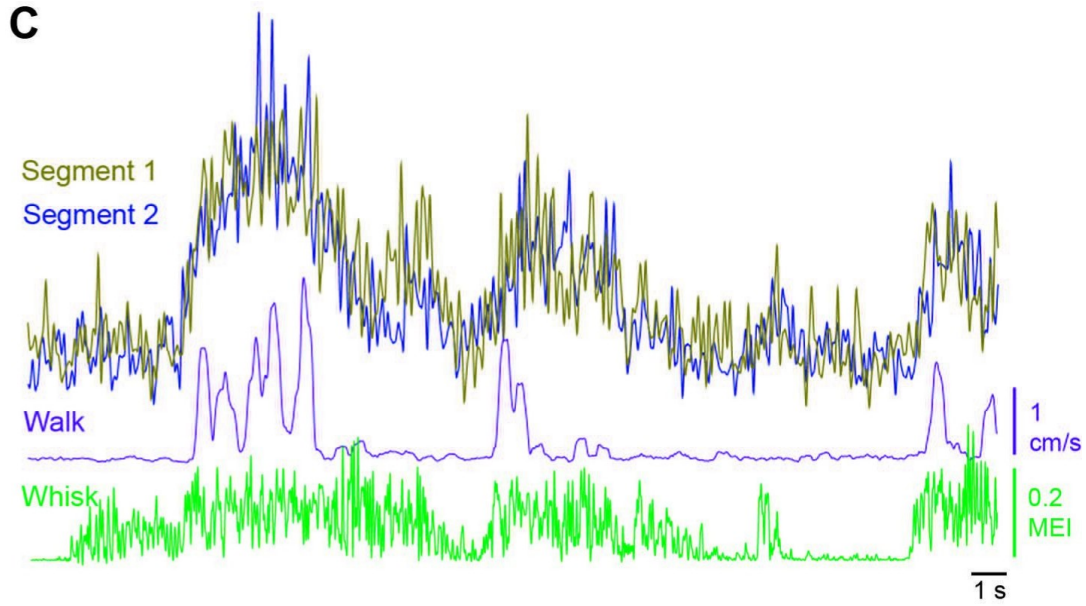
Hirschberg et al. (2017), Fig. 5 (A). Noradrenergic projection-targeting experiment.

Neuromodulatory axons carry a broad cortical state signal

The cortical imaging study recorded cholinergic and noradrenergic axonal segments separated across the dorsal cortex. Many segments showed coordinated activity around changes in arousal and movement. The observation indicated that broad chemical projection systems can transmit a common component of behavioral-state information across distant cortical locations during the same ongoing episode.

The investigators also detected variation among segments that was not fully accounted for by the measured state variables. Broad coordination therefore coexisted with differences within the projection populations. The findings identify both a shared signal and additional activity, avoiding the assumption that every axon from one transmitter system conveys an identical message at every moment.

A drug acting on these systems changes communication within that distributed setting. Receptors, local inputs, and ongoing activity differ across target areas, so a broadly delivered modulator can produce several effects through the receiving circuits. The recordings establish the behavioral context in which such chemical influences are distributed through the awake cortex.



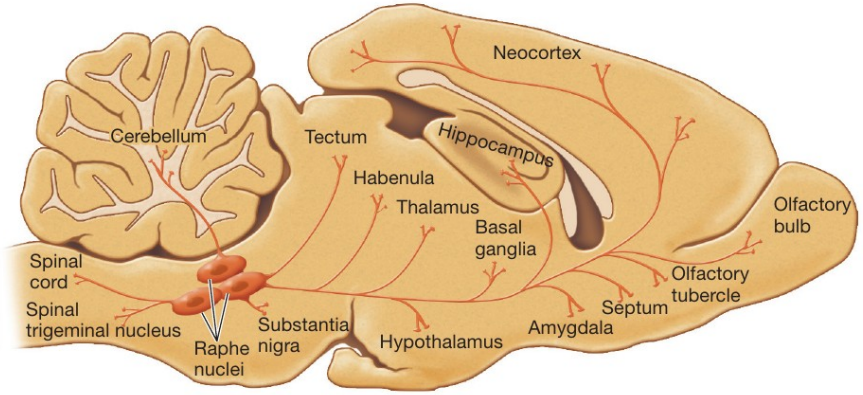
Collins et al. (2023), Fig. 4 (C). Activity along individual labeled axons.

Serotonin acts through several widespread pathways

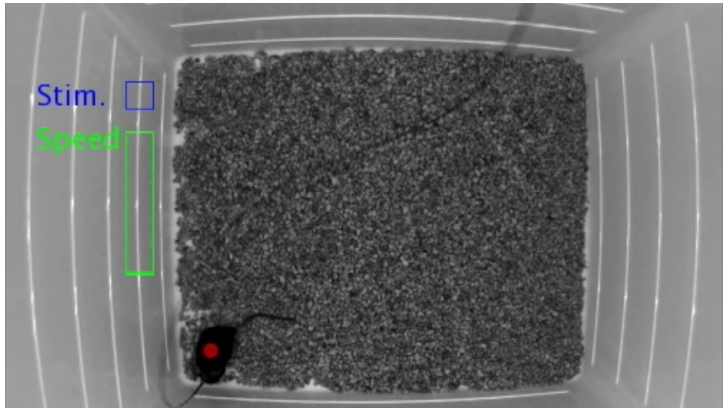
Serotonin, also called **5-hydroxytryptamine** or **5-HT**, is synthesized from tryptophan through enzyme-controlled steps. Major central populations arise in the **raphe nuclei**, a set of brainstem cell groups. Their projections reach many targets, allowing serotonin to influence several functions through circuits that differ in anatomy, receptor expression, and ongoing activity.

Most serotonin receptors communicate through G proteins, but the **5-HT3 receptor** is a ligand-gated cation channel. This distinction creates different cellular responses to the same transmitter. A compound acting at one subtype can therefore affect a different signaling process from a compound that changes serotonin uptake or activates another receptor family.

Selective serotonin reuptake inhibitors prolong transmitter availability by inhibiting its clearance transporter. Their rapid molecular action and the time course of clinical symptom changes are different measurements. Adaptation within cells and circuits contributes to later outcomes, so serotonin pharmacology cannot be explained as one immediate increase in a single universal mood signal.



Carlson and Birkett (2022), Fig. 4.19, p. 113.



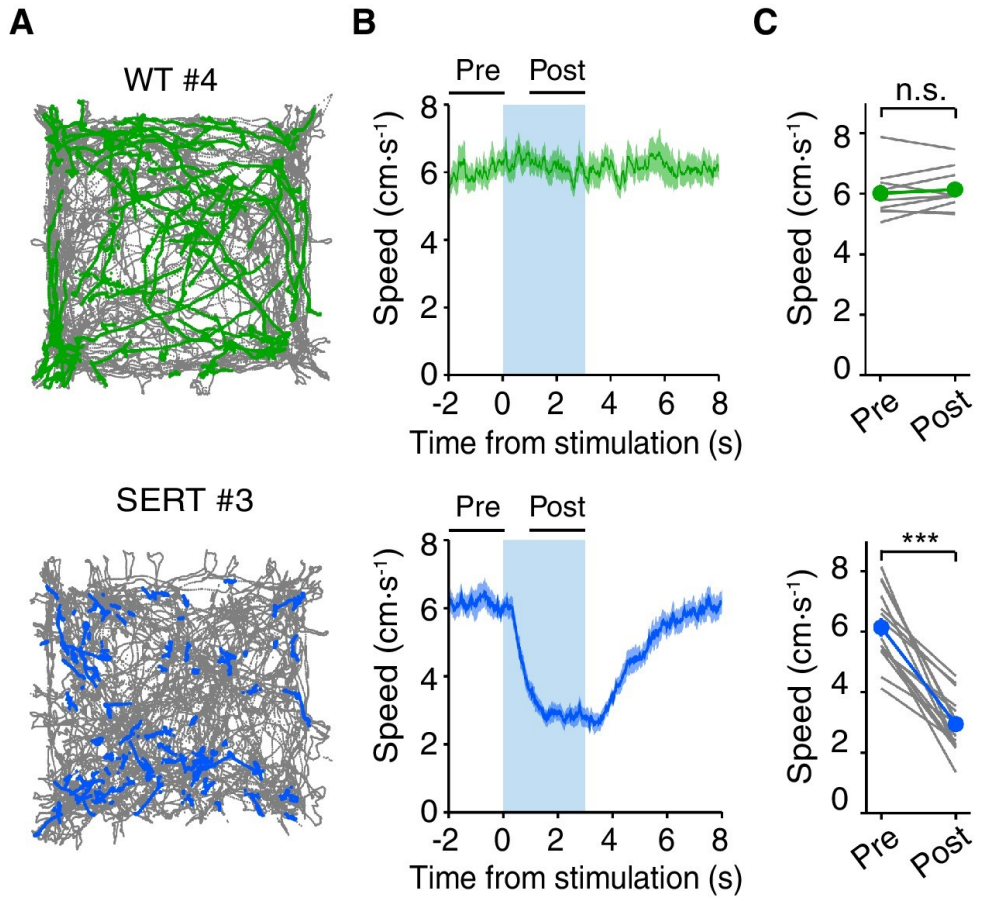
Correia et al. (2017), media1. Mouse movement during serotonin-neuron stimulation.

Serotonin-neuron activation changes movement across time

A serotonin study activated selected dorsal-raphe neurons in mice and measured movement in an open field. Brief activation rapidly reduced spontaneous locomotion under the tested conditions. The investigators separately examined coordination and other behaviors, allowing reduced movement to be interpreted within a defined task rather than equating it automatically with an inability to move.

Repeated daily activation produced a different longer-term outcome: spontaneous locomotion increased across the repeated-treatment period. The direction of the measured response therefore depended on the timescale. A transient change during activation and a persistent adaptation after repeated sessions described different consequences of manipulating the same transmitter-defined neuronal population.

The experiment shows why one short observation cannot stand in for every consequence of repeated chemical-system engagement. Serotonin neurons influence downstream circuits that adapt over time, and behavioral expression depends on the current task and state. Measuring immediate and later responses separates an acute effect from changes that accumulate with a repeated intervention.



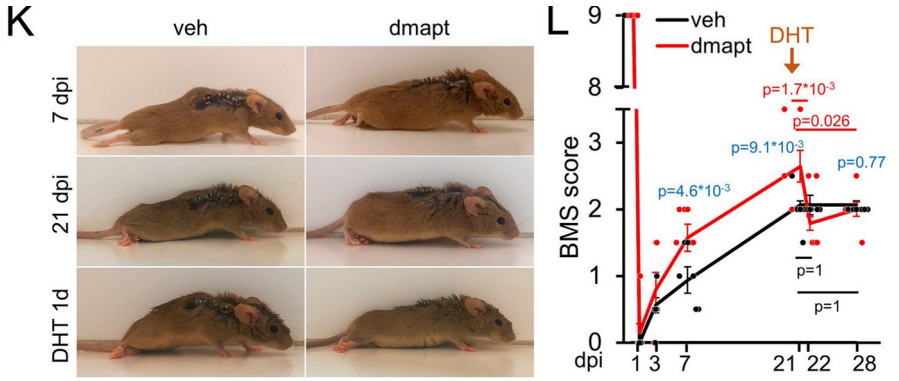
Correia et al. (2017), Fig. 2 (A–C). Locomotion during dorsal raphe stimulation.

Regenerated serotonin pathways aid recovered movement

A spinal-injury study tested **dimethylamino-parthenolide**, or **DMAPT**, a prodrug whose active form changes microtubule modification. A **prodrug** is converted within the body into an active compound. The investigators examined whether treatment could improve axon growth and locomotor recovery after injury, placing a chemical intervention within a defined anatomical repair problem.

The researchers then reduced serotonergic spinal innervation using a selective neurotoxic treatment. The locomotor improvement observed after DMAPT was diminished after this intervention. The comparison tested which regenerated pathway contributed to recovered function, rather than assuming that any increase in axon growth anywhere in the cord would explain the behavioral outcome.

The result connects a drug-induced cellular change with a particular chemical projection system. DMAPT altered growth-related machinery, while descending serotonergic pathways helped carry the functional benefit into spinal circuits controlling movement. The study concerns an experimental injury model and identifies a pathway contribution within that preparation, providing a mechanism to investigate through further pharmacological work.



Leibinger et al. (2023), Fig. 7 (K–L). Movement recovery after serotonin-pathway depletion.



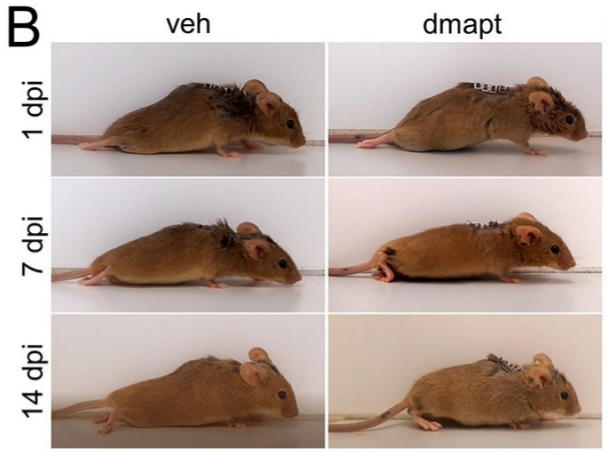
Leibinger et al. (2023), video4. Movement recovery before and after serotonin depletion.

A prodrug changes cellular machinery after injury

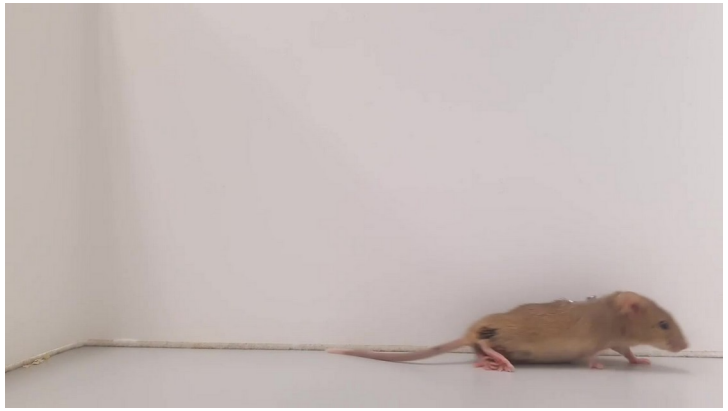
The DMAPT investigators examined **tubulin**, a major component of microtubules, and a chemical modification called **detyrosination**. This modification changes the properties of the intracellular tracks involved in growth. Parthenolide reduced detyrosination, giving the researchers a molecular target through which a drug could alter axonal behavior after nervous-system injury.

Systemic DMAPT treatment promoted regeneration and improved aspects of locomotion in the mouse spinal-injury experiments. The study followed behavior over several post-injury times alongside anatomical measurements of axon growth. These separate outcomes allowed the investigators to examine both the tissue response and the functional changes appearing during the period after treatment.

The intervention differs from directly activating a serotonin receptor. It acts on growth-related machinery, and the resulting repair changes which neural signals can reach spinal targets. The example broadens the sites-of-action account: a drug can affect behavior by altering cellular structure and pathway availability as well as by modifying moment-to-moment synaptic transmission.



Leibinger et al. (2023), Fig. 5 (B). Hindlimb posture after spinal injury.



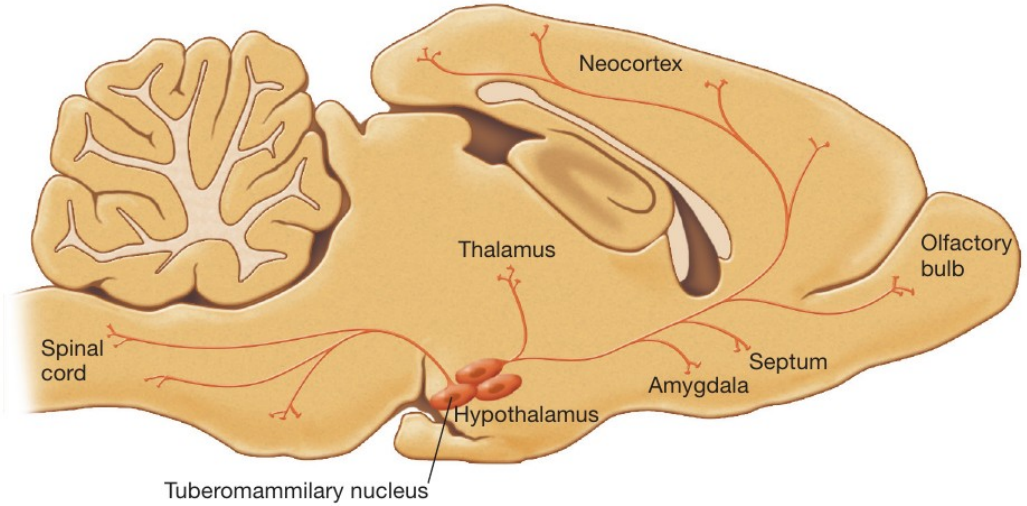
Leibinger et al. (2023), video2. Mouse movement at intervals after spinal injury.

Histamine receptor location changes behavioral effects

Histamine is synthesized from the amino acid histidine. In the brain, histamine-producing neurons arise mainly in the **tuberomammillary nucleus** of the hypothalamus and project widely. Their activity participates in wakefulness and related state regulation, giving this messenger a central role distinct from its actions in peripheral immune and vascular tissues.

Histamine acts through several receptor subtypes, including central **H1** and **H3** receptors. Some H3 receptors regulate transmitter release through presynaptic mechanisms. The receptor's subtype and location determine which cellular process changes after binding, so the same chemical name can describe signaling involved in several different physiological functions across the body.

Some H1-blocking antihistamines enter the brain and cause drowsiness, while compounds with limited central access can have less effect on arousal. The comparison applies distribution to behavior: blocking peripheral histamine-sensitive processes does not require the same degree of brain exposure. Receptor action and access together explain why related drug classes can differ in their central effects.



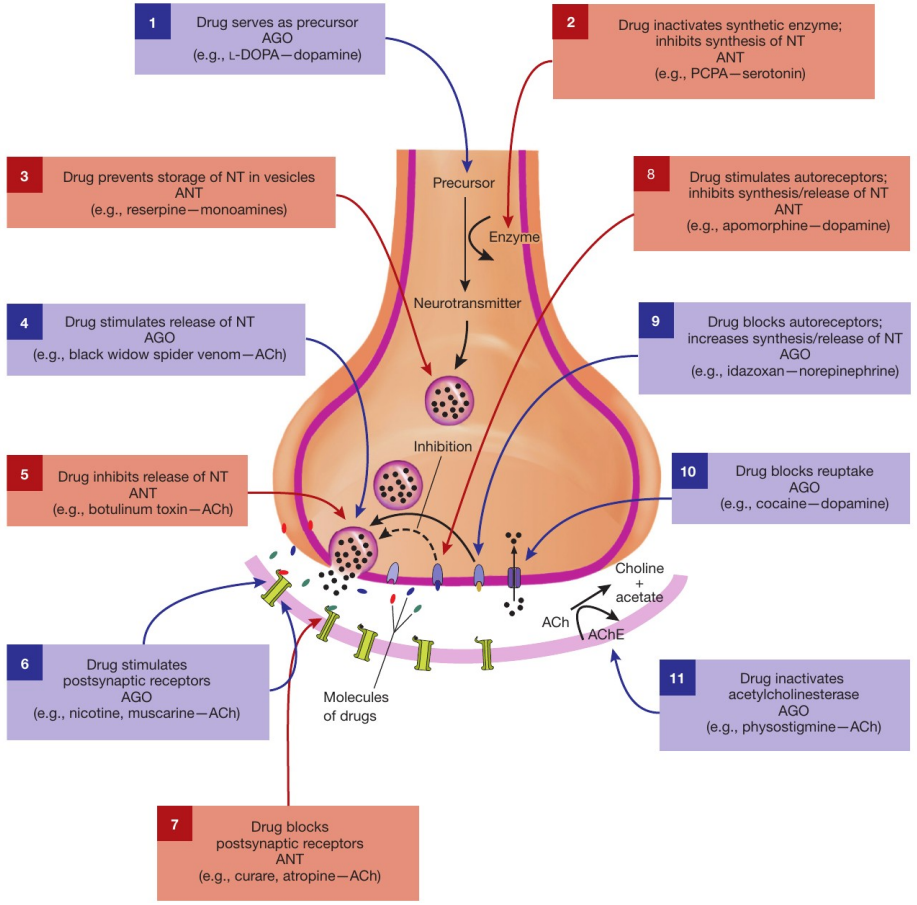
Carlson and Birkett (2022), Fig. 4.21, p. 114.

Peptide transmitters require different packaging

Neuropeptides are chemical messengers composed of short amino-acid chains. Cells synthesize larger precursor proteins, process them, and package the resulting peptides into secretory vesicles. This route differs from making a small classical transmitter through a short enzymatic pathway near the terminal, because peptide supply depends more directly on protein synthesis and processing.

A neuron can release a peptide together with a classical neurotransmitter. The messengers can act at different receptors and on different timescales, producing several components of one neuron's chemical influence. Identifying the cell's peptide marker therefore does not imply that it releases only that peptide or communicates through only one molecular target.

Peptide-sensitive G protein-coupled receptors can change excitability, release, or other cellular processes across a circuit. Examples include opioid peptides, orexin, and melanin-concentrating hormone. Their behavioral consequences depend on where they are released and which populations receive them, applying the same source-target-receptor framework used for the smaller classical transmitter systems.



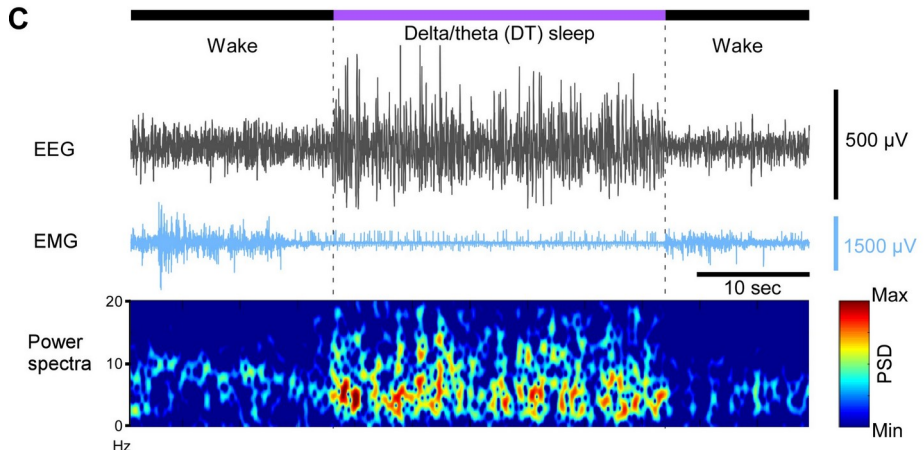
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Loss of two peptide-neuron populations alters sleep states

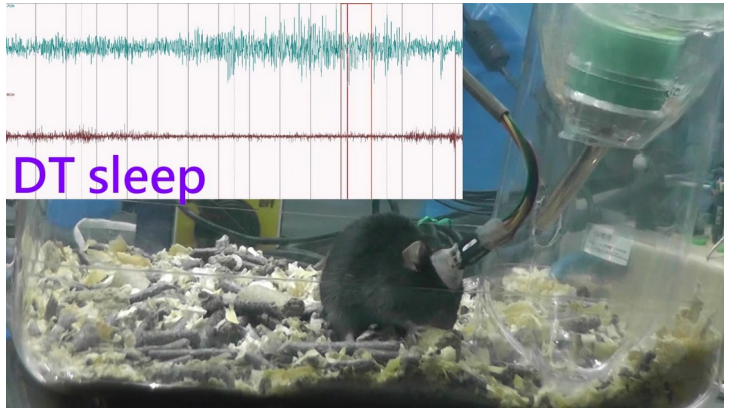
A hypothalamic study examined neurons producing **orexin**, also called **hypocretin**, and neurons producing **melanin-concentrating hormone**, or **MCH**. These populations occupy overlapping anatomical territory but have different contributions to sleep and wakefulness. The investigators generated mice in which both populations could be removed and compared their physiology and behavior with other experimental groups.

Double-ablated mice displayed abnormal sleep attacks with a combination of delta- and theta-frequency EEG activity. The researchers called this state **delta-theta sleep** and distinguished it from ordinary sleep categories using EEG, muscle activity, and behavior. The defined measurements established the altered state rather than relying only on whether the animal appeared still.

Removing a transmitter-defined cell population changes its outputs throughout the network, including any co-released messengers. The study therefore examined how these hypothalamic populations cooperate in state control. Their loss produced a pattern different from treating orexin and MCH as two interchangeable peptide signals acting through one identical pathway in the brain.



Hung et al. (2020), Fig. 3 (C). Delta–theta sleep with low muscle activity.



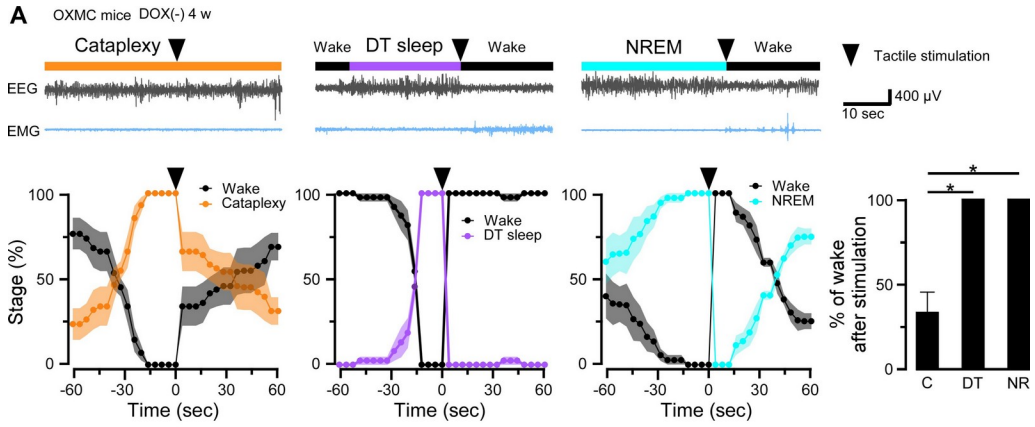
Hung et al. (2020), video1. Delta–theta sleep after two peptide-cell populations are removed.

Cataplexy differs from an ordinary sleep attack

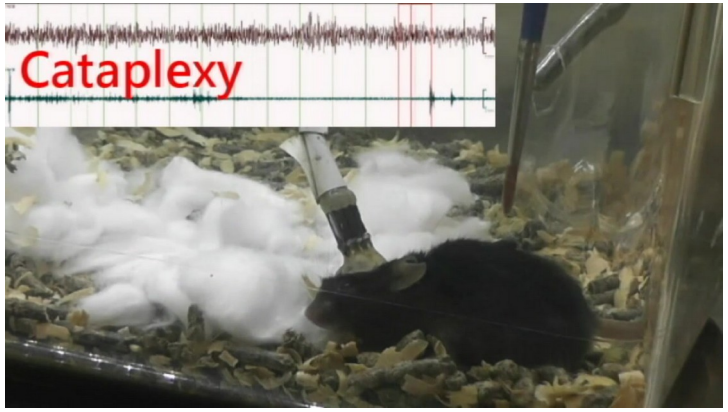
The orexin-MCH study compared delta-theta sleep with **cataplexy**, an episode of markedly reduced muscle tone associated with wake-related brain activity. The conditions can look similar when an animal becomes immobile, so the investigators used physiological signals and responses to tactile stimulation to distinguish the states within the same experimental preparation.

The mice lacking both peptide-neuron populations displayed severe cataplexy as well as the additional sleep-attack state. The comparison indicated that removing MCH neurons changed the outcome beyond loss of orexin neurons alone. The resulting behavior reflected interaction between populations rather than a simple one-to-one mapping from each peptide name to one visible movement.

A drug targeting an orexin receptor and an intervention removing orexin-producing cells act at different levels. One changes signaling at a selected molecular target; the other removes a cellular source and its broader outputs. Keeping those levels explicit helps interpret peptide-system experiments without assuming that every manipulation of a named system produces the same physiological state.



Hung et al. (2020), Fig. 6 (A). Electrical and movement responses to tactile stimulation.



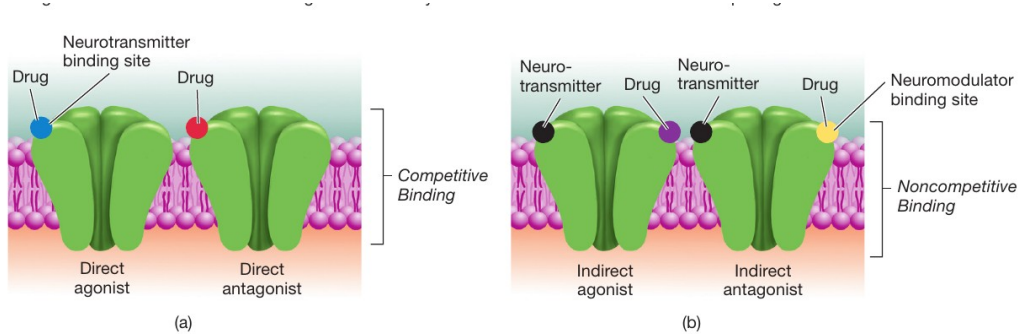
Hung et al. (2020), video2. Tactile stimulation during cataplexy.

Opioid receptors regulate pain and other functions

Opioid receptors include mu, delta, and kappa families activated by endogenous opioid peptides and selected exogenous drugs. These G protein-coupled receptors can reduce cAMP signaling, influence ion channels, and change transmitter release. Their occurrence across central and peripheral circuits gives opioid compounds several possible physiological actions beyond a single pain-related outcome.

Presynaptic opioid signaling can reduce calcium-dependent transmitter release, while postsynaptic actions can increase potassium conductance and reduce excitability. The behavioral result depends on which circuit elements are affected. Inhibiting an inhibitory neuron can disinhibit a downstream population, so a cellular reduction in activity does not always translate into a reduction of every output of the circuit.

Naloxone is an opioid antagonist that can reverse receptor-mediated effects of opioid agonists, including opioid-related respiratory depression. Its mechanism depends on receptor competition rather than repairing every consequence of an exposure. Differences in drug persistence and receptor activation explain why reversing an immediate effect and ending the entire course of an intoxication are distinct physiological outcomes.



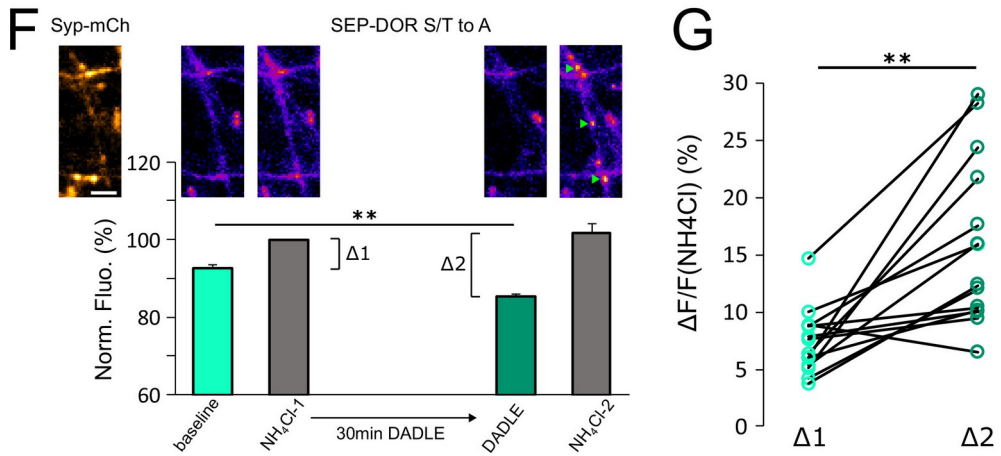
Carlson and Birkett (2022), Fig. 4.8, p. 102.

Receptor subtypes use different routes into tolerance

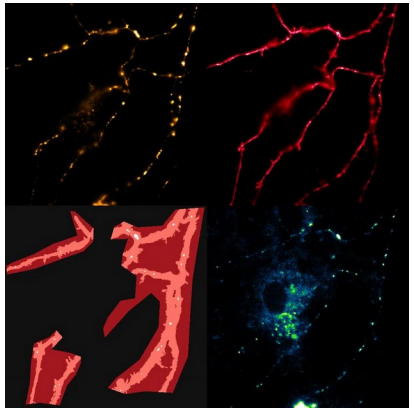
The opioid-tolerance study modified phosphorylation sites on the delta receptor and followed receptor behavior after agonist exposure. **Phosphorylation** adds phosphate groups to selected amino acids and can change interactions with cellular proteins. The experiment tested whether the same modification required for mu-receptor uptake was also necessary for the delta-receptor pathway in axons.

Delta receptors still underwent internalization and developed presynaptic tolerance without the corresponding tail-phosphorylation requirement. This contrasted with the mu-receptor findings in the same general preparation. The comparison identified a subtype-specific difference in how uptake was initiated, while both receptors could ultimately lose surface availability during prolonged activation.

A common physiological outcome can therefore arise through partly different molecular routes. Reducing a kinase-dependent process might alter adaptation for one receptor subtype more strongly than another. The study explains this difference through measured trafficking and release responses, placing receptor identity and intracellular regulation within the mechanism of sustained drug action.



Jullié et al. (2022), Fig. 6 (F–G). Mutant delta receptor recycling and repeated responses.



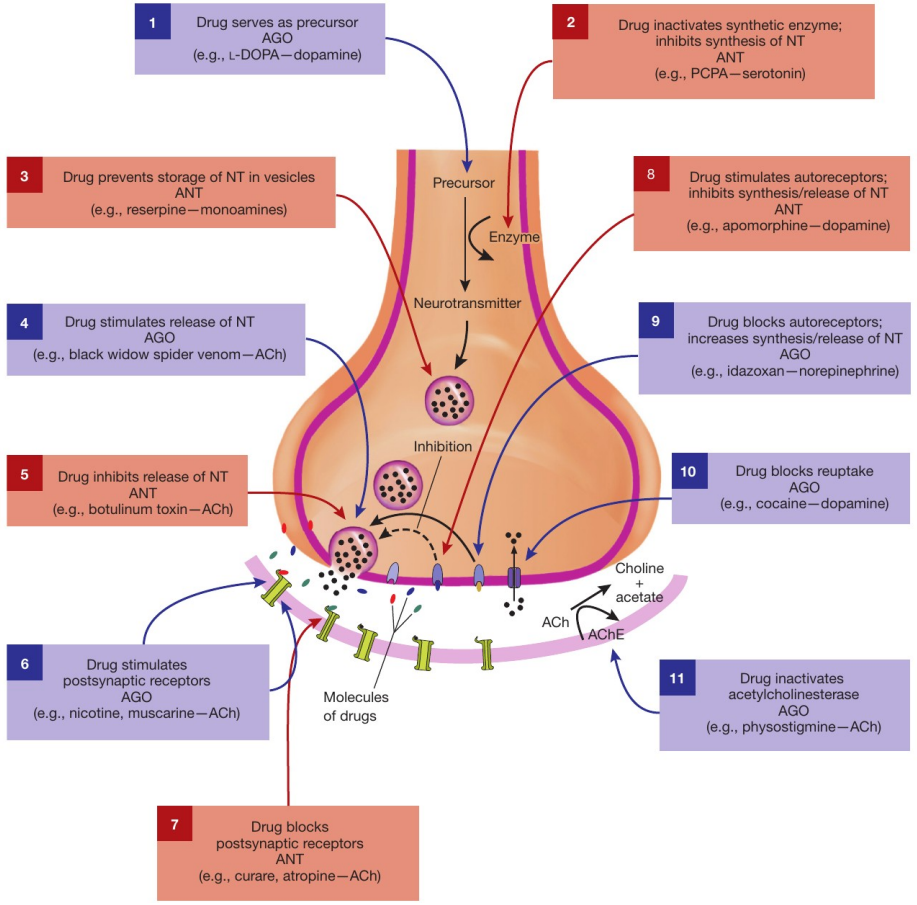
Jullié et al. (2022), fig6-video1. Mutant delta opioid receptor internalization in axons.

Endocannabinoids are synthesized on demand

Endocannabinoids are lipid-derived messengers produced within the body. Major examples include **anandamide** and **2-arachidonoylglycerol**, often abbreviated **2-AG**. Unlike many classical transmitters, they are generally synthesized in response to cellular conditions rather than stored beforehand in conventional synaptic vesicles awaiting release after an action potential reaches a terminal.

The **CB1 receptor** is widely expressed in neural circuits, while **CB2 receptors** are prominent in immune-related signaling and occur in additional cellular contexts. Both are G protein-coupled receptors. An exogenous cannabinoid can recruit parts of this endogenous signaling machinery, but the resulting effect still depends on exposure, receptor location, and the receiving cells.

Enzymes help limit lipid-messenger action after production. **FAAH** participates in anandamide breakdown, while **monoacylglycerol lipase**, or **MAGL**, is a major enzyme clearing 2-AG. Inhibiting synthesis, activating a receptor, and inhibiting degradation intervene at different steps, so they can produce different spatial and temporal patterns of cannabinoid-system engagement.



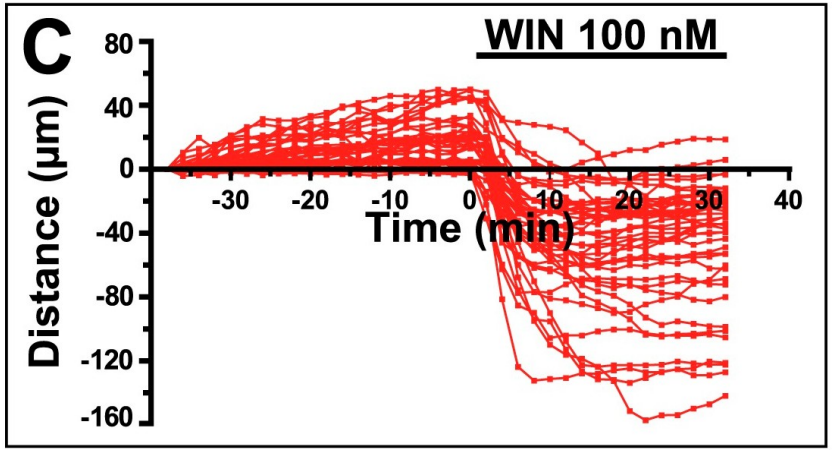
Carlson and Birkett (2022), Fig. 4.7, p. 100.

CB1 activation rapidly changes a neuronal growth cone

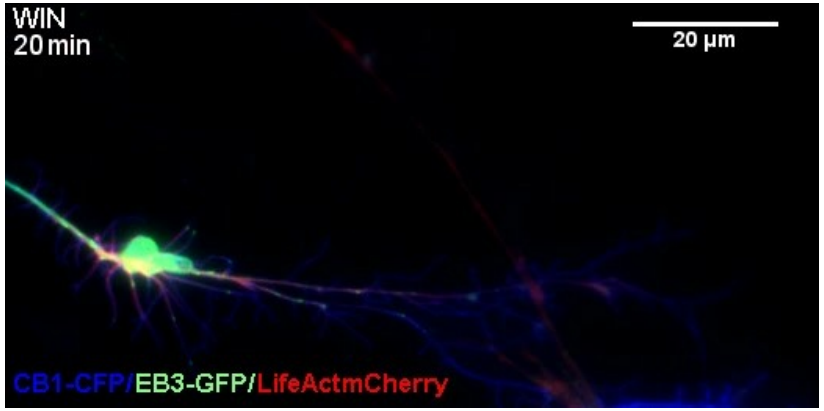
Roland and colleagues examined **growth cones**, dynamic structures at the tips of growing neuronal processes. They labeled actin and cannabinoid receptors in cultured hippocampal neurons, then applied **WIN 55,212-2**, a synthetic cannabinoid agonist. The preparation allowed receptor activation to be followed alongside a directly observable change in neuronal shape over time.

After agonist exposure, actin-rich growth cones retracted under the study's conditions. The response occurred in a cell structure that was already moving and reorganizing before treatment. The experiment therefore measured a change in an ongoing biological process, rather than inferring growth-cone behavior only from a fixed image taken after the exposure.

The result extends cannabinoid signaling beyond a simple description of reduced transmitter release. A receptor can also recruit intracellular machinery that changes the shape and mechanical behavior of a developing neuron. The authors used additional inhibitors and anatomical preparations to identify the pathway from receptor activation toward the cellular contraction producing the retraction.



Roland et al. (2014), Fig. 1 (C). Growth-cone movement during CB1 activation.



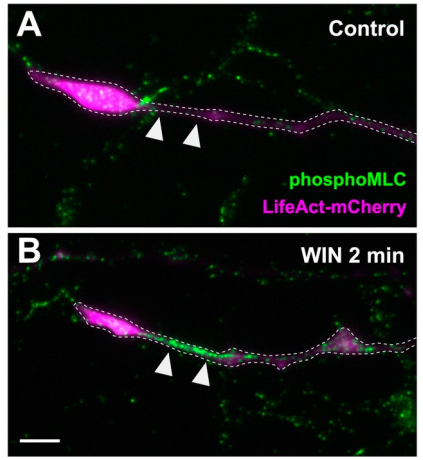
Roland et al. (2014), Video 1. Growth-cone retraction during CB1 activation.

Myosin contributes to cannabinoid-induced retraction

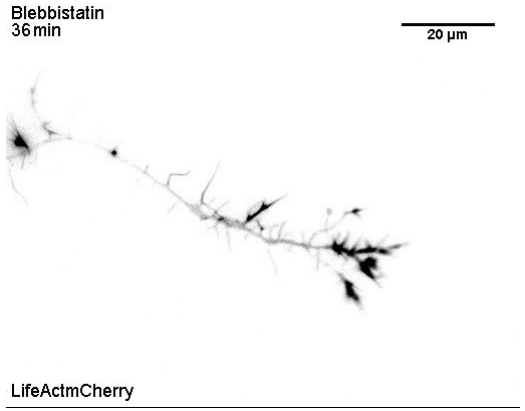
Roland and colleagues tested **non-muscle myosin II**, a motor protein that contracts actin networks within cells. They applied **blebbistatin**, an inhibitor of this motor, before activating cannabinoid receptors. The experiment asked whether receptor-dependent growth-cone retraction required contractile machinery, rather than resulting from transmitter binding alone without an intervening mechanical process.

Myosin inhibition reduced the retraction associated with cannabinoid activation in the tested preparation. The study also examined signaling through selected G proteins, **Rho**, and **ROCK**, proteins that regulate cytoskeletal organization and contraction. These experiments placed the receptor upstream of a cellular pathway capable of generating the observed change in shape.

The pharmacological comparison separated receptor engagement from its downstream effector. Blocking a motor could alter the structural response even while the agonist remained available. This distinction illustrates a general principle of drug action: interfering with a later intracellular step can change the outcome without competing directly for the original ligand's receptor-binding site.



Roland et al. (2014), Fig. 3 (A–B). Actin at a growth cone before and after CB1 activation.



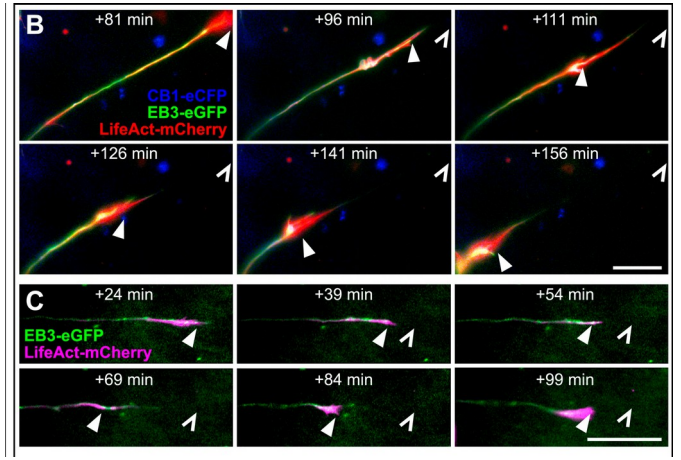
Roland et al. (2014), Video 4. Myosin inhibition before CB1 activation.

Cannabinoid activation also changes growth cones in slices

Roland and colleagues examined growth cones in **organotypic slices**, tissue preparations that preserve part of the original cellular arrangement while being maintained outside the animal. The observed processes included axons growing away from cortical regions. This preparation placed the drug-sensitive structure within a more complex tissue environment than a neuron grown alone on a dish.

The investigators applied the cannabinoid agonist while recording actin-rich growth cones. Retraction also occurred in this tissue preparation, allowing the cellular response to be examined in the presence of neighboring cells and preserved local architecture. The observation extended the range of preparations in which the identified receptor-dependent structural response could be studied.

The comparison matters mechanistically because cellular surroundings influence growth, transport, and mechanical constraints. A response in an isolated cell does not identify every tissue-level consequence, but observing the process in slices tests its operation in a different environment. The shared growth-cone behavior helped place the contractile pathway within developing neural tissue.



Roland et al. (2014), Fig. 4 (B–C). Growth-cone movement in brain slices.



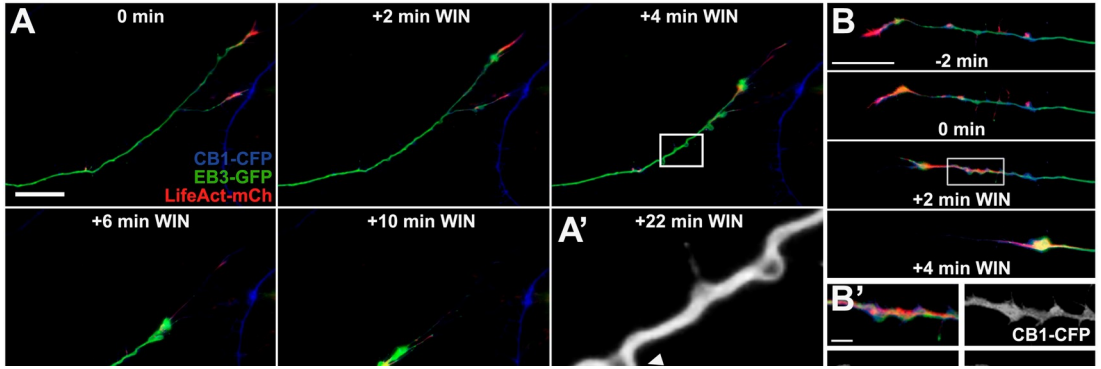
Roland et al. (2014), Video 5. Growth-cone retraction in brain slices.

Cytoskeletal inhibitors separate contraction from transport

The cannabinoid-growth study compared interventions affecting actin and microtubules, two components of the neuronal cytoskeleton. **Nocodazole** disrupts microtubules, while **cytochalasin D** interferes with actin organization. Applying these compounds before cannabinoid stimulation allowed the investigators to examine which parts of the cellular scaffold contributed to the retraction response and which served different roles.

The experiments identified actomyosin contractility as a principal effector of the rapid shape change. A growth cone contains both exploratory actin-rich regions and microtubules extending from the neuronal process, but these structures do not perform identical mechanical tasks. Selective perturbation separated their contributions within the same broader cell compartment.

This is pharmacological analysis of a pathway rather than a comparison of behavioral drug names. The agonist recruited a receptor, intermediate signaling proteins engaged contractile machinery, and the machinery changed the cell's shape. Inhibitors introduced at different positions helped establish the sequence of events connecting the initial chemical exposure with the observed structural outcome.



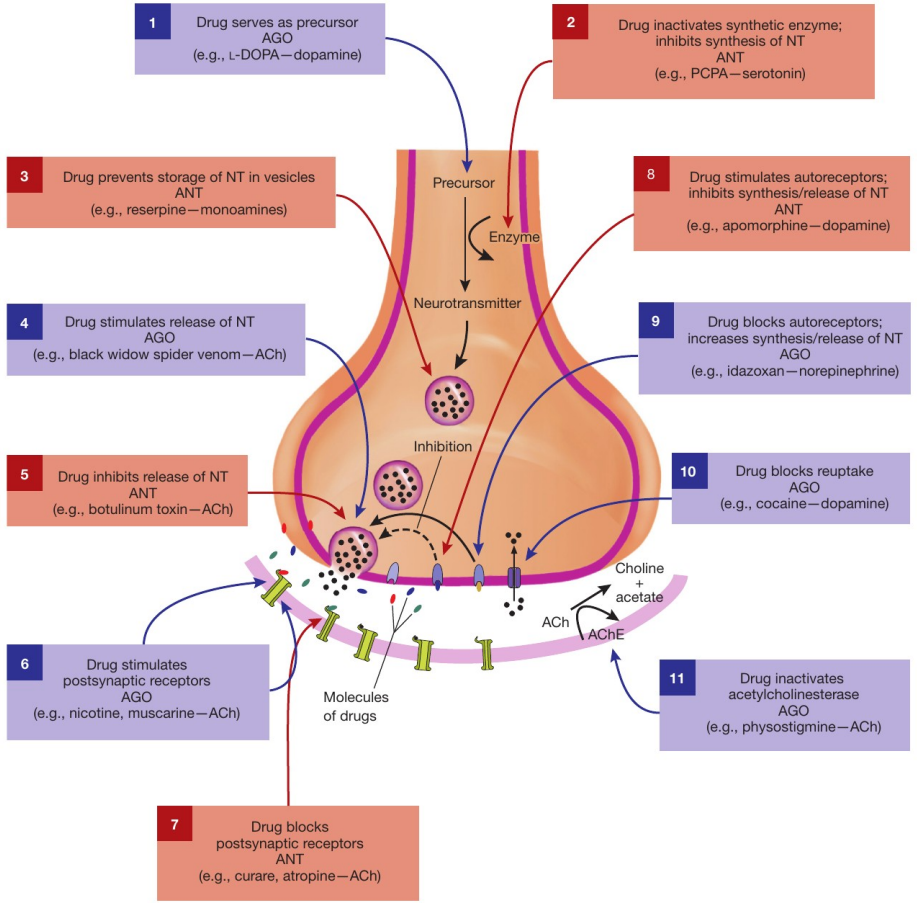
Roland et al. (2014), Fig. 2 (A–B). Cytoskeletal changes during CB1 activation.

Lipid messengers can signal back toward a nerve terminal

At many synapses, a postsynaptic neuron can produce an endocannabinoid after its activity or receptor signaling changes. The lipid messenger reaches presynaptic CB1 receptors and reduces further transmitter release. This **retrograde signaling** carries influence from the receiving side back toward the releasing side, reversing the direction of the conventional transmitter message across that synapse.

CB1 activation can alter presynaptic calcium-channel function and other release-related processes through G proteins. The effect depends on which terminal expresses the receptor. Reducing release from an excitatory terminal and reducing release from an inhibitory terminal produce different consequences for downstream neurons, even though both involve decreased output from the directly affected axon.

Lipid synthesis and degradation help determine when and where this feedback occurs. A receptor agonist delivered from outside the body does not necessarily reproduce that local timing or production pattern. Understanding endogenous signaling therefore requires specifying the source cell, the receptor-bearing target, and the enzymes controlling messenger availability, alongside the receptor's downstream molecular actions.



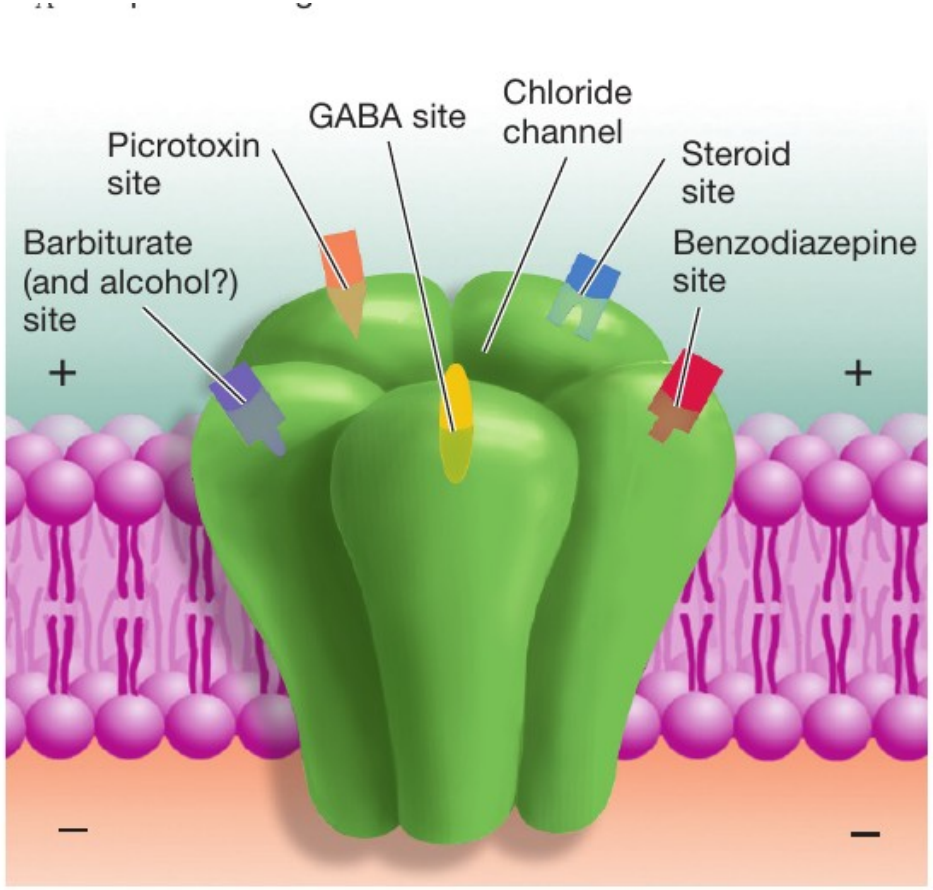
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Chloride gradients determine an anion channel's effect

Opening a chloride-permeable channel moves membrane voltage toward the chloride equilibrium potential, which depends on the ion's distribution across the membrane. Transport proteins help establish that distribution. The same receptor can therefore produce different electrical effects in cells with different chloride gradients, even when it opens a channel in response to the same chemical ligand.

A hyperpolarizing response can reduce firing during the immediate input, but membrane history also changes the availability of voltage-dependent channels. Some sodium channels recover from inactivation during hyperpolarization. A later depolarizing stimulus can therefore encounter a different pool of available channels from the one present before the change in membrane voltage occurred.

These processes distinguish an immediate voltage shift from the cell's response to a subsequent stimulus. Measuring only whether voltage moved downward does not specify every later consequence for excitability. Ion transport, resting potential, channel availability, and stimulus timing together determine how an anion-sensitive pathway changes the neuron's eventual output under the tested conditions.



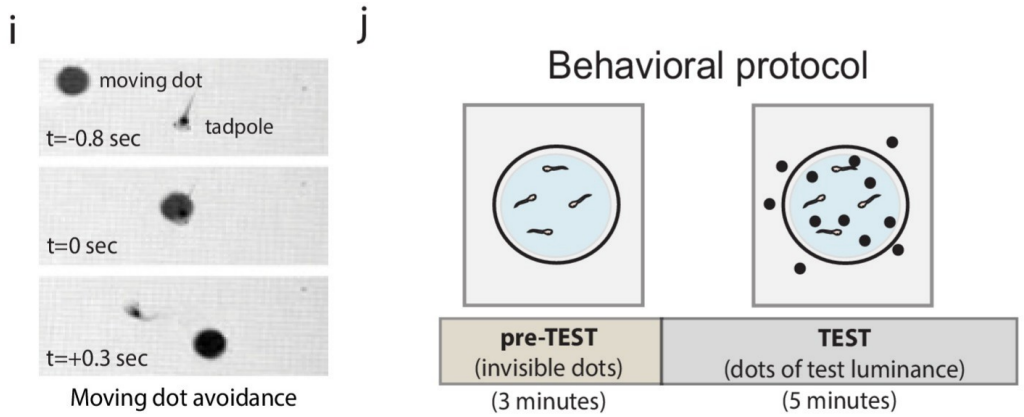
Carlson and Birkett (2022), Fig. 4.10, p. 105.

A cannabinoid study measures visual behavior in tadpoles

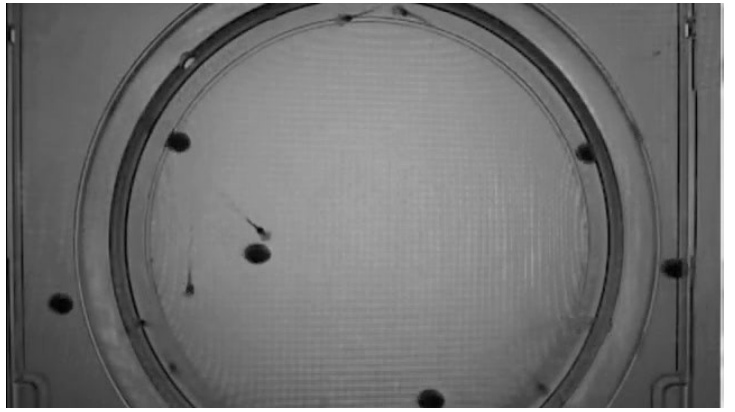
Miraucourt and colleagues examined **retinal ganglion cells**, the output neurons of the retina, and visual behavior in freely swimming **Xenopus** tadpoles. Their behavioral assay presented moving dark dots and measured avoidance. The task provided an observable response to a visual target while the study separately investigated the retinal mechanisms influenced by cannabinoid signaling.

Control tadpoles responded to the dots under brighter **photopic** illumination. The accompanying recording supplies a reference example of the animal and task in that condition. Comparing illumination levels changed the sensory challenge without requiring a different target shape or a different class of behavioral response to be introduced between the experimental conditions.

The investigators combined the behavioral assay with retinal imaging and electrical recordings. Those measurements addressed different stages between a chemical intervention and an action: receptor-sensitive cellular signaling, retinal output, and the animal's response to the visual stimulus. The study therefore tested a pathway from cannabinoid activation toward a defined sensory behavior rather than assigning one global effect to the drug.



Miraucourt et al. (2016), Fig. 8 (i–j). Moving-dot avoidance and behavioral protocol.



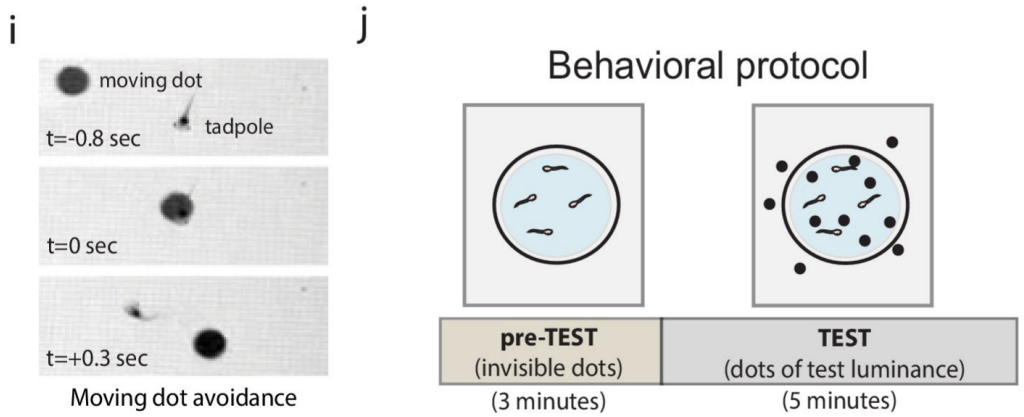
Miraucourt et al. (2016), media1. Control tadpole avoiding moving dots in brighter light.

CB1 activation improves vision under dim illumination

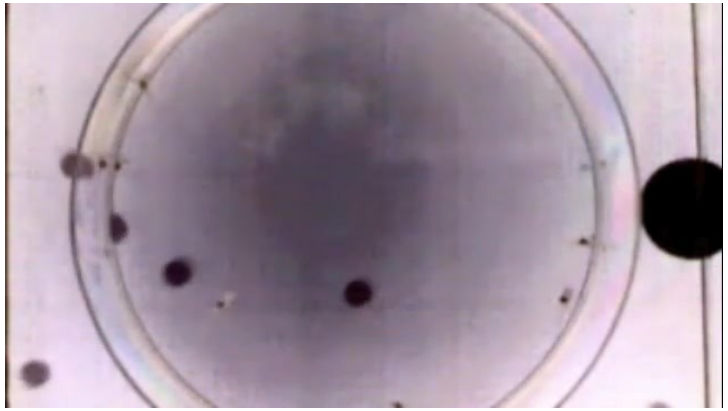
Miraucourt and colleagues treated tadpoles with a cannabinoid agonist and tested dot avoidance under **mesopic** illumination, an intermediate lighting range dimmer than the bright reference condition. Treated animals performed better in the visual detection task under the tested low-light conditions. The result concerned a defined sensory challenge rather than an unspecified increase in every behavior.

The study also measured retinal ganglion-cell responses and identified changes in intrinsic excitability. These cellular findings provided a mechanism to examine alongside improved behavior. A drug-sensitive sensory system can alter the information reaching the brain before downstream circuits select the movement, so the site producing a behavioral change need not be a motor command center.

The original treated-animal recording contains the actual dot-avoidance task under the study's dimmer lighting condition. Its interpretation depends on the control comparisons and retinal measurements. Together, these observations place the chemical effect within a specific sensory pathway and explain how altered neuronal responsiveness can change the probability of detecting an external target.



Miraucourt et al. (2016), Fig. 8 (i–j). Moving-dot avoidance and behavioral protocol.



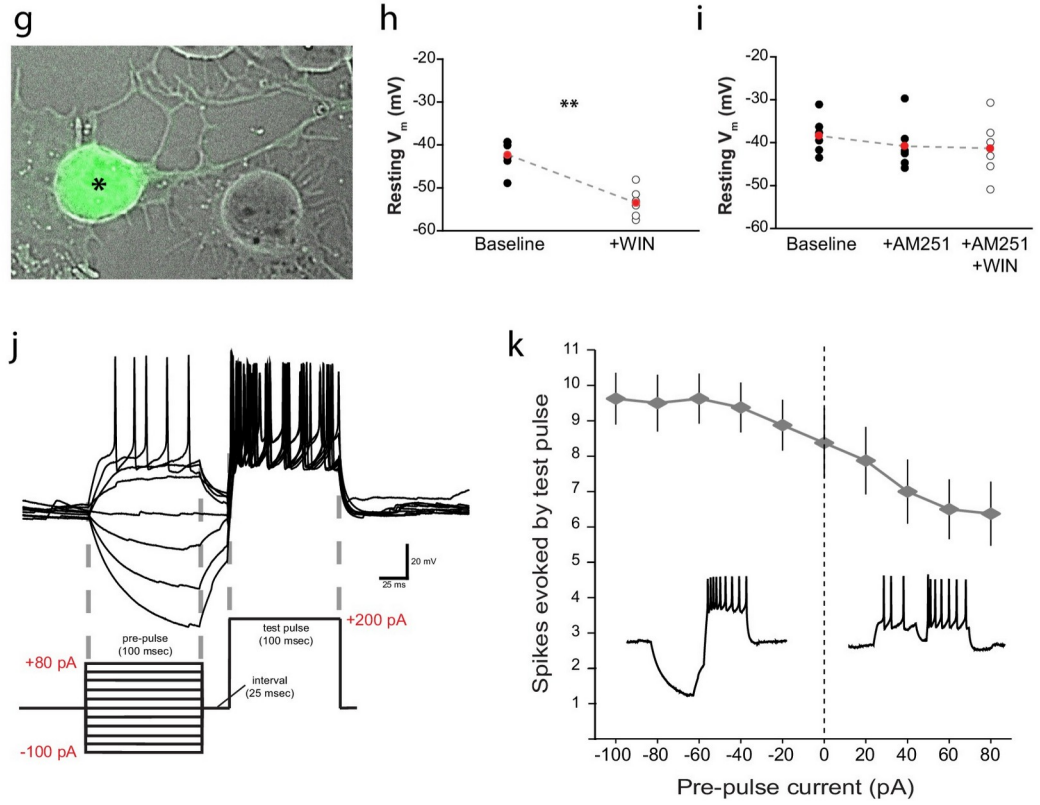
Miraucourt et al. (2016), media3. WIN-treated tadpole avoiding dots in dimmer light.

Cannabinoids change chloride-dependent excitability

The retinal study found that CB1 activation reduced intracellular chloride in ganglion cells through a pathway involving **AMPK**, a cellular signaling enzyme, and **NKCC1**, a membrane ion transporter. The investigators combined imaging with electrical measurements to examine how these changes altered the cells' resting properties and their response to later depolarizing input.

The chloride shift hyperpolarized the resting membrane potential while increasing subsequent action-potential responses under the tested conditions. This combination is possible because resting voltage and later responsiveness are distinct properties. Hyperpolarization can increase the availability of voltage-dependent sodium channels, allowing a later stimulus to recruit a different response from the same cell.

Pharmacological interventions at several points tested the proposed pathway, including glycine receptors, ion transport, and intracellular signaling. CB1 activation reduced intracellular chloride, changed resting voltage, and increased the later firing response. It illustrates how a receptor's behavioral consequence depends on the receiving cell's physiological machinery rather than on a simple label of excitation or inhibition.



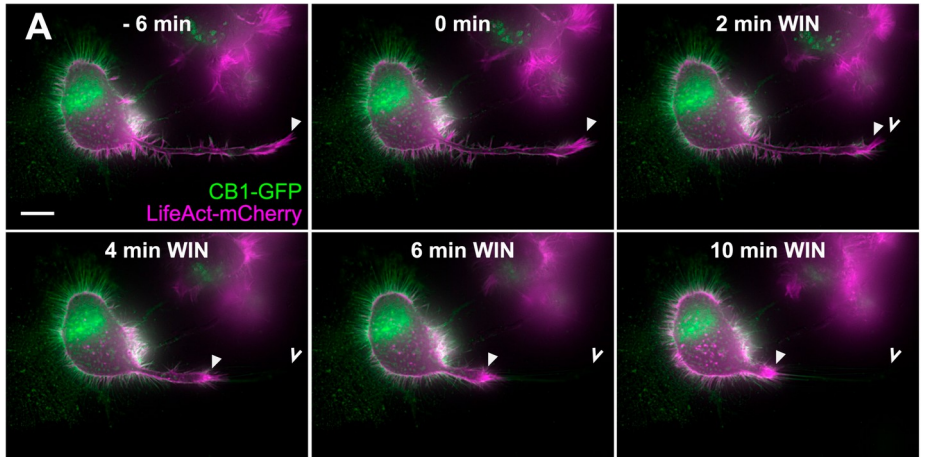
Miraucourt et al. (2016), Fig. 6 (g–k). Chloride measurements and retinal excitability.

Receptor activation changes the mechanical state of a cell

Roland and colleagues extended their cannabinoid experiments to **Neuro2A cells**, a cultured neural cell line with identifiable processes. After agonist exposure, processes retracted and the cells became rounder. The investigators followed actin organization and cell shape during the response, allowing a receptor-sensitive structural change to be measured directly over time in the preparation.

The study also examined mechanical properties and identified a transient increase in cell stiffness associated with the contractile response. Cell stiffness describes resistance to deformation, a property generated partly by the cytoskeleton. The finding therefore placed receptor signaling within measurable cell mechanics rather than treating a changed outline as the only outcome of the exposure.

A chemical signal can modify both communication proteins and the physical organization of a cell. In this preparation, CB1-related signaling recruited actomyosin machinery and changed structure. The experiment gives a concrete example of how a membrane receptor can influence behavior at the cellular level through effectors that generate force and reshape a living biological object.



Roland et al. (2014), Fig. 6 (A). Cell rounding during CB1 activation.



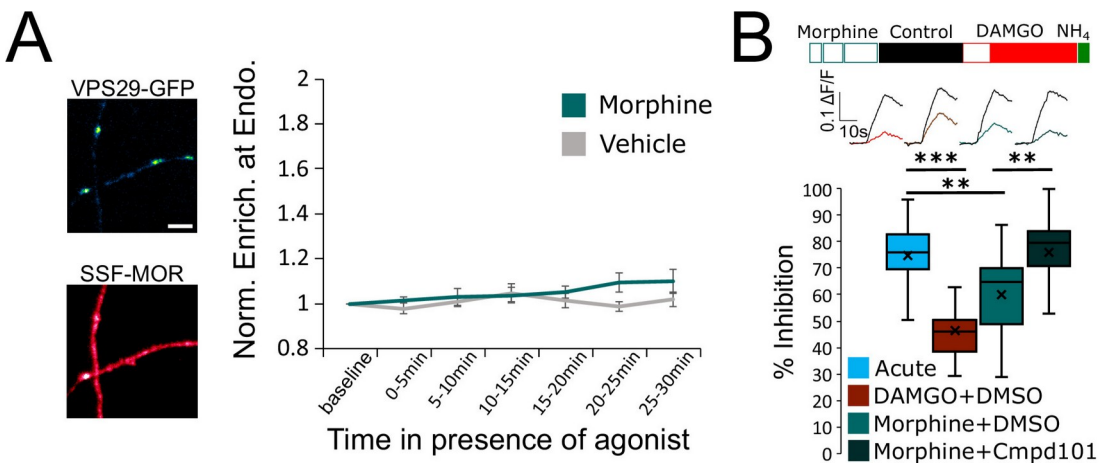
Roland et al. (2014), Video 6. Cell rounding during CB1 activation.

Agonists differ in receptor uptake and cellular tolerance

The opioid-tolerance investigators compared agonists that activated the same receptor system but differed in how strongly they recruited receptor phosphorylation and endocytosis. A **partial agonist** produces a smaller maximum response than a full agonist in a specified assay. Differences between signaling pathways also allow agonists to recruit some measured processes more effectively than others.

In the cultured presynaptic preparation, compounds with less ability to drive phosphorylation-dependent mu-receptor uptake produced less tolerance over the prolonged exposure. The comparison placed ligand properties beside receptor movement and the changing physiological response. Activating the receptor initially and producing later adaptation were therefore related but experimentally separable outcomes of drug treatment.

The result explains why ligand selection can change the time course of a cellular response, even within one receptor family. An agonist's binding, efficacy, and recruitment of regulatory machinery influence what happens after continued exposure. These measurements identify a mechanism in the tested cells without reducing the full set of organism-level opioid effects to one cellular assay.



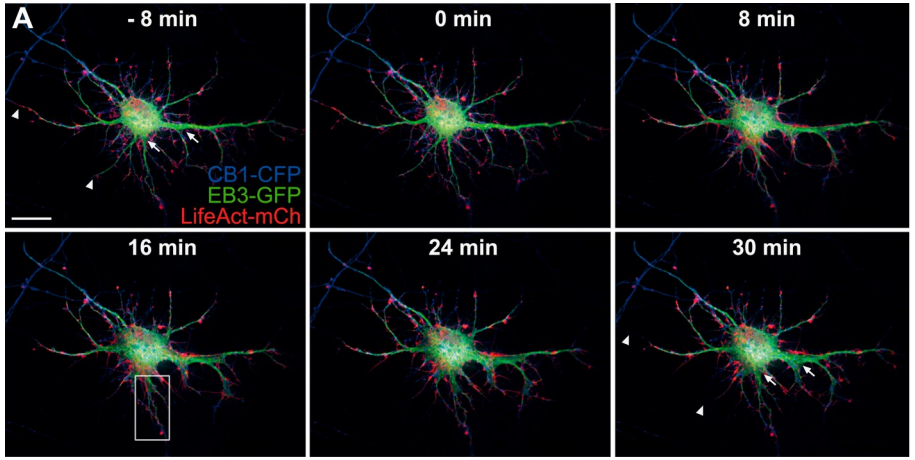
Jullié et al. (2022), Fig. 3 (A–B). Morphine-induced trafficking and presynaptic tolerance.

Cannabinoids reshape several neuronal compartments

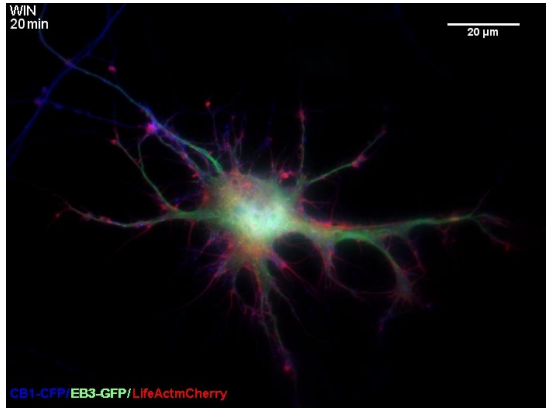
The cannabinoid-growth study also examined the **somatodendritic region**, which includes the cell body and dendrites, in cultured hippocampal neurons. Acute receptor activation changed dendritic shape and actin organization. The response extended the structural mechanism beyond the axonal growth cone, placing the same signaling system within several recognizable compartments of a developing neuron.

The investigators distinguished rapid remodeling from effects measured after longer exposure or interference with contractile machinery. The timescale of treatment changed which structural outcome was being examined. A receptor-driven process can therefore contribute to an immediate mechanical response and to later neuronal organization through related downstream components operating over different periods.

The study brings drug action back to a complete causal sequence: a compound reaches a receptor, the receptor recruits intracellular signaling, effectors alter cellular function, and the changed cell contributes to a larger neural system. Identifying each step explains why access, target location, exposure history, and receiving-cell machinery all matter when describing how chemicals change behavior.



Roland et al. (2014), Fig. 7 (A). Somatic and dendritic changes during CB1 activation.



Roland et al. (2014), Video 8. Somatic and dendritic changes during CB1 activation.

Key takeaways

Exposure determines access. Absorption, distribution, metabolism, and excretion govern the amount of drug reaching a target. Membrane permeability and transporters also determine access to receptors inside cells.

Binding and response are separate properties. Affinity describes binding, potency describes the amount needed for an effect, and efficacy describes the response a drug can produce. Agonists activate receptors; antagonists prevent activation.

Synaptic drugs act at several steps. Synthesis, vesicle storage, release, receptor activation, reuptake, and enzymatic breakdown each provide targets for changing neurotransmission.

Repeated exposure changes drug responses. Tolerance reduces a response to the same exposure, whereas sensitization increases it. Receptor trafficking and cellular adaptations can change signaling during repeated treatment.

Transmitter identity does not specify one behavior. Glutamate, GABA, acetylcholine, monoamines, peptides, and lipid messengers act through distinct receptors and pathways. Receptor location and circuit target determine the resulting cellular and behavioral effects.

Experiments connect targets to behavior. Identify the manipulated molecule, the affected cell, and the measured response. Receptor imaging, transmitter sensors, pathway targeting, and behavioral recordings reveal different steps in a drug's action.

Carlson & Birkett (2022); cited primary studies